Nontrivial Summability and the Split in UOM and 2T-LS: Completion-Sector Leak, Classicality, and Theoretical Derivation

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Abstract

This note studies two coupled structures in Universe-Originating Measurement (UOM): *(i)* the nontrivial summability of the Brown–York (BY) increments $\{\Theta_k\}$ and *(ii)* the split as exposure of a completion sector into the next aeon's source. The leak term in the SK/HJ wall balance is formulated as a completion-sector export flux controlled by a signed odd split amplitude and carried by hidden Stinespring/commutant degrees of freedom. The note derives the continued-sector RP coefficient $\alpha(\Lambda)$, a parity-based rank-one wall-match normal form, a projected Gaussian affine realization of the odd split mode, a centered/squeeze-dominated specialization of that affine dynamics, a positive decomposition of the split-driving leak into fresh-injection and transport pieces, a low-rank projected stock law, and an exported characterization interface for the canonical odd representative used downstream in $\mathbf{2T}_{\mathrm{Main}}$. Where UOM does not determine a unique completion Hamiltonian, export projector, or one-wall/two-wall matching, the corresponding steps are left as explicit *hypotheses*. Classicality is assigned to the coarse-grained accessible ledger, whereas RP/OS oddness is assigned to the analytic-continued completion sector.

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1 Scope and inputs

Primary sources (internal). We take as authoritative the UOM main manuscript and the 2T-LS rigorous translation note. The simulators are used only as *phenomenological inputs*: they probe reduced scalar closures of the wall balance and help identify which qualitative leak behaviors do or do not produce split. They are not treated as the primary source of theoretical definitions.

What "nontrivial summability" means here. There are two levels:

1. **Within one aeon:** beyond the trivial telescoping identity that follows from the bookkeeping update, the observed phenomenon is that the *implicit one-wall* Λ*-solve* can dynamically adjust the cutoff trajectory so as to avoid overshoot *at all finite ticks* while still exhausting the curvature budget:

$$0 < \Theta_k < R_k^{\mathrm{abs}} \text{ for all finite } k, \qquad R_k^{\mathrm{abs}} \downarrow 0, \qquad \sum_{k\geq 0} \Theta_k = |R_0|\,.$$

   This is what we call *nontrivial (exact) summability* in this note; the "flip" is an asymptotic boundary point of the solvable one-wall channel rather than a finite threshold crossing.

2. **Across aeons:** in the exhaustion/terminal branch the total available wall-work is summable (UOM Theorem `thm:ads5-static`), yielding a terminal cutoff Λ_∞ and decoupled tail behavior.

What "split" means here. Operationally, a split is the end-of-aeon event in which the construction produces a *new* dS source branch and a *new* AdS receiver branch. In UOM this is treated as compatible with BY charge conservation across a pillbox around the split region (UOM App. `app:energy-split`). In 2T-LS, the split/continuation has a mathematically clean signature: a parity cocycle ω whose nontrivial holonomy obstructs a global "splitting" (sectioning) of a $\mathbb{Z}_2$-cover (2T-LS Theorem `thm:non-splitting`).

2 Mathematical objects: states, channels, and per-tick scalars

2.1 State spaces and admissible coarse views

Let $\mathcal{H}_{\mathrm{tot}}$ denote the total microscopic Hilbert space of the composite (source patches, commutant/bath bookkeeping, receiver degrees of freedom, and any bulk/shadow auxiliaries retained). We write $\mathcal{D}(\mathcal{H})$ for density matrices on $\mathcal{H}$.

UOM frequently avoids assuming a literal tensor factorization for "accessible" degrees of freedom and instead uses operator-algebra restrictions and conditional expectations. A key UOM definition (label `def:eta-bridge-leak`) introduces:

- a conditional expectation $\mathbb{E}_{\mathrm{ptr}}$ onto a pointer algebra, and
- a commutant record map $\mathcal{R}_k$ with commutative range (classical record algebra),

together with a bridge/leak instrument decomposition $\Phi_k = \mathcal{B}_k + \mathcal{L}_k$ of the tick CPTP map.

2.2 Source-side QND dephaser and exported payload ΔI_k

The source-side mechanism is a pointer-aligned QND absorption channel with generator of the form (UOM label `lem:exp-tail`):

$$\mathcal{L}(t)\rho \;=\; -\frac{1}{2}\sum_a \gamma_a(t)\,[L_a,[L_a,\rho]], \qquad L_a = f_a(H_{\mathrm{sp},\Delta}), \qquad [L_a, H_{\mathrm{sp},\Delta}] = 0. \tag{1}$$

Across a tick window $I_k = [t_-, t_+]$ in a steady-actuator regime one obtains exponential contraction of the modular free energy $F(t) = S(\rho_t \| \sigma_\Delta)$ relative to a band-limited BD/KMS anchor σ_Δ:

$$F(t_+) \leq e^{-\beta_k} F(t_-), \qquad \Delta I_k := F(t_-) - F(t_+) \geq (1 - e^{-\beta_k}) F(t_-). \tag{2}$$

In reduced scalar closures this appears as:

$$\Delta I_k \approx (1 - e^{-\beta_k})\, F_k, \qquad F_{k+1} = F_k - \Delta I_k, \tag{3}$$

where F_k is represented by the scalar source stock `F_dS` in `aeon_driver.py`, and where β_k is computed from a schedule factor times the source Hubble scale and (in v1) a band-fraction modifier.

Matrix-element form (for later classicality bounds). The closed form of the dephasing channel on the band is (UOM label `eq:app-phi-closed`)

$$\Phi_\Delta^{(k)} = \exp\Big\{ -\tfrac{1}{2} \sum_a \Gamma_a^{(k)} \, \mathrm{ad}_{L_a}^2 \Big\}, \qquad \Gamma_a^{(k)} = \int_{\mathbb{R}} \gamma_a^{(k)}(t)\, dt. \tag{4}$$

In a common eigenbasis of $\{L_a\}$, $\mathrm{ad}_{L_a}^2$ acts diagonally on matrix elements, yielding suppression of off-diagonals by factors $\exp(-\Gamma_{ij}^{(k)})$ with $\Gamma_{ij}^{(k)} = \frac{1}{2}\sum_a \Gamma_a^{(k)} (\ell_a(i) - \ell_a(j))^2$.

2.3 Receiver-side BY rectifier increment Θ_k

UOM defines the BY anomaly/rectifier increment (UOM label `eq:theta-def`) as

$$\Theta_k := \gamma\, \Delta\langle R^2 \rangle_k \;=\; \underbrace{\frac{aG_4}{12\pi}}_{=\gamma} (8\pi G_4)^2 \left\langle |\mathcal{K}|^2 \right\rangle_k C_{\mathrm{LoG}} \, \frac{A_k^2}{\sigma_{t,k}\, \sigma_{x,k}^3}. \tag{5}$$

Equivalently, in the welded/throughput form (UOM label `eq:R2`) one writes a per-tick BY penalty

$$\Phi_{\mathrm{BY},k} = \gamma (8\pi G_4)^2 \, C_{\mathrm{LoG}} \, \frac{\Delta I_k}{s_{t,k}\, s_{x,k}^3 \, G(s_{t,k}, s_{x,k}; \Lambda_k)}. \tag{6}$$

In reduced phenomenological closures, Θ_k is evaluated by a discretized Euclidean BY kernel integrating over a frequency grid and an S^3 harmonic grid, with:

- welded widths $s_t(\Lambda) = \alpha_t/\Lambda$ and $s_x(\Lambda) = \alpha_x/(v_{\mathrm{geom}}\Lambda)$,
- an evanescent suppression across the brane gap d_{sep} as $\exp(-d_{\mathrm{sep}} q)$ with $q = \sqrt{\omega^2 + k^2 + m_g^2}$,
- screening $K = q/(q + r(\Lambda)k^2)$ with $r(\Lambda) \simeq \ell_{\mathrm{AdS5}}/2 + C_{m3}\Lambda^2$,
- degeneracy weights $D_\ell = (\ell + 1)^2$.

Remark 2.1 (Linearity in ΔI_k and the effective kernel $K(\Lambda)$). Both the manuscript and the code implement Θ_k as *linear* in the exported payload ΔI_k when the welded shape family is fixed: the reduced closure chooses an amplitude A_{eff} so that (heuristically) "signal power" equals ΔI_k, and then integrates a quadratic response which is proportional to $A_{\mathrm{eff}}^2 \propto \Delta I_k$. Thus one can write

$$\Theta_k = K(\bar{\Lambda}_k; d_k, L_{4,k}, \mathrm{locks})\, \Delta I_k, \tag{7}$$

for a (state-dependent) kernel factor K extracted numerically as $\Theta_k/\Delta I_k$. This factorization is explicit in UOM's receiver evolution summary (label `par:uom-main-138`).

3 Nontrivial summability within an aeon

3.1 Curvature bookkeeping and exact summability

UOM's receiver-side coarse evolution tracks a signed curvature variable (AdS negative, dS positive) or equivalently its magnitude $|R|$. The simulators encode the update in one of two equivalent ways:

- **v1 (magnitude bookkeeping):** $R^{\text{abs}}_{k+1} = R^{\text{abs}}_k - \Theta_k$ with $R^{\text{abs}}_0 = |R_0|$.
- **v2 (signed bookkeeping):** $R_{k+1} = R_k + \Theta_k$ with $R_0 = -|R_0| < 0$.

In either case, $\Theta_k \geq 0$ drives the receiver toward the flip.

Remark 3.1 (Legacy "first-crossing" convention). For comparison with UOM's manuscript language, one sometimes declares a flip at the first N such that $S_N := \sum_{k=0}^{N} \Theta_k \geq |R_0|$. In the exact-summability regimes observed in the implicit Λ-solver experiments (no-leak and fixed-work-percentage (i.e. proportional) leak models), one instead finds $S_N < |R_0|$ for all finite N but $\lim_{N\to\infty} S_N = |R_0|$. Thus there is no finite "flip tick": the flip is realized as an asymptotic boundary point of the solvable one-wall channel.

Proposition 3.2 (Telescoping identity (exact curvature-budget summation))**.** *Assume the within-aeon update is $R^{\text{abs}}_{k+1} = R^{\text{abs}}_k - \Theta_k$ without additional clamps and that $\Theta_k \geq 0$. Then for any N,*

$$\sum_{k=0}^{N} \Theta_k = R^{\text{abs}}_0 - R^{\text{abs}}_{N+1}. \tag{8}$$

In particular, if the update runs indefinitely with $R^{\text{abs}}_{N+1} \downarrow 0$ as $N \to \infty$ (no overshoot / no sign crossing), then the series is exactly *summable:*

$$\sum_{k=0}^{\infty} \Theta_k \;=\; |R_0|\,. \tag{9}$$

Derivation and mechanism. **(i) Exact telescoping.** Summing the within-aeon update $R^{\text{abs}}_{k+1} = R^{\text{abs}}_k - \Theta_k$ from $k = 0$ to N gives

$$R^{\text{abs}}_{N+1} = R^{\text{abs}}_0 - \sum_{k=0}^{N} \Theta_k,$$

which is (8). This is not an approximation: it is the exact discrete bookkeeping of the receiver curvature budget used by the reduced one-wall closures.

(ii) Why the *no-leak solvable branch* produces $R^{\text{abs}}_k \downarrow 0$ (hence exact summability). The nontrivial experimental observation is not (8) but that, in the no-leak and fixed-work-percentage-leak regimes where the implicit move equation is solvable, one gets a trajectory with *no finite flip/crossing tick*:

$$0 < \Theta_k < R^{\text{abs}}_k \quad \text{for all finite } k, \qquad \text{and} \qquad R^{\text{abs}}_k \downarrow 0.$$

This is a feedback consequence of the *implicit* Λ-solve. At tick k the solver does *not* choose Θ_k directly; it chooses Λ_{k+1} as a real root of the energy-balance residual $f_k(\Lambda) = 0$ (cf. (17)). The resulting Θ_k is then the BY-kernel output at that Λ_{k+1}, i.e. a function $\Theta_k = \Theta(\Lambda_{k+1}; \Delta I_k, d_k, L_{4,k}, \text{locks})$ which is linear in ΔI_k but is strongly suppressed at large Λ and/or large separation d_k by the evanescent factor $\exp(-d_k q)$ in the kernel (section 2.3). Thus, when R^{abs}_k becomes small, the solver can keep

the trajectory feasible (no overshoot) by moving to a slightly larger cutoff so that Θ_k becomes correspondingly smaller. In the solvable no-leak branch this yields an infinite "micro-step" regime in which $\Lambda_{k+1} - \Lambda_k \to 0$ and $\Theta_k \to 0$ while still maintaining $\Theta_k < R_k^{\mathrm{abs}}$ *until the budget itself vanishes*.

Formally, the exact-summability regime is the one-wall continuation in which (a) the move equation remains solvable for all ticks (no split) and (b) exported payload does not terminate before the feasibility boundary is reached ("enough coherence"). In that regime the only terminal point is $R^{\mathrm{abs}} = 0$, reached asymptotically with $\Theta_k \to 0$, and taking $N \to \infty$ in (8) yields (9).

(iii) Why constant/additive leaks spoil exact summability (and create splits). Additive leaks enter as a positive shift Q_{leak} in the residual (17). Once R_k^{abs} is small enough that the residual minimum becomes strictly positive, the real root disappears and the local quadratic continuation produces a complex-conjugate pair ("two complex solutions"); this is the split mechanism discussed in section 4.4. The trajectory then terminates before R_k^{abs} reaches 0, and (9) fails exactly by the nonzero remainder R_∞^{abs} in (8). □

Remark 3.3 (Why this is still nontrivial). The identity (8) is algebraic once the update law is fixed. The *nontrivial* content found in the simulator experiments is that the *implicit one-wall* Λ*-solve* dynamically adjusts the cutoff trajectory so as to *avoid overshoot* when a real solution exists: the solver selects Λ_{k+1} satisfying the discrete energy-balance residual (17), and in the no-leak / fixed-work-percentage (proportional) leak regimes this leads to $0 < \Theta_k < R_k^{\mathrm{abs}}$ for all ticks and hence $R_k^{\mathrm{abs}} \downarrow 0$ while $S_N \uparrow |R_0|$. By contrast, additive/rigid leak models can destroy the existence of real solutions for the Λ-move equation, producing a residual minimum strictly above tolerance (numerically: loss of bracketing/root and appearance of an "imaginary root" diagnostic), which is interpreted as the onset of a split (see section 4.4 and section 8).

3.2 A sufficient summability bound from QND exponential tails

In the steady-actuator regime, UOM provides a geometric/exponential tail for ΔI_k (label `eq:tail-geom` in `lem:exp-tail`):

$$\Delta I_k \simeq F_0\,(1 - e^{-\beta})\,e^{-\beta k}, \qquad \sum_{k\geq 0} \Delta I_k = F_0. \tag{10}$$

If $K(\bar{\Lambda}_k)$ is bounded above by $K_{\max}$ over the relevant tick range, then

$$\sum_{k\geq 0} \Theta_k \;=\; \sum_{k\geq 0} K(\bar{\Lambda}_k)\Delta I_k \;\leq\; K_{\max} \sum_{k\geq 0} \Delta I_k \;=\; K_{\max} F_0. \tag{11}$$

This gives a clean *sufficient* summability condition and makes explicit the two control levers: (i) the available coherence stock F_0 and (ii) the magnitude of K along the receiver trajectory.

4 Wall balance and the implicit Λ-solver

4.1 Continuous wall RG balance

UOM's SK/HJ wall balance is written as (UOM labels `eq:wall-ode` and `eq:ads5-wall-ode`)

$$\mu_{\mathrm{RG}}(\Lambda)\,\dot{\Lambda}(t) = (\mathcal{P}_{\mathrm{in}} - \mathcal{P}_{\mathrm{out}})(t) - \dot{\mathcal{F}}_{\mathrm{rec}}(t), \tag{12}$$

and in integrated/budget form

$$\Delta\lambda \leq \int W^{\mathrm{avail}}(t)\,dt, \qquad \lambda'(\Lambda) = \mu_{\mathrm{RG}}(\Lambda). \tag{13}$$

Reduced one-wall closures treat the tick as an implicit step that chooses Λ_{k+1} such that a discrete analogue of this balance holds.

4.2 Geometry-side free energy surrogate and discrete balance

The phenomenological one-wall closures use a geometry-side surrogate

$$E_{\text{geom}}(\Lambda, L_4; \kappa) \;=\; \zeta\, \frac{A + \kappa\, a_{\text{cft}}\, \Lambda^2}{L_4}, \qquad A = \frac{M_5^3}{\ell_{\text{AdS5}}}. \tag{14}$$

The discrete step compares E_{geom} between tick endpoints, with L_4 updated from the curvature bookkeeping $L_4 = \sqrt{12/\,|R|}$.

4.3 Wall-work kernel

The one-wall closure uses a wall penalty (work) kernel $J_{\text{wall}}(\Lambda_k \to \Lambda)$ obtained by numerically evaluating an integral along the log-Λ path. Heuristically, it implements the tick budget

$$J_{\text{wall}}(\Lambda_k \to \Lambda_{k+1}) \approx E_{\text{geom}}(\Lambda_k, L_{4,k}) - E_{\text{geom}}(\Lambda_{k+1}, L_{4,k+1}), \tag{15}$$

and the open-system closure generalizes this to include a nonnegative leak term Q_{leak} (UOM text around `eq:ads5-wall-ode`):

$$E_{\text{geom}}(\Lambda_k, L_{4,k}) - E_{\text{geom}}(\Lambda_{k+1}, L_{4,k+1}) \approx J_{\text{wall}}(\Lambda_k \to \Lambda_{k+1}) + Q_{\text{leak,k}}. \tag{16}$$

4.4 Implicit solve and "root vs split"

Define the residual at a candidate $\Lambda \geq \Lambda_k$:

$$f(\Lambda) := J_{\text{wall}}(\Lambda_k \to \Lambda) + Q_{\text{leak}}(\Lambda) - \Big(E_{\text{geom}}(\Lambda_k, L_{4,k}) - E_{\text{geom}}(\Lambda, L_4(\Lambda))\Big). \tag{17}$$

- **Move step:** if a root exists, choose Λ_{k+1} so that $f(\Lambda_{k+1}) = 0$.
- **Fallback + split (v2):** if no root is bracketed, minimize $f(\Lambda)$ over a grid (optionally refine). If the *minimized residual is strictly positive* (exceeds tolerance), trigger a split event at $\Lambda_\star$; if the minimized residual is nonpositive, accept an approximate move with slack.

This is the operational meaning of the split diagnostic used in phenomenological one-wall solvers.

Remark 4.1 (Bracketing failure $\neq$ loss of a real move solution). The implementations typically seed the sign-change search at a slightly increased cutoff $\Lambda_{\text{lo}} = \Lambda_k(1 + \varepsilon_\Lambda)$ (e.g. $\varepsilon_\Lambda = 10^{-9}$) to avoid the trivial fixed point. In late-time regimes where the true solution satisfies $\Lambda_{k+1} - \Lambda_k = o(\varepsilon_\Lambda \Lambda_k)$, the root can lie inside the *micro-step* interval $[\Lambda_k, \Lambda_{\text{lo}}]$ and is therefore missed by the bracketing loop (which only evaluates f for $\Lambda \geq \Lambda_{\text{lo}}$). The subsequent grid-min fallback can then return Λ_{lo} with a very small residual, even though a real root exists.

In phenomenological implementations this issue appears whenever $f(\Lambda_k) < 0$ but $f(\Lambda_k(1+\varepsilon_\Lambda)) > 0$: then a real solution exists inside the excluded micro-step interval. Therefore a split should be diagnosed by the *positivity of the minimized residual* (equivalently, under local convexity, $f(\Lambda_\star) > 0$ at the minimizer), not by "no bracket found" alone.

Remark 4.2 ("Two complex solutions" as the analytic continuation of the move equation). In reduced numerical closures, loss of a real move solution is naturally viewed as the emergence of two complex-conjugate solutions of an (approximately) quadratic move equation. One may therefore use a local diagnostic based on the second derivative f'' in the variable $\ell = \log \Lambda$.

Indeed, if locally

$$f(\ell) \approx r + \tfrac{1}{2} f''(\ell_\star)\,(\ell - \ell_\star)^2, \qquad r := f(\ell_\star) > 0,$$

then the equation $f(\ell) = 0$ has no real solutions but has complex solutions

$$\ell = \ell_\star \pm i\sqrt{\frac{2r}{f''(\ell_\star)}}.$$

Thus the split trigger "$\min f > 0$" (no feasible one-wall move) is naturally equivalent to "the real root pair has moved off the real axis", and the proxy estimates the imaginary displacement scale. This matches the stated interpretation of the split as the point where the one-wall continuation ceases to be physically realizable by a real cutoff move.

4.5 Receiver-wall leakage Q_{leak} and the RG-complement (split ancilla) sector

Overview. UOM has an information-theoretic "leakage" notion (bridge/leak instrument, Definition `def:eta-bridge-leak`), but the leak term relevant for the one-wall Λ-solver is different: it is a *receiver-wall energy complement* appearing in the SK/HJ balance. In the open-system version of that balance one may include a nonnegative term $Q_{\text{leak}} \geq 0$ so that

$$\Delta E_{\text{geom},k} \;=\; J_{\text{wall},k} \;+\; Q_{\text{leak},k}, \qquad Q_{\text{leak},k} \geq 0, \tag{18}$$

equivalently $W^{\text{avail}} = W_{\text{RG}} + Q_{\text{leak}}$ in integrated form.

Interpretation used below. Throughout the rest of the note we interpret $Q_{\text{leak},k}$ as the tickwise energy flux into the hidden completion sector B, namely $Q_{\text{leak},k} = \Delta E_{B,k}$, with B prepared by the welded history through the quadratic SK data. The detailed derivation of this statement, the Gaussian recursion for the B-covariance, and the resulting scalar split criterion are given in section 9.

Placement of RP/OS diagnostics. UOM's active KR receiver is OS-positive and has vanishing RP block on the welded family, so any nontrivial RP witness must be assigned to the analytic-continued completion sector rather than to the healthy receiver itself. This placement is the central reason the current receiver-tangent proxy cannot be the fundamental leak law.

5 Split in UOM: conservation, diagnostics, and dilation picture

5.1 Charge conservation across a split

UOM app. `app:energy-split` states: enclose the split region in a small Gaussian pillbox and integrate the Hamiltonian constraint. With a static UV boundary, the Brown–York charge change is $\Delta Q_{\text{BY}} = 0$ and the pillbox balance reproduces the wall ODE projected onto Σ_Λ. In short: the split is a redistribution of quasi-local energy compatible with global conservation.

5.2 Split identification diagnostic

UOM treats $\sigma_{\text{eff}} \to \sigma_\star$ (equivalently $\Lambda_{4,\text{R}} \to 0^+$) as a *diagnostic* of approach in the one-wall channel; in "Model A" the split onset is identified instead by the first crossing of a two-wall vs one-wall free-energy difference $\Delta\mathcal{F}(\Lambda)$ (UOM assumption `ass:split-ident` and app. `app:split-check`).

Reduced one-wall closures implement an *analogous* idea: if the one-wall balance residual $f(\Lambda)$ cannot be made small within tolerance, they interpret the failure as a split and use $\Lambda_\star$ minimizing the residual as the split point. This is a computational proxy for "two-channel (two-wall) becomes favorable".

5.3 No-cloning resolution via Stinespring dilation

UOM makes the split mechanism compatible with no-cloning by interpreting it as exposing part of a Stinespring dilation ancilla as dynamical brane sector (UOM remark `rem:uom-main-007`): the two outputs (new dS source and new AdS receiver) are correlated and do not carry independent copies of the UV state. Operationally, UOM uses an algebraic notion of sub-cutoff equivalence (UOM definition `def:uom-main-002`) which does not require a microscopic factorization.

6 Split in 2T-LS: parity cocycle and non-splitting obstruction

We recall only the pieces needed for the UOM split interpretation.

6.1 Square-loop and parity cocycle ω

In 2T-LS, one defines a typed syntax Proc_5 of record-level morphisms with generators (`time`, `rg`) and a "square" constructor $\text{sq}(f; t, u, r)$ with $f \in \{\text{diag}, \text{bridge}\}$ (2T-LS definition `def:Proc5`). One then defines:

- the canonical "to-corner" paths $\text{toCorner}_f(t, u, r)$ and
- the *square-loop* $\text{sqLoop}(t, u, r)$ comparing diag vs bridge procedures (2T-LS definition `def:sqloop`).

The parity cocycle ω assigns 0 to diag squares and 1 to bridge squares, extends additively mod 2 under composition, and is invariant under inverses and casts (2T-LS definition `def:limited-scope-theorem-translation`

The key lemma is that ω is well-defined and detects the square-loop:

$$\omega(\text{sqLoop}(t, u, r)) = 1 \qquad (t, u, r), \tag{19}$$

which is 2T-LS lemma `lem:omega`.

6.2 $\mathbb{Z}_2$-cover and non-splitting

Define the $\mathbb{Z}_2$-cover $\pi : \widetilde{\mathcal{G}} \to \mathcal{G}$ whose objects are pairs (r, b) with $b \in \mathbb{Z}_2$ and whose morphisms $(r, b) \to (s, c)$ are those base morphisms $p : r \to s$ with $c = b + \omega(p) \bmod 2$ (2T-LS definition `def:limited-scope-theorem-translation-clean-004`).

Splitting $\Leftrightarrow$ coboundary (2T-LS). 2T-LS makes the relationship between splittings and coboundaries explicit (lemma `lem:split-cob`): there exists a splitting if and only if there exists a function $\sigma : \text{Sys} \to \mathbb{Z}_2$ such that $\omega(p) = \sigma(r) + \sigma(s) \pmod 2$ for every morphism $p : r \to s$ in $\mathcal{G}$. Equivalently, an *odd loop* ($\omega(p) = 1$ for some loop $p : r \to r$) obstructs splitting (corollary `cor:odd-loop`).

Then:

Theorem 6.1 (Non-splitting of the $\mathbb{Z}_2$-cover (2T-LS))**.** *Under assumptions **(A1)–(A6)** of the 2T-LS paper, if* Sys *is nonempty then the cover does not admit a splitting; equivalently, ω is not a coboundary on $\mathcal{G}$ (2T-LS theorem* `thm:non-splitting`*).*

The obstruction can already be witnessed by the two parallel "to-corner" morphisms p_{diag} and p_{bridge} whose ω-values differ (2T-LS proposition `prop:two-way-obstruction`).

Interpretation for UOM. At the level of *record-accessible* descriptions, the split is precisely the place where one must choose between two continuations that are not homotopic within a purely diagonal (classical) sector; the $\mathbb{Z}_2$ holonomy is the formal encoding of that "odd" choice. This matches the proposed identification of the split with the odd cocycle generator.

7 Coarse-graining classicality reframing (analysis of the proposal)

This section formalizes (and slightly sharpens) the proposed split reframing given in the prompt.

7.1 One global accessibility/coarse-graining map

Fix a tick/aeon scale index k and define a CPTP *accessibility* (or RG coarse-graining) map

$$\mathcal{C}_k : \mathcal{D}(\mathcal{H}_{\text{tot}}) \to \mathcal{D}(\mathcal{H}_{\text{acc},k}), \qquad \rho_k^{\text{RG}} := \mathcal{C}_k(\varrho_k^{\text{tot}}). \tag{20}$$

The point is that "accessibility" is defined *after* applying $\mathcal{C}_k$; no claim is made that the microscopic 5D state ever becomes diagonal.

UOM-compatible construction. UOM already provides the natural building blocks for $\mathcal{C}_k$:

1. patchwise pointer-aligned QND dephasing,
2. conditional expectation $\mathbb{E}_{\text{ptr}}$ onto the pointer algebra,
3. a commutant record map $\mathcal{R}_k$ with commutative range,
4. an admissible bridge $\mathcal{B}_k$ into the receiver code subspace, followed by any "export" selection of RG-visible observables (e.g. DC anomaly coordinates).

Composing these yields a single global CPTP map of the form (20).

7.2 Classicality defect at RG scale

Fix on $\mathcal{H}_{\text{acc},k}$ a preferred "ledger" basis (or commutative subalgebra) that defines what "diagonal" means. Define the *classicality defect* by trace distance to the diagonal set:

$$\text{cl}_k := \inf_{\sigma \in \text{Diag}(\mathcal{H}_{\text{acc},k})} \left\| \rho_k^{\text{RG}} - \sigma \right\|_1 . \tag{21}$$

This is a legitimate operational requirement: it quantifies how well an observer restricted to the RG-accessible degrees of freedom can approximate the state by a classical probability distribution.

A computable surrogate. While (21) is the clean definition, in practice one often uses

$$\mathrm{Off}_k := \left\|\rho_k^{\mathrm{RG}} - \mathrm{diag}(\rho_k^{\mathrm{RG}})\right\|_1, \tag{22}$$

which satisfies $\mathrm{cl}_k \le \mathrm{Off}_k$ by feasibility of $\mathrm{diag}(\rho_k^{\mathrm{RG}})$. (Note: for the trace norm, $\mathrm{diag}(\rho)$ need not be the unique optimizer in all dimensions; we will only use Off_k as a controlled upper bound and as a diagnostic aligned with dephasing.)

7.3 End-of-aeon classicality: where the bound is tightest

Under pointer-basis dephasing, off-diagonals are suppressed multiplicatively in the pointer eigenbasis. Combined with data-processing (CPTP contractivity of distinguishability), one expects the smallest classicality defect to occur after:

1. enough QND time has accumulated so that coherence is exponentially small,
2. patch reconciliation has produced a consistent global record basis, and
3. the accessibility map $\mathcal{C}_k$ has "forgotten" phases (projects/averages onto commutative output).

This motivates defining *diagonal ledger points* at the *end of an aeon*:

$$\inf_{\sigma\in\mathrm{Diag}} \left\|\rho_{n,\mathrm{end}}^{\mathrm{RG}} - \sigma\right\|_1 \le \varepsilon_n, \qquad \varepsilon_n \downarrow 0. \tag{23}$$

It is consistent with UOM that the diagonal approximation is tightest late in the aeon and not necessarily after each tick.

Lemma 7.1 (Off-diagonal suppression under pointer dephasing (diagnostic bound))**.** *Let Φ be a pointer-basis QND dephasing channel of the form (4) on a finite-dimensional space $\mathcal{H}_{\mathrm{acc},k}$ with pointer eigenbasis $\{|i\rangle\}$. Let Π_{diag} denote the pinching (conditional expectation) onto the diagonal in this basis: $\Pi_{\mathrm{diag}}(\rho) = \mathrm{diag}(\rho)$. Then for $A := \rho - \Pi_{\mathrm{diag}}(\rho)$ (zero diagonal) the matrix elements obey*

$$(\Phi(A))_{ij} = e^{-\Gamma_{ij}} A_{ij} \qquad (i \ne j)$$

with $\Gamma_{ij} \ge 0$ determined by the $\Gamma_a^{(k)}$ and pointer gaps. Consequently, if $\Gamma_{\min} := \min_{i\ne j} \Gamma_{ij}$ then

$$\|\Phi(A)\|_F \le e^{-\Gamma_{\min}} \|A\|_F \qquad \Longrightarrow \qquad \mathrm{Off}(\Phi(\rho)) \le \sqrt{d}\, e^{-\Gamma_{\min}} \mathrm{Off}(\rho), \tag{24}$$

where $d = \dim \mathcal{H}_{\mathrm{acc},k}$, $\|\cdot\|_F$ is the Frobenius norm, and $\mathrm{Off}(\rho) = \|\rho - \mathrm{diag}(\rho)\|_1$.

Remark 7.2 (Backflow, leakage, and the "tail + leak" corrections)*.* If the effective reduced dynamics is not exactly Markovian on the accessible space (e.g. due to memory kernels) or if the tick map includes a leakage branch $\mathcal{L}_k$ outside the commutative record/code sector, then (24) is modified schematically to

$$\mathrm{Off}_{k+1} \lesssim e^{-\gamma_k} \mathrm{Off}_k + (\text{backflow tail}) + (\text{leakage}),$$

matching the qualitative structure suggested in the prompt. UOM explicitly controls backflow tails in its source-side analysis (e.g. App. `app:dS-collapse`, "Exponentially decaying backflow") and treats leakage via the instrument decomposition in `def:eta-bridge-leak`.

7.4 Split as an odd lift / Stinespring refactorization

Let $\rho^{\mathrm{RG}}_{n,\mathrm{end}}$ denote the end-of-aeon accessible state. Model the split by a (generally non-unique) *section* (odd lift)

$$\mathrm{sec}_{\Pi} : \rho^{\mathrm{RG}}_{n,\mathrm{end}} \mapsto x_{n+1,\mathrm{start}} \in \mathcal{S}_{\mathrm{SK}}, \tag{25}$$

where $\mathcal{S}_{\mathrm{SK}}$ is the space of Schwinger–Keldysh (two-time) states for the next aeon, including a new receiver initialized in a pure "vacuum-like" code state together with a new source state. Operationally, sec_{Π} is best understood as a Stinespring refactorization/isometry on an enlarged space rather than as "recohering the same subsystem".

Connection to 2T-LS. The non-uniqueness / parity of the lift is exactly what the 2T-LS cocycle ω formalizes: the choice between diag vs bridge corner procedures has intrinsic one-bit holonomy and cannot be absorbed into a single global state label on records. Thus the split is the natural locus of the odd cocycle.

7.5 Split ancilla B and coherent birth: what drives it in UOM

This subsection develops the split proposal from the prompt into a UOM-faithful identification of *(i)* what the "newborn" sector B is and *(ii)* why it is generically coherent even if the inherited ledger sector A is (approximately) classical/QND at the end of the aeon.

Subsystem identification: inherited ledger A vs completion sector B

UOM's microscopic 5D evolution is unitary; CPTP structure only appears after discarding degrees of freedom. Therefore the split is *not* a CPTP map acting on the old receiver alone. The clean reading of the Stinespring/refactorization picture is:

- A is the *inherited receiver ledger sector*: the commuting family of record variables that constitute the end-of-aeon accessible "classical" description (e.g. a DC anomaly/trace coordinate together with any other commuting registers retained by $\mathcal{C}_{n,\mathrm{end}}$).
- B is the *completion / environment sector* that makes the receiver export isometric before reduction and that, at the split, is reclassified as (part of) the *new dS brane Hilbert space* (new source).

UOM offers two canonical realizations of such a completion sector:

1. **Receiver Stinespring completion.** UOM explicitly enlarges the receiver code by a Weyl compensator σ so that the welded export is an isometry, and the trace anomaly appears only after tracing out σ (UOM `par:uom-main-040`–`par:uom-main-044` and `par:uom-main-077`). In this reading, B may include σ (and allied completion modes such as the brane-local boundary fluctuation $\mathcal{S}$ used in the portal sector).
2. **Commutant/purifying partner.** On the source side, UOM's intrinsic bath is the commutant/purifying sector: patch marginals can look KMS/thermal because the global state is purified across a complementary algebra (UOM `subsec:source-qnd-commutant` and `subsec:commutant-stinespring` In this reading, B is the canonical partner sector that becomes dynamical as the next aeon's source.

Both choices implement the same structural point: B is *not a new knob* but "whatever sector was previously integrated out" in the reduced description and is fixed by the same 5D SK saddle.

Classical-control split unitary (ledger inheritance) and the coherence target

To keep the inherited ledger classical while creating a coherent newborn source, the split must be QND on the ledger:

$$U_{\text{split}} \;=\; \sum_i \Pi_i^{(A)} \otimes U_i^{(B)}, \qquad [U_{\text{split}}, \Pi_i^{(A)} \otimes \mathbf{1}_B] = 0 \quad \forall i, \tag{26}$$

where $\{\Pi_i^{(A)}\}$ are orthogonal projectors onto the end-of-aeon ledger basis (or, equivalently, minimal projections of the commutative record algebra). If ρ_A is diagonal in this basis (or $\mathrm{cl}_{n,\mathrm{end}} \ll 1$), then A remains (approximately) diagonal after the split and can be viewed as *classical control*.

Recoherence, in the operational sense relevant for UOM's "recohere → QND dephase" picture, is a condition on B: let Π_Δ denote the pointer/record projection used in the next aeon (band/pointer basis on the newborn source). Then the newborn source must satisfy

$$\left\| \rho_{\text{start}}^{(B)} - \Pi_\Delta(\rho_{\text{start}}^{(B)}) \right\|_1 \;>\; 0, \tag{27}$$

otherwise the next aeon's QND absorber produces no new record increment (and no positive "entropy production" in the Spohn/QND sense).

What UOM fixes about the welded source $\int J \cdot \mathcal{O}$

UOM's microscopic receiver action contains an explicit welded source term (UOM `par:uom-main-077`):

$$S_{\text{weld}}[J] \;=\; \int_{\Sigma_{\text{rec}}} d^4x \, \sqrt{-\gamma} \, \big(J \cdot \mathcal{O}_{\text{LoG}}\big), \qquad J \equiv J_k \text{ (tick-localized LoG envelope).} \tag{28}$$

Two points are essential for the split proposal:

1. J **is not arbitrary.** In UOM it is a *compiled/welded* image of the source-side payload into the receiver trace channel, regulated by the receiver cutoff (UOM `par:uom-main-080`, and "welding/regulation" step in the proof of `lem:ImW-noise`). In the portal sector, J is explicitly identified as a LoG projection of the Brown–York trace response (UOM `eq:JLoG-def`): $J_{\text{LoG}} = \kappa_{\text{HS}}(\text{LoG} \star T^\mu{}_\mu{}^{(\text{BY})})$.

2. J **is a classical control parameter at the coarse-grained level.** Once the ledger A is diagonal, J is effectively a function of the ledger history (the accepted payload/trace record), so the controlled form (26) is the natural way to package "history-dependent preparation" of the newborn sector B.

Quadratic generators and the squeeze picture: what UOM provides (and what it implies)

UOM evaluates the *Schwinger–Keldysh influence functional* on the welded source and integrates out bulk+brane fields in the Gaussian sector at *quadratic order in* J (UOM `par:uom-main-080`):

$$W[J] = \frac{1}{2} \int \frac{d^4k}{(2\pi)^4} \, J(-k) \, \big(G_R(k) + i\,\mathcal{N}(k)\big) \, J(k) + O(J^3), \tag{29}$$

with G_R retarded and $\mathcal{N} \geq 0$ a KMS/noise kernel (hence $\Im W \geq 0$; UOM `lem:ImW-noise`). Equivalently, in Keldysh (cl, q) variables,

$$W[\tau] \;=\; -\langle \tau_{\text{q}}, G^{\text{R}} \tau_{\text{cl}} \rangle \;+\; \frac{i}{2} \langle \tau_{\text{q}}, N_\Lambda \tau_{\text{q}} \rangle \;+\; O(\tau^3),$$

as in the proof of UOM `lem:ImW-noise`.

Implication for split-state preparation. To quadratic order, the completion sector excited by J is *Gaussian*. For bosonic degrees of freedom, any Gaussian state can be prepared from a vacuum-like reference by a product of a displacement (linear drive) and a symplectic transformation (squeezing/Bogoliubov):

$$U_i^{(B)} \sim D(\alpha_i)\, S(\zeta_i), \qquad S(\zeta) = \exp\Big(\tfrac{1}{2}\sum_{\alpha,\beta} \zeta_{\alpha\beta}\, b_\alpha^\dagger b_\beta^\dagger - \text{h.c.}\Big), \tag{30}$$

with multi-mode creation/annihilation operators b_α on B. In this representation:

- the **linear** welded source term (28) supplies the displacement/drive (coherent component), and
- the **quadratic** SK kernels in (29) fix the Gaussian covariance, hence the effective squeeze/Bogoliubov parameters.

UOM does not presently write the squeeze generator as an operator $H_{B,\text{sq}}$; rather, it provides the quadratic influence data $(G_R, \mathcal{N})$ and the regulated on-shell quadratic functionals (Fisher and noise forms) from which one can reconstruct an equivalent Gaussian unitary description if one chooses an explicit mode basis on B.

Contact with $H^{(i)}_{B,\text{drive}}$ and $H^{(i)}_{B,\text{sq}}$. If one insists on a Hamiltonian/stroboscopic representation of the conditional preparation unitaries $U_i^{(B)}$, a minimal parametrization consistent with (30) is

$$U_i^{(B)} \simeq \mathcal{T}\exp\Big(-i\int dt\, \big(H_B^{(0)}(t) + H^{(i)}_{B,\text{drive}}(t) + H^{(i)}_{B,\text{sq}}(t)\big)\Big), \tag{31}$$

where the drive is linear in fields/operators and the squeeze part is quadratic in canonical modes:

$$H^{(i)}_{B,\text{drive}}(t) = \int d^3x\, \big(J_i(t,x)\,\mathcal{O}_B(t,x) + \overline{J_i(t,x)}\,\mathcal{O}_B^\dagger(t,x)\big),$$

$$H^{(i)}_{B,\text{sq}}(t) = \frac{i}{2}\sum_{\alpha,\beta}\Big(\kappa^{(i)}_{\alpha\beta}(t)\, b_\alpha^\dagger b_\beta^\dagger - \overline{\kappa^{(i)}_{\alpha\beta}(t)}\, b_\alpha b_\beta\Big) + \sum_{\alpha,\beta} h^{(i)}_{\alpha\beta}(t)\, b_\alpha^\dagger b_\beta.$$

In UOM terms, J_i is supplied by the welded/compiled source history, and the quadratic kernels determining $\kappa^{(i)}$ and $h^{(i)}$ are encoded by the same Gaussian SK reduction data that define $W[J]$.

Why analytic continuation naturally produces squeezing. UOM's AdS→dS analytic continuation maps a Euclidean on-shell quadratic functional to an oscillatory Hartle–Hawking wavefunctional (UOM `subsec:analytic-continuation`):

$$\Psi_{\text{HH}}[\phi_{(0)}] \propto e^{-S_{\text{os}}^{(\text{EAdS})}[\phi_{(0)}]} \longrightarrow e^{iS_{\text{eff}}^{(\text{dS})}[\phi_{(0)}]}.$$

For free/linearized modes this continuation is precisely the mechanism by which an "in" vacuum becomes a squeezed state with respect to an "out" (static-patch/pointer) basis: the quadratic phase encodes the Bogoliubov mixing. Thus UOM's existing quadratic (Gaussian) infrastructure already supplies the natural home for the "quadratic generators" in a split-as-recoherence picture.

8 Phenomenological inputs from existing simulations

The simulator family is useful here only as a reduced phenomenology of the wall balance. Three lessons are robust and theory-relevant.

8.1 The exhaustion branch is physically meaningful

The one-wall solver can remain real for all finite ticks and realize exact summability:

$$0 < \Theta_k < R_k^{\mathrm{abs}} \quad \text{for all finite } k, \qquad R_k^{\mathrm{abs}} \downarrow 0, \qquad \sum_{k\geq 0} \Theta_k = |R_0|\,.$$

This is the phenomenological evidence for the *exhaustion branch*: split is not guaranteed, and proportional or no-leak closures can stay on the solvable one-wall channel indefinitely.

8.2 Residual positivity is the meaningful split diagnostic

The theory-level split criterion is the positivity of the minimized discrete residual,

$$\min_{\Lambda\geq\Lambda_k} f_k(\Lambda) > 0,$$

not mere failure of numerical bracketing and not the magnitude $\min|f|$. This is exactly the criterion compatible with the wall-balance interpretation of split as loss of a real one-wall continuation.

8.3 The current RP proxy is only a phenomenological probe

The presently implemented quantity $Q_{\mathrm{rg}} = \chi J_{\mathrm{RP}}/\beta_{\mathrm{dS}}$ is a useful reduced probe of when a one-wall scalar closure becomes hard to realize, but it is not a derived completion-sector leak. The key reason is structural: it is built from a receiver-side welded tangent, whereas the healthy KR receiver is OS-positive and has vanishing RP block on the welded family. The simulations therefore motivate two qualitative conclusions, but do not yet provide the microscopic leak law:

1. a leak capable of forcing split must be history-conditioned and non-proportional to the shrinking payload;
2. any RP-based witness must be placed on the analytic-continued completion sector rather than on the healthy receiver tangent.

9 Completion-sector leak model and split criterion

This section separates what is directly derived from UOM from the additional hypotheses required to turn those ingredients into a concrete split-driving leak model. The new organizing principle is: the primary hidden variable is a *signed odd split amplitude* a_k, while the positive stock $S_k := a_k^2$ is the quantity that can enter scalar observables and leak laws. We reserve R_k^{abs} for the remaining curvature budget and

$$\mathfrak{r}_k := \min_{\Lambda\geq\Lambda_k} f_k(\Lambda)$$

for the minimized one-wall residual.

9.1 Odd split mode, output parity, and the rank-one normal form

The only "leak" that directly competes with wall work in the Λ-solver is the open-system modification of the wall energy balance allowed explicitly in UOM's SK/HJ wall equation:

$$\Delta E_{\mathrm{geom},k} \;=\; J_{\mathrm{wall},k} \;+\; Q_{\mathrm{leak},k}, \qquad Q_{\mathrm{leak},k} \geq 0, \tag{32}$$

This is the welded wall-balance form used in the earlier summability analysis. This is a *receiver-side energy complement* term. It must not be conflated with UOM's information-theoretic bridge/leak instrument from the UOM main manuscript.

What is already derived in UOM. Four ingredients are fixed by the manuscript:

1. **Wall balance.** The reduced one-wall description may contain a nonnegative complement Q_{leak} as in (32).

2. **Quadratic welded SK data.** In the Gaussian sector,

$$W_\Lambda[J] = \frac{1}{2}\int J\,(G_{R,\Lambda} + i\,\mathcal{N}_\Lambda)\,J + O(J^3), \qquad \mathcal{N}_\Lambda \geq 0.$$

3. **Healthy receiver.** The active KR receiver has vanishing welded RP block, so a receiver-tangent RP penalty is not the fundamental home of split-driving oddness.

4. **Continued RP defect.** The analytic-continued covariance has a negative OS mode with

$$\lambda_{\min}(\mu_{ps}, \Lambda) = -\alpha(\Lambda)\mu_{ps}^2 + O(\mu_{ps}^4), \qquad \alpha(\Lambda) \sim \kappa_d\,\Lambda^{d+2}H^{-d}.$$

Assumption 9.1 (Split-mode observability)**.** Near split onset there exists a one-dimensional odd line L_k with local coordinate $a_k \in \mathbb{R}$ such that the split involution acts by

$$a_k \longmapsto -a_k.$$

The true two-wall advantage

$$\Delta F_k := \Delta\mathcal{F}_{2\text{w}-1\text{w},k}$$

and the minimized one-wall residual

$$\mathfrak{r}_k := \min_{\Lambda \geq \Lambda_k} f_k(\Lambda)$$

are smooth even scalar outputs of this same local mode. Equivalently, the 2T Γ-odd descendant direction and the UOM continued negative mode $u_-(\Lambda_k)$ are assumed to represent the same active split mode up to a smooth local identification.

Lemma 9.1 (Evenness of the minimized residual)**.** *Suppose the pre-minimized one-wall residual is a smooth function $f_k(\Lambda, a)$ satisfying*

$$f_k(\Lambda, -a) = f_k(\Lambda, a),$$

and suppose there is a unique local minimizer $\Lambda_*(a)$ near $a = 0$ such that

$$\partial_\Lambda f_k(\Lambda_*(a), a) = 0, \qquad \partial_\Lambda^2 f_k(\Lambda_*(0), 0) > 0.$$

Then

$$\mathfrak{r}_k(a) := f_k(\Lambda_*(a), a)$$

is a smooth even function of a.

Proof. By the implicit function theorem, $\Lambda_*(a)$ is smooth near $a = 0$. Since $f_k(\Lambda, a)$ is even in a, the stationarity equation for a and $-a$ is identical; uniqueness of the local minimizer gives

$$\Lambda_*(-a) = \Lambda_*(a).$$

Therefore

$$\mathfrak{r}_k(-a) = f_k(\Lambda_*(-a), -a) = f_k(\Lambda_*(a), a) = \mathfrak{r}_k(a),$$

so $\mathfrak{r}_k$ is smooth and even. □

Proposition 9.2 (Rank-one normal form near split onset)**.** *Under assumption 9.1, the two scalar outputs admit local even expansions*

$$\Delta F_k(a) = \mu_k^{(2\mathrm{w})} + h_k a^2 + O(a^4), \tag{33}$$

$$\mathfrak{r}_k(a) = \mu_k^{(1\mathrm{w})} + m_k a^2 + O(a^4), \tag{34}$$

with $h_k, m_k > 0$ on the active split line.

Proof. Smooth even functions of one real variable have Taylor series containing only even powers. The constants $\mu_k^{(2\mathrm{w})}$ and $\mu_k^{(1\mathrm{w})}$ are the baseline detunings of the two-wall and one-wall channels. Positivity of h_k and m_k expresses that the selected mode is active in both outputs. □

Corollary 9.3 (Onset wall-match in rank one)**.** *On the onset manifold,*

$$\mu_k^{(1\mathrm{w})} = \mu_k^{(2\mathrm{w})} = 0,$$

so

$$\Delta F_k = h_k a_k^2 + O(a_k^4), \qquad \mathfrak{r}_k = m_k a_k^2 + O(a_k^4).$$

Eliminating a_k^2 gives

$$\Delta F_k = \frac{h_k}{m_k}\,\mathfrak{r}_k + O(\mathfrak{r}_k^2). \tag{35}$$

9.2 Low-rank generalization

If the active split sector is m-dimensional, with odd amplitude vector $a_k \in \mathbb{R}^m$, parity gives quadratic normal forms

$$\Delta F_k(a) = \mu_k^{(2\mathrm{w})} + a^\top H_k a + O(\|a\|^4), \tag{36}$$

$$\mathfrak{r}_k(a) = \mu_k^{(1\mathrm{w})} + a^\top M_k a + O(\|a\|^4), \tag{37}$$

with $H_k, M_k \succeq 0$ on the active split subspace.

Assumption 9.2 (Low-rank comparability)**.** On the active split subspace there exist $c_k^-, c_k^+ > 0$ such that

$$c_k^- M_k \preceq H_k \preceq c_k^+ M_k.$$

Proposition 9.4 (Low-rank wall-match comparability)**.** *Assume assumptions 9.1 and 9.2 and restrict to the onset manifold $\mu_k^{(1\mathrm{w})} = \mu_k^{(2\mathrm{w})} = 0$. Then for sufficiently small $\mathfrak{r}_k$ there exists $C_k > 0$ such that*

$$c_k^-\,\mathfrak{r}_k - C_k \mathfrak{r}_k^2 \le \Delta F_k \le c_k^+\,\mathfrak{r}_k + C_k \mathfrak{r}_k^2. \tag{38}$$

Proof. On onset,

$$\Delta F_k = a_k^\top H_k a_k + O(\|a_k\|^4), \qquad \mathfrak{r}_k = a_k^\top M_k a_k + O(\|a_k\|^4).$$

The quadratic-form comparison gives

$$c_k^- a_k^\top M_k a_k \le a_k^\top H_k a_k \le c_k^+ a_k^\top M_k a_k.$$

Since $\mathfrak{r}_k$ is itself quadratic in a_k to leading order, the quartic remainders are $O(\mathfrak{r}_k^2)$, which yields (38). □

9.3 Gaussian realization of the odd mode and the leak stock law

The theorem path above does *not* rely on a KMS-subtracted matrix. In particular, we do not use

$$\Sigma_{B,k} := \left(C_{B,k} - C^{\mathrm{KMS}}_{\Lambda_k}\right) + m_{B,k} m_{B,k}^{\top}$$

as a positive object, because it need not be positive semidefinite. When a safe matrix observable is needed in the realization layer, the natural PSD quantity is

$$M_{B,k} := C_{B,k} + m_{B,k} m_{B,k}^{\top} \succeq 0. \tag{39}$$

Assumption 9.3 (Gaussian completion realization)**.** For each admissible cutoff Λ in the welded regime, the quadratic influence functional $W_\Lambda[J]$ admits a Gaussian dilation, unique up to symplectic equivalence, by a bosonic completion sector B_Λ with canonical quadratures X_Λ and positive quadratic Hamiltonian

$$H_{B,\Lambda} = \frac{1}{2} X_\Lambda^{\top} h_\Lambda X_\Lambda, \qquad h_\Lambda \succeq 0.$$

Assumption 9.4 (Dominant odd-mode projection)**.** Inside the Gaussian completion of assumption 9.3, there exists a smooth unit vector

$$u_k := u_-(\Lambda_k)$$

spanning the active odd line and separated from the orthogonal sector by a local spectral gap. We write

$$P_k^{\perp} := I - u_k u_k^{\top}$$

for the orthogonal projector.

Assumption 9.5 (Classical-control gate)**.** There exists a scalar gate $g_{\mathrm{cl}}(k) \in [0,1]$, determined by the accessible classicality defect cl_k or Off_k, such that $g_{\mathrm{cl}}(k) \ll 1$ at a fresh-aeon start and $g_{\mathrm{cl}}(k) \to 1$ late in the aeon.

Proposition 9.5 (Projected Gaussian affine dynamics)**.** *Under assumptions 9.3 and 9.4, one may choose a finite effective completion-mode basis in which the completion quadratures obey a linear-Gaussian recursion*

$$X_{k+1} = A_k X_k + B_k j_k^{\mathrm{ctl}} + \nu_k, \tag{40}$$

with

$$\mathbb{E}[\nu_k \mid \mathcal{F}_k] = 0, \qquad \mathbb{E}[\nu_k \nu_k^{\top} \mid \mathcal{F}_k] = Q_k \succeq 0.$$

Consequently the PSD second moment

$$M_{B,k} = C_{B,k} + m_{B,k} m_{B,k}^{\top}$$

obeys

$$M_{B,k+1} = A_k M_{B,k} A_k^{\top} + Q_k + \textit{drift terms}. \tag{41}$$

Projecting onto the odd line via

$$a_k := u_k^{\top} X_k$$

gives the scalar affine law

$$a_{k+1} = \rho_k a_k + \beta_k + \xi_k + \varepsilon_k^{\mathrm{mix}}, \tag{42}$$

where

$$\rho_k := u_{k+1}^{\top} A_k u_k, \qquad \beta_k := u_{k+1}^{\top} B_k j_k^{\mathrm{ctl}},$$

$$\xi_k := u_{k+1}^\top \nu_k, \qquad \varepsilon_k^{\mathrm{mix}} := u_{k+1}^\top A_k P_k^\perp X_k.$$

Moreover,

$$\mathbb{E}[\xi_k \mid \mathcal{F}_k] = 0, \qquad \mathbb{E}[\xi_k^2 \mid \mathcal{F}_k] = u_{k+1}^\top Q_k u_{k+1} \geq 0.$$

If assumption 9.5 also holds and the welded shape family is fixed, then

$$\mathbb{E}[\xi_k^2 \mid \mathcal{F}_k] = \eta_k \, g_{\mathrm{cl}}(k) \, \Delta I_k + O(\Delta I_k^2, \delta\Lambda_k^2), \tag{43}$$

with $\eta_k \geq 0$.

Proof. The quadratic SK influence functional determines a Gaussian CPTP map on the finite active completion-mode sector, hence a linear-Gaussian state-space realization of the form (40); (41) is the standard update for the PSD second moment. Decompose $X_k = u_k a_k + P_k^\perp X_k$, apply $\langle u_{k+1}, \cdot \rangle$ to (40), and collect the projected transport, deterministic drive, noise, and orthogonal-mode contamination to obtain (42). The mean-zero and positive-variance statements for ξ_k follow directly from the projected Gaussian noise covariance $u_{k+1}^\top Q_k u_{k+1} \geq 0$. Finally, at fixed welded shape family one has $A_{\mathrm{eff},k}^2 \propto \Delta I_k$, so the fresh covariance injection into the odd mode is linear in the current payload at leading order; the gate $g_{\mathrm{cl}}(k)$ suppresses this injection near a fresh-aeon start and yields (43). □

Assumption 9.6 (Centered odd preparation)**.** Near split onset, the active odd mode is prepared either branch-symmetrically or in a squeeze-dominated regime, so the projected deterministic displacement vanishes and orthogonal-mode contamination is higher order:

$$\beta_k = 0, \qquad \varepsilon_k^{\mathrm{mix}} = O_2.$$

Corollary 9.6 (Centered odd-mode law)**.** *Under assumptions 9.3 to 9.6, the projected affine law* (42) *reduces to*

$$a_{k+1} = \rho_k a_k + \xi_k + O_2, \tag{44}$$

with $\mathbb{E}[\xi_k \mid \mathcal{F}_k] = 0$ *and variance scaling* (43).

Stochastic/Gaussian realization layer. At theorem level the positive stock is simply

$$S_k := a_k^2.$$

The general projected Gaussian dynamics need not be centered. Under assumption 9.6, however, the physically relevant late-time regime is the one in which the odd mode carries no leading deterministic displacement and the fresh effect enters through its variance. In that centered specialization it is natural to use the conditional second moment

$$S_k := \mathbb{E}[a_k^2 \mid \mathcal{F}_k].$$

Proposition 9.7 (Odd-mode stock law)**.** *Under assumptions 9.3 to 9.6, the conditional stock*

$$S_k := \mathbb{E}[a_k^2 \mid \mathcal{F}_k]$$

satisfies

$$S_{k+1} = t_k \, S_k + \eta_k \, g_{\mathrm{cl}}(k) \, \Delta I_k + O(S_k \Delta I_k, \Delta I_k^2, \delta\Lambda_k^2), \qquad t_k = \rho_k^2. \tag{45}$$

Proof. Square (44):

$$a_{k+1}^2 = \rho_k^2 a_k^2 + 2\rho_k a_k \xi_k + \xi_k^2 + O_2.$$

Taking conditional expectation and using $\mathbb{E}[\xi_k \mid \mathcal{F}_k] = 0$ removes the mixed term. Substituting (43) gives (45). □

Assumption 9.7 (UV comparability of energetic stiffness)**.** There exist constants $c_-, c_+ > 0$ and a monotone increasing energetic stiffness $\widetilde{\alpha}(\Lambda)$ of the active odd mode such that, for all sufficiently large Λ,

$$c_- \, \alpha(\Lambda) \le \widetilde{\alpha}(\Lambda) \le c_+ \, \alpha(\Lambda). \tag{46}$$

This is the precise place where the note passes from what is derived in UOM to a physical hypothesis. The coefficient $\alpha(\Lambda)$ is derived from the continued OS problem; the statement that the energetic stiffness of the split mode is comparable to it is not yet a theorem.

With the stock S_k as the actual state variable, the positive split-driving export flux is modeled by

$$Q_{\mathrm{leak},k}^{\mathrm{model}} = \widetilde{\alpha}'(\Lambda_k)\, \delta\Lambda_k \, S_k \; + \; \widetilde{\alpha}(\Lambda_k)\, \eta_k \, g_{\mathrm{cl}}(k)\, \Delta I_k \; + \; O_2, \tag{47}$$

where $\delta\Lambda_k := \Lambda_{k+1} - \Lambda_k$. The first term is the transport cost of carrying already-prepared odd stock to larger cutoff; the second is the fresh injection due to the current payload. Under assumption 9.7, this gives the scaling comparison

$$Q_{\mathrm{leak},k}^{\mathrm{model}} \asymp \alpha'(\Lambda_k)\, \delta\Lambda_k \, S_k \; + \; \alpha(\Lambda_k)\, \eta_k \, g_{\mathrm{cl}}(k)\, \Delta I_k. \tag{48}$$

Current RP proxy reinterpreted. The quantity

$$Q_{\mathrm{rg}} = \chi J_{\mathrm{RP}} / \beta_{\mathrm{dS}}$$

therefore remains useful only as a phenomenological probe. It may correlate with a_k^2 or with the stock S_k, but it is not the fundamental leak law because the healthy receiver itself has no quadratic RP block on the welded family.

9.4 Approximate split condition from the odd-stock leak law

Let $R_k^{\mathrm{abs}} := |R_k|$. In the micro-step regime the geometry-side surrogate gives

$$\Delta E_{\mathrm{geom},k} \approx \frac{\zeta\,(A + c_\Lambda \Lambda_k^2)}{2\sqrt{12\, R_k^{\mathrm{abs}}}}\, \Theta_k \; - \; \frac{2\zeta\, c_\Lambda \Lambda_k \sqrt{R_k^{\mathrm{abs}}}}{\sqrt{12}}\, \delta\Lambda,$$

while

$$J_{\mathrm{wall},k} \approx \mu_{\mathrm{RG}}(\Lambda_k)\, \delta\Lambda, \qquad \mu_{\mathrm{RG}}(\Lambda) \simeq \mu_0 + \mu_2 \Lambda^2 > 0.$$

Hence the local one-wall threshold remains

$$Q_{\mathrm{leak},k} \gtrsim \frac{\zeta\,(A + c_\Lambda \Lambda_k^2)}{2\sqrt{12\, R_k^{\mathrm{abs}}}}\, \Theta_k. \tag{49}$$

Since

$$\Theta_k = K(\Lambda_k)\, \Delta I_k \sim K_0\, \Lambda_k^{-p}\, e^{-2 d_{\mathrm{sep}}(\Lambda_k)\Lambda_k}\, \Delta I_k, \qquad p \in \{10, 14\},$$

substituting (47) gives

$$\widetilde{\alpha}'(\Lambda_k)\,\delta\Lambda_k\,S_k \;+\; \widetilde{\alpha}(\Lambda_k)\,\eta_k\,g_{\mathrm{cl}}(k)\,\Delta I_k \gtrsim \frac{\zeta\,(A + c_\Lambda \Lambda_k^2)}{2\sqrt{12\,R_k^{\mathrm{abs}}}}\,K(\Lambda_k)\,\Delta I_k. \tag{50}$$

This is the corrected split threshold. It shows the two late-time branches transparently:

1. if S_k remains negligible, only the fresh term remains, and the exhaustion branch can survive;
2. if S_k has accumulated, the transport term can dominate even when ΔI_k is already tiny.

Under assumption 9.7 and $\alpha(\Lambda) \sim \kappa_d \Lambda^{d+2} H^{-d}$, the transport term scales like

$$\widetilde{\alpha}'(\Lambda_k)\,\delta\Lambda_k\,S_k \asymp \Lambda_k^{d+1}\,\delta\Lambda_k\,S_k,$$

which is exactly the mechanism by which late split can beat the shrinking fresh payload.

9.5 Nondegeneracy clause for the normalized seam coefficient

The split threshold (50) is a statement about when the late-time odd stock can overcome the shrinking fresh payload. It is *not* by itself a theorem that the normalized seam coefficient c_{Λ_k} on the canonical compressed welded LoG line is nonzero. Any theorem proving $c_{\Lambda_k} \neq 0$ must therefore come from a separate overlap or transport hypothesis, not from the threshold inequality alone. Downstream, this clause is absorbed into the delayed selected-line package rather than carried as a separate package summand.

Definition 9.8 (Unit-source seam overlap scalar and no-destructive-cancellation margin)**.** Fix an accepted tick k. Assume the canonical compressed welded LoG line is defined and let

$$\tau_{\Lambda_k}^{\mathrm{unit}}$$

be a detector-factorized unit welded odd source. Define the raw unit-source seam overlap scalar by

$$\widetilde{c}_{\Lambda_k} := \left\langle\!\!\left\langle \Pi_{\mathrm{phys},\Lambda_k} \mathsf{T}_{\Lambda_k}\big[\tau_{\Lambda_k}^{\mathrm{unit}}\big],\, \delta C_{\Lambda_k}^{\mathrm{LoG}} \right\rangle\!\!\right\rangle_{C_{*,\Lambda_k}}. \tag{51}$$

A *no-destructive-cancellation margin* at Λ_k is any real number

$$\mu_{\Lambda_k} > 0$$

such that

$$|\widetilde{c}_{\Lambda_k}| \geq \mu_{\Lambda_k}. \tag{52}$$

On a connected accepted-branch segment I, a *branchwise no-destructive-cancellation margin* is a constant $\mu_I > 0$ such that

$$|\widetilde{c}_\Lambda| \geq \mu_I \qquad \forall\,\Lambda \in I. \tag{53}$$

Theorem 9.9 (Nondegeneracy criterion for c_{Λ_k})**.** *Fix an accepted tick k. Assume:*

1. *a detector-factorized unit welded odd source at Λ_k;*
2. *the unit-source seam-faithfulness package, namely:*
   (a) *the raw seam pairing factorizes as*

$$\widetilde{s}_{\Lambda_k}(f, y) = \sigma_{\mathrm{fill}}(f)\,\widetilde{c}_{\Lambda_k}\,\overline{\mathrm{obs}}(y) \qquad \forall f, \forall y; \tag{54}$$

(b) *whenever* $\widetilde{c}_{\Lambda_k} \neq 0$ *and* $\overline{\mathrm{obs}}(y) \neq 0$*, the compressed welded LoG line is transversely physical, dominant-line realization holds, and the normalized seam coefficient*

$$c_{\Lambda_k} := \kappa_{\Lambda_k}^{-1/2} \widetilde{c}_{\Lambda_k} \tag{55}$$

is defined;

3. *a no-destructive-cancellation margin at* Λ_k *in the sense of definition 9.8;*

4. *a witness* y *with*

$$\overline{\mathrm{obs}}(y) \neq 0.$$

Then:

1. *the raw seam pairing is nonzero for every filler tag* f*:*

$$\widetilde{s}_{\Lambda_k}(f, y) \neq 0;$$

2. *the canonical compressed welded LoG line is transversely physical at* Λ_k*;*

3. *dominant-line realization holds at* Λ_k*;*

4. *the normalized seam coefficient is defined and nonzero:*

$$c_{\Lambda_k} \neq 0.$$

If, in addition, for some filler tag f *the source* $\tau_{\Lambda_k}^{f,y}$ *is representable on the characterization layer and admits a source-compatible descendant realization, then the corresponding descendant representative satisfies*

$$\rho_{\Lambda_k}^{\mathrm{dom}}\big(J_{\Lambda_k}^{f,y}\big) \neq 0.$$

Proof. The margin (52) implies $\widetilde{c}_{\Lambda_k} \neq 0$. Together with $\overline{\mathrm{obs}}(y) \neq 0$, the factorization (54) gives $\widetilde{s}_{\Lambda_k}(f, y) \neq 0$ for every filler tag f, proving item (1). Item (2), item (3), and the definition and nonvanishing of c_{Λ_k} are exactly the remaining conclusions of the assumed unit-source seam-faithfulness package under the same nondegeneracy hypothesis. The final sentence is the corresponding witness-to-descendant comparison step under representability and source-compatible descendant realization. □

Corollary 9.10 (Branchwise sign-stable nondegeneracy)**.** *Let* I *be a connected accepted-branch segment. Assume:*

1. *the branchwise normalized seam coefficient*

$$\Lambda \mapsto c_\Lambda$$

is continuous on I*;*

2. *a branchwise no-destructive-cancellation margin on* I *in the sense of definition 9.8.*

Then

$$c_\Lambda \neq 0 \qquad \forall\, \Lambda \in I.$$

In particular, the sign of c_Λ *is constant on* I*. If there also exist a fixed LS witness* y *with* $\overline{\mathrm{obs}}(y) \neq 0$ *and a branchwise source-compatible descendant realization satisfying*

$$\rho_\Lambda^{\mathrm{dom}}\big(J_\Lambda^{f,y}\big) = \sigma_{\mathrm{fill}}(f)\, c_\Lambda\, \overline{\mathrm{obs}}(y),$$

then the sign of $\rho_\Lambda^{\mathrm{dom}}(J_\Lambda^{f,y})$ *is constant on* I *as well.*

Proof. By (53), the scalar $\widetilde{c}_\Lambda$ never vanishes on I. Hence neither does c_Λ, because the normalization only rescales the raw overlap scalar by a positive branchwise factor. A continuous nonzero real-valued function on a connected segment cannot change sign, proving the constancy of $\operatorname{sgn}(c_\Lambda)$. The final statement follows immediately from the displayed branchwise comparison identity and the fixed nonzero scalar $\overline{\mathrm{obs}}(y)$. □

Remark 9.11 (Logical separation from the split threshold). The hypotheses of theorem 9.9 and corollary 9.10 are intentionally separate from (50). The threshold governs the emergence of split-side stock; the no-destructive-cancellation margin governs overlap of the unit welded odd source with the canonical compressed dominant line. Accordingly, the present note treats $c_\Lambda \neq 0$ as the transport/overlap clause later imported into the delayed selected-line package rather than as a consequence of the split threshold.

10 Canonical Odd Spectral Selection on the Compressed Welded LoG Channel

Fix an accepted tick k on a welded admissible branch. This section isolates the compressed welded LoG channel and proves that the continued odd quadratic form canonically selects a negative rank-one odd line on that channel. The transport of the descendant class into that line is recorded separately in theorem 12.20; here it is used only in corollary 10.13.

The macro source-side temporal envelope is DoG, the macro source-side spatial envelope is LoG, the micro receiver temporal envelope is DoG, and the micro receiver spatial envelope is Gaussian. By Corollary 3.1, the welded envelope is LoG in both time and space, with scales adding in quadrature.

Definition 10.1 (Canonical welded LoG profile)**.** Let t_k be the accepted tick time and let $\Omega_* \in S^3$ denote the common spatial center selected by the co-centering/history-lock mechanism. Let

$$s_{t,k} := \sqrt{\Sigma_k^2 + \sigma_{t,k}^2}, \qquad s_{x,k} := \sqrt{\rho_k^2 + \sigma_{x,k}^2}$$

be the effective welded scales. Define the normalized welded LoG pulse by

$$u_{\Lambda_k}^{\mathrm{LoG}}(t,\Omega) := \mathcal{N}_k \, \frac{d^2}{dt^2} g_{s_{t,k}}(t-t_k) \, \Delta_{S^3} g_{s_{x,k}}(\Omega,\Omega_*), \tag{56}$$

where $\mathcal{N}_k > 0$ is chosen so that $\|u_{\Lambda_k}^{\mathrm{LoG}}\|_{L^2(\mathbb{R}\times S^3)} = 1$. We write

$$\mathcal{H}_{\Lambda_k}^{\mathrm{LoG}} := \mathbb{R}\, u_{\Lambda_k}^{\mathrm{LoG}} \tag{57}$$

for the compressed welded LoG channel.

Proposition 10.2 (Exact welded LoG form)**.** *The normalized profile $u_{\Lambda_k}^{\mathrm{LoG}}$ of definition 10.1 is exactly the welded macro/micro profile produced by:*

1. *macro temporal DoG convolved with micro temporal DoG;*
2. *macro spatial LoG convolved with micro spatial Gaussian.*

In particular the welded profile is LoG in time and LoG in space, with scales adding in quadrature.

Proof. The temporal and spatial welded identities are exactly the two statements of Corollary 3.1. After multiplying the factors and normalizing in L^2, one obtains (56). □

Proposition 10.3 (Band-edge locking and unique center)**.** *Assume the accepted tick lies on an admissible band-locked welded branch in the sense of Definition 3.2. Then:*

1. *the effective welded widths satisfy*

$$s_{t,k} = \frac{\alpha_t^*}{\Lambda_k}, \qquad s_{x,k} = \frac{\alpha_x^*}{v\,\Lambda_k},$$

   with $\alpha_t^*, \alpha_x^* > 0$ independent of k along a fixed admissible branch;

2. *the macro and micro temporal centers coincide at t_k, and the macro and micro spatial centers coincide at Ω_*;*

3. *once the receiver trace sector carries a nonzero inherited LoG background, the absolute center Ω_* is uniquely locked for all subsequent accepted ticks.*

Consequently, after normalization, the admissible welded pulse space at fixed accepted tick is the one-dimensional line $\mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}$.

Proof. The width lock is the band-locking clause of Definition 3.2. The temporal and spatial co-centering statements are ($\mathrm{C_t}$) and ($\mathrm{C_x}$), and the persistence of the spatial center is exactly Proposition 3.5.

Therefore, at fixed accepted tick, the shape, widths, and center are fixed. The only remaining freedom is an overall real amplitude. After L^2-normalization, that freedom reduces to the sign $\pm u^{\mathrm{LoG}}_{\Lambda_k}$. Hence the admissible welded channel is precisely the one-dimensional real line $\mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}$. □

Proposition 10.4 (Smooth dependence along a branch)**.** *Along any admissible branch on which $\Lambda \mapsto \alpha_t^*, \alpha_x^*$ are constant and the locked center Ω_* is fixed, the map*

$$\Lambda \longmapsto u^{\mathrm{LoG}}_{\Lambda}$$

is C^1 in $L^2(\mathbb{R} \times S^3)$. Consequently the rank-one orthogonal projector

$$P^{\mathrm{LoG}}_{\Lambda} := u^{\mathrm{LoG}}_{\Lambda} \otimes u^{\mathrm{LoG}}_{\Lambda} \tag{58}$$

is C^1 along the branch.

Proof. Along the branch one has $s_t(\Lambda) = \alpha_t^*/\Lambda$ and $s_x(\Lambda) = \alpha_x^*/(v\Lambda)$, so both scales depend C^∞-smoothly on Λ. The Gaussian heat kernel and its time and spatial derivatives depend smoothly on the scale parameter, hence the unnormalized welded LoG profile depends C^1-smoothly on Λ in L^2. Its L^2-norm is strictly positive, so normalization preserves C^1-dependence. Formula (58) then gives a C^1 projector-valued map. □

Proposition 10.5 (Well-defined restriction of the continued quadratic form)**.** *Assume the welded linear-response/Gaussian radiative-sector package of Paragraph 3.5 and Theorem E.1. Then the analytically continued odd quadratic response form is well-defined on the compressed welded channel $\mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}$.*

Proof. By Paragraph 3.5, the welded source enters the Gaussian radiative sector through the linear response map T_Λ, and Theorem E.1 supplies the finite-cutoff bounded Gaussian response package on the welded envelope class. By proposition 10.3, the compressed welded channel is a one-dimensional subspace of that admissible class. Restricting any continuous quadratic form to a one-dimensional subspace is automatic. Hence the analytically continued odd quadratic response form restricts canonically to $\mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}$. □

Definition 10.6 (Compressed continued odd quadratic coefficient)**.** Fix a witness window $\mu_{ps} > 0$ in the accepted regime. Let

$$\mathfrak{q}^{\mathrm{cont}}_{\Lambda_k,\mu_{ps}} : \mathcal{H}^{\mathrm{LoG}}_{\Lambda_k} \times \mathcal{H}^{\mathrm{LoG}}_{\Lambda_k} \to \mathbb{R}$$

denote the analytically continued odd quadratic response form restricted to the compressed welded LoG channel. Since $\mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}$ is one-dimensional, there is a unique scalar $\beta_k \in \mathbb{R}$ such that

$$\mathfrak{q}^{\mathrm{cont}}_{\Lambda_k,\mu_{ps}}(X,Y) = \beta_k \, \langle X, u^{\mathrm{LoG}}_{\Lambda_k}\rangle_{L^2} \langle Y, u^{\mathrm{LoG}}_{\Lambda_k}\rangle_{L^2} \qquad \forall X, Y \in \mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}. \tag{59}$$

Equivalently,

$$\beta_k = \mathfrak{q}^{\mathrm{cont}}_{\Lambda_k,\mu_{ps}}\big(u^{\mathrm{LoG}}_{\Lambda_k}, u^{\mathrm{LoG}}_{\Lambda_k}\big).$$

Proposition 10.7 (Rigorous control of the compressed coefficient)**.** *Assume:*

1. *the smooth RP degradation law of Theorem D.1;*
2. *the approximate intertwiner bound of Lemma E.2;*
3. *the temporal DoG commutator bound of Lemma E.3;*
4. *the exponentially decaying memory-tail package of Lemma C.2 and Proposition C.1.*

Then there exist constants $C_{\mathrm{int}}, C_{\mathrm{DoG}}, C_{\mathrm{mem}}, C_4 > 0$ and $\eta > 0$, depending only on the fixed welded family and not on k, such that

$$\beta_k = -\,\alpha(\Lambda_k)\,\mu_{ps}^2 \;+\; r_k, \tag{60}$$

with remainder bounded by

$$|r_k| \le C_{\mathrm{int}}\,\varepsilon_k^{\eta} \;+\; C_{\mathrm{DoG}}\,\Phi(\sigma_{t,k}\Omega_{\mathrm{gap}}) \;+\; C_{\mathrm{mem}}\,e^{-\Delta_k/\tau_b} \;+\; C_4\,\mu_{ps}^4. \tag{61}$$

Proof. By Theorem D.1, the ideal analytically continued Gaussian OS problem gives $\lambda_{\min}(\mu_{ps}, \Lambda_k) = -\alpha(\Lambda_k)\mu_{ps}^2 + O(\mu_{ps}^4)$ on the welded Gaussian reference kernel. Restricting that quadratic form to the canonical normalized welded LoG profile $u^{\mathrm{LoG}}_{\Lambda_k}$ produces the leading term $-\alpha(\Lambda_k)\mu_{ps}^2$, with the $O(\mu_{ps}^4)$ correction bounded by $C_4\mu_{ps}^4$.

The regulated intertwiner contribution is bounded by Lemma E.2, the temporal DoG contribution is bounded by Lemma E.3, and the intrinsic KMS bath contributes only the Born-Nakajima-Zwanzig memory tail controlled by Lemma C.2 and Proposition C.1. Since $u^{\mathrm{LoG}}_{\Lambda_k}$ is normalized, each of these operator bounds transfers directly to the scalar quadratic coefficient. Summing the four contributions gives (60)–(61). □

Definition 10.8 (Certified compressed spectral regime)**.** We say that an accepted tick k lies in the *certified compressed spectral regime* if

$$\alpha(\Lambda_k)\,\mu_{ps}^2 > 2\Big(C_{\mathrm{int}}\,\varepsilon_k^{\eta} \;+\; C_{\mathrm{DoG}}\,\Phi(\sigma_{t,k}\Omega_{\mathrm{gap}}) \;+\; C_{\mathrm{mem}}\,e^{-\Delta_k/\tau_b} \;+\; C_4\,\mu_{ps}^4\Big). \tag{62}$$

Theorem 10.9 (Finite-cutoff canonical odd spectral selection on the compressed welded LoG channel)**.** *Assume propositions 10.2 to 10.5 and 10.7. Let k be an accepted tick in the certified compressed spectral regime of definition 10.8. Then:*

1. *the compressed welded channel $\mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}$ is a canonical one-dimensional odd line;*

2. *the compressed continued odd quadratic form is represented by the rank-one self-adjoint operator*

$$K^{\mathrm{LoG}}_{\Lambda_k} := \beta_k \, P^{\mathrm{LoG}}_{\Lambda_k}, \tag{63}$$

with $\beta_k < 0$*;*

3. *the unique negative spectral projector of* $K^{\mathrm{LoG}}_{\Lambda_k}$ *is*

$$P^{\mathrm{LoG}}_{-,\Lambda_k} = P^{\mathrm{LoG}}_{\Lambda_k} = u^{\mathrm{LoG}}_{\Lambda_k} \otimes u^{\mathrm{LoG}}_{\Lambda_k}; \tag{64}$$

4. *along any admissible branch, the projector* $P^{\mathrm{LoG}}_{-,\Lambda}$ *is* C^1*, hence the compressed active odd line is canonically and smoothly transported.*

Proof. By proposition 10.3, the admissible welded channel at fixed accepted tick is exactly the one-dimensional line $\mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}$. By definition 10.6, the compressed continued odd quadratic form on that line is represented by the scalar β_k, equivalently by the self-adjoint rank-one operator (63).

By proposition 10.7,

$$\beta_k \le -\,\alpha(\Lambda_k)\mu_{ps}^2 \;+\; \Big(C_{\mathrm{int}}\,\varepsilon_k^{\eta} \;+\; C_{\mathrm{DoG}}\,\Phi(\sigma_{t,k}\Omega_{\mathrm{gap}}) \;+\; C_{\mathrm{mem}}\,e^{-\Delta_k/\tau_b} \;+\; C_4\,\mu_{ps}^4\Big).$$

Under the certified regime inequality (62), the bracketed term is strictly smaller than $\frac{1}{2}\alpha(\Lambda_k)\mu_{ps}^2$. Therefore

$$\beta_k < -\frac{1}{2}\,\alpha(\Lambda_k)\mu_{ps}^2 < 0.$$

Thus $K^{\mathrm{LoG}}_{\Lambda_k}$ has exactly one negative eigenvalue, namely β_k, and its unique negative spectral subspace is the entire line $\mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}$. Since $P^{\mathrm{LoG}}_{\Lambda_k} = u^{\mathrm{LoG}}_{\Lambda_k} \otimes u^{\mathrm{LoG}}_{\Lambda_k}$, the negative spectral projector is exactly (64).

Finally, proposition 10.4 gives C^1-dependence of $\Lambda \mapsto u^{\mathrm{LoG}}_{\Lambda}$, hence also of $\Lambda \mapsto P^{\mathrm{LoG}}_{-,\Lambda}$. This proves the canonical branchwise transport of the compressed active odd line. □

Corollary 10.10 (Canonical compressed odd projector)**.** *For each accepted tick in the certified compressed spectral regime, the compressed, co-centered, band-locked welded LoG channel carries the canonical rank-one projector*

$$P^{\mathrm{LoG}}_{-,\Lambda_k}$$

onto its negative odd line. Equivalently,

$$\mathcal{H}^{\mathrm{LoG}}_{\Lambda_k} = \operatorname{ran} P^{\mathrm{LoG}}_{-,\Lambda_k}.$$

Proof. This is exactly theorem 10.9: the theorem identifies the negative spectral projector with the rank-one projector $u^{\mathrm{LoG}}_{\Lambda_k} \otimes u^{\mathrm{LoG}}_{\Lambda_k}$, whose range is $\mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}$. □

Corollary 10.11 (Completion-sector realization of the compressed projector)**.** *Assume assumption 9.3. Then, for each accepted tick in the certified compressed spectral regime, any Gaussian completion realization identifies the compressed projector* $P^{\mathrm{LoG}}_{-,\Lambda_k}$ *with a unique rank-one active odd line in the completion sector. Equivalently, there exists a unit vector*

$$u_-(\Lambda_k)$$

spanning that realized line such that

$$P^{\mathrm{LoG}}_{-,\Lambda_k} = u_-(\Lambda_k) \otimes u_-(\Lambda_k)$$

in the chosen realization.

Proof. Under assumption 9.3, the finite active completion-mode sector is defined only up to symplectic equivalence. A one-dimensional real line is invariant under such equivalences as a subspace, even though a basis vector on that line is not canonical before normalization. By theorem 10.9, the compressed odd line is already canonically fixed abstractly by the projector $P^{\mathrm{LoG}}_{-,\Lambda_k}$. Choosing the unique unit vector on that line in a given realization yields the stated formula, and changing the completion realization changes only the coordinate representative, not the underlying rank-one line. □

Remark 10.12 (Scope of the compressed spectral theorem). Theorem theorem 10.9 determines the odd line only on the compressed welded LoG channel. It does not classify the full completion-sector odd subspace, and the completion-side statement of corollary 10.11 is only a realization statement for this rank-one projector.

Corollary 10.13 (Nontrivial descendant hit of the compressed dominant line)**.** *Assume theorem 12.20. Then for each accepted tick k, there exists a representative*

$$J^{\mathrm{desc}}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$$

whose response has nonzero component on the canonically selected compressed dominant odd line. Equivalently, the transported descendant sector maps nontrivially into the compressed line fixed by $P^{\mathrm{LoG}}_{-,\Lambda_k}$.

Proof. This is exactly item (1) of theorem 12.20. □

11 Raw Odd Selector and Scheme-Stable Affine Output

This section isolates the raw selector step on the odd descendant side of the UOM-to-Main interface used in Definition 4.2. At fixed finite cutoff, it records the Γ-odd affine output determined by a chosen sufficiently local anomaly representative and its same-scheme and inter-scheme translation laws.

Convention on the two $\mathbb{Z}_2$ gradings. The cohomological degree for the Q_t-complex is written with bars:

$$(\cdot)_{\bar{0}}, \qquad (\cdot)_{\bar{1}}.$$

Physical parity refers to the swap involution Γ, with eigenspaces

$$\Gamma F = \pm F.$$

Thus the selector output constructed below is cohomologically degree 0 but physically odd. Unless bar-degree notation is used explicitly, "odd" below means Γ-odd rather than Q_t-degree 1.

Work at one accepted tick k and one branch value u at cutoff Λ_k. Use the raw UOM descendant package of theorems 6.9 and 6.11. In particular:

1. a sufficiently local good monotone representative

$$\mathcal{O}^{(0)}_{\Lambda_k}(u);$$

2. its canonical anomaly representative

$$\mathcal{A}^{(1)}_{\Lambda_k}(u) := Q^{\mathrm{phys}}_{u,\Lambda_k}\mathcal{O}^{(0)}_{\Lambda_k}(u);$$

3. a swap involution
$$\Gamma : \big(\mathcal{F}^{\Lambda_k}_{\text{s.loc}}(\mathcal{X}_k)\big)_{\bar{\imath}} \to \big(\mathcal{F}^{\Lambda_k}_{\text{s.loc}}(\mathcal{X}_k)\big)_{\bar{\imath}} \qquad (\bar{\imath} \in \{\bar{0}, \bar{1}\})$$
preserving sufficient locality and cohomological degree, and commuting with Q_t:
$$\Gamma Q_t = Q_t \Gamma;$$

4. swap-evenness of the canonical core,
$$\Gamma \mathcal{O}^{(0)}_{\Lambda_k}(u) = \mathcal{O}^{(0)}_{\Lambda_k}(u), \qquad \Gamma \mathcal{A}^{(1)}_{\Lambda_k}(u) = \mathcal{A}^{(1)}_{\Lambda_k}(u);$$

5. swap covariance of the fixed-scheme physical radial differential on the chosen monotone sector,
$$\Gamma Q^{\text{phys}}_{u,\Lambda_k} = Q^{\text{phys}}_{u,\Lambda_k} \Gamma;$$

Items (1) and (2) are contained in theorem 6.9; item (3) uses definition 6.1 and lemma 6.2; item (4) uses proposition 6.4; and item (5) uses proposition 6.3.

Fix a chosen Γ-stable selector class of sufficiently local odd representatives of the same Q_t-cohomology class as $\mathcal{A}^{(1)}_{\Lambda_k}(u)$, and restrict to representatives for which the exact descent equation
$$Q^{\text{phys}}_{u,\Lambda_k} \mathcal{O}^{(0)}_{\Lambda_k}(u) + Q_t \widetilde{\Xi}^{(1)}_{\Lambda_k}(u) = \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u) \tag{65}$$
admits at least one sufficiently local even solution $\widetilde{\Xi}^{(1)}_{\Lambda_k}(u) \in \big(\mathcal{F}^{\Lambda_k}_{\text{s.loc}}(\mathcal{X}_k)\big)_{\bar{0}}$.

Definition 11.1 (Selector class and canonical odd kernel channel)**.** Let
$$\mathcal{C}_{\text{sel},\Lambda_k}(u) \subset \big(\mathcal{F}^{\Lambda_k}_{\text{s.loc}}(\mathcal{X}_k)\big)_{\bar{1}}$$
be the chosen Γ-stable class of sufficiently local odd representatives of the same degree-one Q_t-cohomology class as $\mathcal{A}^{(1)}_{\Lambda_k}(u)$ for which (65) admits a sufficiently local even solution. For $\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u) \in \mathcal{C}_{\text{sel},\Lambda_k}(u)$, define its swap-even and swap-odd parts by
$$\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,\pm}(u) := \frac{1}{2}\Big(\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u) \pm \Gamma \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u)\Big).$$
Define the canonical odd kernel channel by
$$\mathcal{K}^{-}_{\Lambda_k} := \ker(Q_t) \cap \big(\mathcal{F}^{\Lambda_k}_{\text{s.loc}}(\mathcal{X}_k)\big)_{\bar{0}} \cap \ker(\Gamma + \text{id}).$$

Proposition 11.2 (Odd defect of a selector representative is always Q_t-exact)**.** *Let $\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u) \in \mathcal{C}_{\text{sel},\Lambda_k}(u)$. Then:*

1. *$\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,+}(u)$ and $\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u)$ are sufficiently local;*

2. *$\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,+}(u)$ is a swap-even representative of the same Q_t-cohomology class as $\mathcal{A}^{(1)}_{\Lambda_k}(u)$;*

3. $$\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u) \in \text{im}(Q_t) \cap \big(\mathcal{F}^{\Lambda_k}_{\text{s.loc}}(\mathcal{X}_k)\big)_{\bar{1}} \cap \ker(\Gamma + \text{id}).$$

Proof. Sufficient locality is preserved by linear combinations, proving item (1).

Because $\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u)$ and $\mathcal{A}^{(1)}_{\Lambda_k}(u)$ represent the same Q_t-cohomology class, there exists $Y_{\Lambda_k}(u) \in \big(\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(\mathcal{X}_k)\big)_{\bar{0}}$ such that

$$\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u) - \mathcal{A}^{(1)}_{\Lambda_k}(u) = Q_t Y_{\Lambda_k}(u).$$

Taking the swap-even part and using $\Gamma Q_t = Q_t \Gamma$ and $\Gamma \mathcal{A}^{(1)}_{\Lambda_k}(u) = \mathcal{A}^{(1)}_{\Lambda_k}(u)$, we obtain

$$\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,+}(u) - \mathcal{A}^{(1)}_{\Lambda_k}(u) = Q_t Y_{\Lambda_k,+}(u),$$

so item (2) follows.

Taking the swap-odd part of the same identity gives

$$\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u) = Q_t Y_{\Lambda_k,-}(u).$$

By construction, $\Gamma \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u) = -\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u)$, which proves item (3). □

Definition 11.3 (Selected odd descendant affine class)**.** For $\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u) \in \mathcal{C}_{\mathrm{sel},\Lambda_k}(u)$, define the associated raw odd selector output by

$$\mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u; \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}\big) := \Big\{ \Xi^-_{\Lambda_k}(u) \in \big(\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(\mathcal{X}_k)\big)_{\bar{0}} \;\Big|\; \Gamma \Xi^-_{\Lambda_k}(u) = -\Xi^-_{\Lambda_k}(u), \quad Q_t \Xi^-_{\Lambda_k}(u) = \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u) \Big\}.$$

Remark 11.4 (Dependence only on the swap-odd defect)*.* By theorem 11.5, the affine class $\mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u; \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}\big)$ depends only on the swap-odd defect $\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u)$.

Theorem 11.5 (Selector decomposition into even representative channel and odd output)**.** *Let $\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u) \in \mathcal{C}_{\mathrm{sel},\Lambda_k}(u)$, and let $\widetilde{\Xi}^{(1)}_{\Lambda_k}(u)$ be any sufficiently local even solution of (65). Define*

$$\widetilde{\Xi}^{(1)}_{\Lambda_k,\pm}(u) := \frac{1}{2}\Big(\widetilde{\Xi}^{(1)}_{\Lambda_k}(u) \pm \Gamma \widetilde{\Xi}^{(1)}_{\Lambda_k}(u)\Big).$$

Then:

1. *the swap-even and swap-odd channels decouple as*

$$Q^{\mathrm{phys}}_{u,\Lambda_k} \mathcal{O}^{(0)}_{\Lambda_k}(u) + Q_t \widetilde{\Xi}^{(1)}_{\Lambda_k,+}(u) = \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,+}(u), \tag{66}$$

$$Q_t \widetilde{\Xi}^{(1)}_{\Lambda_k,-}(u) = \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u); \tag{67}$$

2. *the raw odd selector output $\mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u; \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}\big)$ contains the odd projection of any full solution; in particular it is nonempty;*

3. *it is an affine space over the canonical odd kernel channel:*

$$\mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u; \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}\big) = \widetilde{\Xi}^{(1)}_{\Lambda_k,-}(u) + \mathcal{K}^-_{\Lambda_k}.$$

Proof. Apply Γ to (65). Using $\Gamma Q^{\mathrm{phys}}_{u,\Lambda_k} = Q^{\mathrm{phys}}_{u,\Lambda_k}\Gamma$ on the chosen monotone sector and $\Gamma \mathcal{O}^{(0)}_{\Lambda_k}(u) = \mathcal{O}^{(0)}_{\Lambda_k}(u)$, we obtain

$$Q^{\mathrm{phys}}_{u,\Lambda_k} \mathcal{O}^{(0)}_{\Lambda_k}(u) + Q_t \Gamma \widetilde{\Xi}^{(1)}_{\Lambda_k}(u) = \Gamma \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u).$$

Adding and subtracting this equation from (65) yields (66) and (67), proving item (1).

Item (2) is immediate from item (1): the odd projection $\widetilde{\Xi}^{(1)}_{\Lambda_k,-}(u)$ of any full solution belongs to the set in definition 11.3.

Now let $\Xi^-_{\Lambda_k,1}(u)$ and $\Xi^-_{\Lambda_k,2}(u)$ lie in $\mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u;\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}\big)$. Subtracting their defining equations gives

$$Q_t\big(\Xi^-_{\Lambda_k,1}(u)-\Xi^-_{\Lambda_k,2}(u)\big)=0.$$

Because both are Γ-odd, their difference lies in $\mathcal{K}^-_{\Lambda_k}$. Conversely, if $K^-_{\Lambda_k}\in\mathcal{K}^-_{\Lambda_k}$, then

$$Q_t\big(\widetilde{\Xi}^{(1)}_{\Lambda_k,-}(u)+K^-_{\Lambda_k}\big)=\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u),$$

and the sum remains Γ-odd. This proves item (3). □

Corollary 11.6 (The two selector channels)**.** *At fixed cutoff, the selector problem splits into exactly two channels.*

1. ***Kernel channel.*** *If $\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u)=0$, then*

$$\mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u;\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}\big)=\mathcal{K}^-_{\Lambda_k}.$$

   In particular, the canonical anomaly representative $\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u)=\mathcal{A}^{(1)}_{\Lambda_k}(u)$ produces only the canonical odd kernel sector.

2. ***Representative channel.*** *If $\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u)\neq 0$, then the raw odd selector output is the translated affine space*

$$\widetilde{\Xi}^{(1)}_{\Lambda_k,-}(u)+\mathcal{K}^-_{\Lambda_k}$$

   solving

$$Q_t\Xi^-_{\Lambda_k}(u)=\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u).$$

Thus the selector output is determined by the canonical odd kernel channel together with the swap-odd part of the chosen anomaly representative.

Proof. This is an immediate specialization of theorem 11.5. □

Remark 11.7 (Kernel activity versus representative-driven activity)*.* By corollary 11.6, the selector output is either the canonical odd kernel channel $\mathcal{K}^-_{\Lambda_k}$ or a translate of that channel by a particular solution of $Q_t\Xi^-_{\Lambda_k}(u)=\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u)$.

Proposition 11.8 (Same-scheme affine translation of selector outputs)**.** *Let*

$$\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u),\qquad \widehat{\mathcal{A}}^{(1)}_{\Lambda_k}(u)\in\mathcal{C}_{\mathrm{sel},\Lambda_k}(u)$$

be selector representatives in the same finite-cutoff scheme, and suppose there exists an even sufficiently local cochain $Y_{\Lambda_k}(u)$ such that

$$\widehat{\mathcal{A}}^{(1)}_{\Lambda_k}(u)-\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}(u)=Q_tY_{\Lambda_k}(u). \tag{68}$$

Define

$$Y^-_{\Lambda_k}(u):=\frac{1}{2}\Big(Y_{\Lambda_k}(u)-\Gamma Y_{\Lambda_k}(u)\Big).$$

Then

$$\mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u;\widehat{\mathcal{A}}^{(1)}_{\Lambda_k}\big)=\mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u;\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}\big)+Y^-_{\Lambda_k}(u).$$

If $Y'_{\Lambda_k}(u)$ is another primitive satisfying (68), then

$$Y'^{-}_{\Lambda_k}(u) - Y^{-}_{\Lambda_k}(u) \in \mathcal{K}^{-}_{\Lambda_k}.$$

In particular, if $\widehat{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u) = \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u)$, then

$$\mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u;\widehat{\mathcal{A}}^{(1)}_{\Lambda_k}\big) = \mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u;\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}\big).$$

Proof. Taking swap-odd parts of (68) gives

$$\widehat{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u) - \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u) = Q_t Y^{-}_{\Lambda_k}(u).$$

Let $\Xi^{-}_{\Lambda_k}(u) \in \mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u;\widetilde{\mathcal{A}}^{(1)}_{\Lambda_k}\big)$. Then

$$Q_t \Xi^{-}_{\Lambda_k}(u) = \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u),$$

so

$$Q_t\big(\Xi^{-}_{\Lambda_k}(u) + Y^{-}_{\Lambda_k}(u)\big) = \widehat{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u).$$

Because both terms are Γ-odd, the sum lies in $\mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k}\big(u;\widehat{\mathcal{A}}^{(1)}_{\Lambda_k}\big)$. This proves one inclusion, and the reverse inclusion follows by symmetry with $-Y^{-}_{\Lambda_k}(u)$.

If $Y'_{\Lambda_k}(u)$ is another primitive, then $Q_t(Y'_{\Lambda_k}(u) - Y_{\Lambda_k}(u)) = 0$. Taking swap-odd parts yields

$$Q_t\big(Y'^{-}_{\Lambda_k}(u) - Y^{-}_{\Lambda_k}(u)\big) = 0.$$

Since the difference is also Γ-odd, it lies in $\mathcal{K}^{-}_{\Lambda_k}$. If $\widehat{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u) = \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k,-}(u)$, then one may choose $Y^{-}_{\Lambda_k}(u) = 0$, and the last claim follows. □

Theorem 11.9 (Scheme-stable affine translation of the odd selector output)**.** *Let S and S' be cohomologically admissibly comparable finite-cutoff schemes on the same chosen monotone sector. Assume in addition that:*

1. *the canonical anomaly representatives $\mathcal{A}^{(1)}_{k,S}(u)$ and $\mathcal{A}^{(1)}_{k,S'}(u)$ are swap-even;*
2. *there exist chosen Γ-stable selector classes*

$$\mathcal{C}^{S}_{\mathrm{sel},\Lambda_k}(u), \qquad \mathcal{C}^{S'}_{\mathrm{sel},\Lambda_k}(u),$$

and chosen selector representatives

$$\widehat{\mathcal{A}}^{(1)}_{k,S}(u) \in \mathcal{C}^{S}_{\mathrm{sel},\Lambda_k}(u), \qquad \widehat{\mathcal{A}}^{(1)}_{k,S'}(u) \in \mathcal{C}^{S'}_{\mathrm{sel},\Lambda_k}(u),$$

and an even sufficiently local cochain $Y_k(u)$ such that

$$\widehat{\mathcal{A}}^{(1)}_{k,S'}(u) - \widehat{\mathcal{A}}^{(1)}_{k,S}(u) = \mathcal{A}^{(1)}_{k,S'}(u) - \mathcal{A}^{(1)}_{k,S}(u) + Q_t Y_k(u). \tag{69}$$

Define the odd comparison primitive

$$Y^{-}_{k}(u) := \frac{1}{2}\Big(Y_k(u) - \Gamma Y_k(u)\Big).$$

Then

$$\mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k,S'}\big(u;\widehat{\mathcal{A}}^{(1)}_{k,S'}\big) = \mathfrak{X}^{\mathrm{sel},-}_{\Lambda_k,S}\big(u;\widehat{\mathcal{A}}^{(1)}_{k,S}\big) + Y^{-}_{k}(u).$$

If $Y'_k(u)$ is another comparison primitive satisfying (69) for the same chosen pair of representatives, then

$$Y'^{-}_{k}(u) - Y^{-}_{k}(u) \in \mathcal{K}^{-}_{\Lambda_k}.$$

Proof. Taking swap-odd parts of (69) and using the swap-evenness of the canonical anomaly representatives gives

$$\widehat{\mathcal{A}}^{(1)}_{k,S',-}(u) - \widehat{\mathcal{A}}^{(1)}_{k,S,-}(u) = Q_t Y_k^-(u).$$

The same affine-fiber argument as in proposition 11.8 therefore gives the translation statement by $Y_k^-(u)$, and the same odd-kernel argument gives the ambiguity claim for alternative primitives. □

Remark 11.10 (Selector output handed to the downstream notes). The raw selector datum is the affine class

$$\mathfrak{x}^{\mathrm{sel},-}_{\Lambda_k}(u; \widetilde{\mathcal{A}}^{(1)}_{\Lambda_k})$$

attached to the chosen selector representative. Its ambiguity descent, transported dominant-response value, and weak-interface image are treated in theorems 12.20, 13.16 and 13.56.

Remark 11.11 (Relation to the Hodge-normalized layer). The Hodge-normalized comparison layer for admissible alternative representatives is treated in theorem 6.10. The present section uses only raw sufficiently local selector classes.

12 Polarized Characterization Interface

This section records the finite-cutoff characterization data extracted from the split/leak construction and used in the 2T Main assumption package Definition 4.2. In this note, an "accepted tick" denotes the local accepted-branch slice at cutoff Λ_k.

For each accepted tick k, write

$$a_{\Lambda_k} := a_k, \qquad S_{\Lambda_k} := a^2_{\Lambda_k}.$$

The variable a_{Λ_k} is the distinguished odd split coordinate identified in sections 9.1 and 9.3; the even quantity S_{Λ_k} is the positive stock that enters the leak law and the split threshold.

Definition 12.1 (Characterization algebra, operational triviality, and even null ideal)**.** Fix an accepted tick k and let $E^-_{\Lambda_k}$ denote the physically odd sufficiently local generator space at cutoff Λ_k, i.e. the odd sector for the transported physical involution Γ. Let $E^+_{\Lambda_k}$ denote the corresponding physically even sufficiently local observable space. Define the core characterization algebra by

$$\mathcal{O}^{\mathrm{char}}_{\Lambda_k} := \mathrm{alg}\{a_{\Lambda_k},\, S_{\Lambda_k}\}. \tag{70}$$

Define the operationally interface-trivial odd subspace by

$$\mathcal{T}^{\mathrm{op}}_{\Lambda_k} := \overline{\left\{T \in E^-_{\Lambda_k} : \{T, O\}_{\mathrm{P},\Lambda_k} = 0 \ \forall O \in \mathcal{O}^{\mathrm{char}}_{\Lambda_k}\right\}}. \tag{71}$$

Define the even interface-null ideal by

$$\mathcal{N}^{\mathrm{even}}_{\Lambda_k} := \overline{\left\{E \in E^+_{\Lambda_k} : \{E, O\}_{\mathrm{P},\Lambda_k} = 0 \ \forall O \in \mathcal{O}^{\mathrm{char}}_{\Lambda_k}\right\}}. \tag{72}$$

Remark 12.2 (Do not identify the characterization observable with the generator). The odd observable a_{Λ_k} characterizes the active completion-sector channel, but it is not itself the odd Hamiltonian generator class appearing in item **(M4)** of Definition 4.2. That generator is denoted $J^{\mathrm{act}}_{\Lambda_k} \in E^-_{\Lambda_k}$ below. The quotient statement belongs to $[J^{\mathrm{act}}_{\Lambda_k}]$, not to $[a_{\Lambda_k}]$.

Definition 12.3 (Intrinsic covariance-tangent response map)**.** Fix an accepted tick k. Let $\mathcal{T}^{\mathrm{cov}}_{\Lambda_k}$ denote the finite-cutoff receiver covariance-tangent space at the UOM reference Gaussian state ρ_{*,Λ_k}, equipped with the Fisher pairing

$$\langle\!\langle \delta C, \delta C' \rangle\!\rangle_{C_{*,\Lambda_k}} := \tfrac{1}{2}\,\mathrm{tr}\big(C^{-1}_{*,\Lambda_k}\delta C\, C^{-1}_{*,\Lambda_k}\delta C'\big).$$

For admissible band-limited receiver probes f, g, let

$$\mathcal{Q}^{\Lambda_k}_{f,g} \in E^{+}_{\Lambda_k}$$

denote the corresponding sufficiently local quadratic observable, normalized so that on centered Gaussian states with covariance C,

$$\big\langle \mathcal{Q}^{\Lambda_k}_{f,g} \big\rangle_{\rho(C)} = \langle f, Cg\rangle.$$

For $F \in E^{-}_{\Lambda_k}$, define $\mathcal{R}_{\Lambda_k}(F) \in \mathcal{T}^{\mathrm{cov}}_{\Lambda_k}$ by the matrix elements

$$\big\langle f, \mathcal{R}_{\Lambda_k}(F) g\big\rangle := \Big\langle \{F, \mathcal{Q}^{\Lambda_k}_{f,g}\}_{\mathrm{P},\Lambda_k}\Big\rangle_{\rho_{*,\Lambda_k}}.$$

We call

$$\mathcal{R}_{\Lambda_k} : E^{-}_{\Lambda_k} \to \mathcal{T}^{\mathrm{cov}}_{\Lambda_k}$$

the *intrinsic covariance-tangent response map.*

Proposition 12.4 (Intrinsic definition and completion independence)**.** *The map $\mathcal{R}_{\Lambda_k}$ of definition 12.3 is continuous and linear. It depends only on the finite-cutoff UOM Peierls package and the UOM covariance/Fisher package. If a Gaussian completion realization is chosen, then $\mathcal{R}_{\Lambda_k}(F)$ is represented by the same completion-sector covariance tangent that one obtains by linearizing the completion quadratures. Thus Gaussian completion is a realization of the response map, not extra structure required to define it.*

Proof. For fixed $F \in E^{-}_{\Lambda_k}$, the finite-cutoff UOM Peierls package makes

$$\delta_F := \{F, -\}_{\mathrm{P},\Lambda_k}$$

a continuous derivation on the sufficiently local algebra. Applying δ_F to the quadratic probes $\mathcal{Q}^{\Lambda_k}_{f,g}$ and evaluating in the reference Gaussian state produces a continuous symmetric bilinear form in f, g. In the UOM Gaussian radiative sector, such bilinear forms are exactly covariance tangents at C_{*,Λ_k}, equipped with the Fisher metric. Linearity and continuity in F are inherited directly from the Peierls bracket. No completion coordinates are used in this construction.

If a Gaussian completion is later chosen, the same second moments are encoded by the covariance matrix of the completion quadratures. The bilinear form defined above and the completion-sector covariance tangent therefore represent the same infinitesimal response object in two different coordinate systems. □

Definition 12.5 (Physical intrinsic covariance-response map)**.** Fix an accepted tick k. Let

$$\mathcal{T}^{\mathrm{phys}}_{\Lambda_k} := \Pi_{\mathrm{phys},\Lambda_k}\mathcal{T}^{\mathrm{cov}}_{\Lambda_k} \subset \mathcal{T}^{\mathrm{cov}}_{\Lambda_k}.$$

Define the physical intrinsic covariance-response map by

$$R^{\mathrm{phys}}_{\Lambda_k} := \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k} : E^{-}_{\Lambda_k} \to \mathcal{T}^{\mathrm{phys}}_{\Lambda_k}.$$

Proposition 12.6 (Physical intrinsic response map)**.** *The map $R^{\mathrm{phys}}_{\Lambda_k}$ of definition 12.5 is continuous and linear.*

Proof. This is immediate from proposition 12.4 and the continuity and linearity of the finite-cutoff physical projector $\Pi_{\mathrm{phys},\Lambda_k}$. □

Proposition 12.7 (Reduction of operational triviality to the active odd coordinate)**.** *Fix an accepted tick k. If $F \in E^-_{\Lambda_k}$ satisfies*

$$\{F, a_{\Lambda_k}\}_{\mathrm{P},\Lambda_k} = 0,$$

then $F \in \mathcal{T}^{\mathrm{op}}_{\Lambda_k}$.

Proof. Since $S_{\Lambda_k} = a^2_{\Lambda_k}$ and the Peierls bracket is a derivation,

$$\{F, S_{\Lambda_k}\}_{\mathrm{P},\Lambda_k} = \{F, a^2_{\Lambda_k}\}_{\mathrm{P},\Lambda_k} = 2\, a_{\Lambda_k}\, \{F, a_{\Lambda_k}\}_{\mathrm{P},\Lambda_k} = 0.$$

The algebra $\mathcal{O}^{\mathrm{char}}_{\Lambda_k} = \mathrm{alg}\{a_{\Lambda_k}, S_{\Lambda_k}\}$ is generated by a_{Λ_k} and S_{Λ_k}, and the Peierls bracket acts on it by the Leibniz rule. Hence $\{F, O\}_{\mathrm{P},\Lambda_k} = 0$ for every $O \in \mathcal{O}^{\mathrm{char}}_{\Lambda_k}$, proving $F \in \mathcal{T}^{\mathrm{op}}_{\Lambda_k}$. □

Assumption 12.1 (Active odd generator and intrinsic-response factorization)**.** For each accepted tick k, there exist a distinguished active odd generator

$$J^{\mathrm{act}}_{\Lambda_k} \in E^-_{\Lambda_k}$$

and a nonzero physical covariance tangent

$$C^{\mathrm{act}}_{\Lambda_k} \in \mathcal{T}^{\mathrm{phys}}_{\Lambda_k}$$

and a continuous nonzero linear functional

$$\rho_{\Lambda_k} : E^-_{\Lambda_k} \to \mathbb{R}$$

normalized by

$$\rho_{\Lambda_k}(J^{\mathrm{act}}_{\Lambda_k}) = 1,$$

such that

$$R^{\mathrm{phys}}_{\Lambda_k}(J^{\mathrm{act}}_{\Lambda_k}) = C^{\mathrm{act}}_{\Lambda_k} \neq 0$$

and

$$R^{\mathrm{phys}}_{\Lambda_k}(F) = \rho_{\Lambda_k}(F)\, C^{\mathrm{act}}_{\Lambda_k} \qquad \forall F \in E^-_{\Lambda_k} \tag{73}$$

holds on the strong sector under consideration.

Remark 12.8 (Strong normalization). If one further chooses the Fisher-unit normalization $\widehat{C}^{\mathrm{act}}_{\Lambda_k} := \|C^{\mathrm{act}}_{\Lambda_k}\|^{-1}_{C_{*,\Lambda_k}}\, C^{\mathrm{act}}_{\Lambda_k}$, then (73) sharpens to

$$R^{\mathrm{phys}}_{\Lambda_k}(F) = \widehat{\rho}_{\Lambda_k}(F)\, \widehat{C}^{\mathrm{act}}_{\Lambda_k}$$

with $\widehat{\rho}_{\Lambda_k} := \|C^{\mathrm{act}}_{\Lambda_k}\|_{C_{*,\Lambda_k}}\, \rho_{\Lambda_k}$. The export below does not require this extra normalization; it only needs the scalar response coefficient ρ_{Λ_k} relative to a fixed nonzero active tangent $C^{\mathrm{act}}_{\Lambda_k}$.

Assumption 12.2 (Physical-response sufficiency)**.** For each accepted tick k, the operationally trivial odd directions are exactly the kernel of the physical intrinsic covariance-response map:

$$\ker R^{\mathrm{phys}}_{\Lambda_k} = \mathcal{T}^{\mathrm{op}}_{\Lambda_k}.$$

Equivalently, an odd generator has zero physical intrinsic covariance response if and only if it is operationally trivial on the full core characterization algebra $\mathcal{O}^{\mathrm{char}}_{\Lambda_k}$.

Remark 12.9 (Why the strong package must be stated intrinsically)*.* The earlier quotient-valued formulation of the strong package in $E^+_{\Lambda_k}/\mathcal{N}^{\mathrm{even}}_{\Lambda_k}$ is not stable. Because the Peierls bracket here is the ordinary derivation bracket and $S_{\Lambda_k} = a^2_{\Lambda_k}$, the even observable $\{F, a_{\Lambda_k}\}_{\mathrm{P},\Lambda_k}$ already Poisson-commutes with both a_{Λ_k} and S_{Λ_k}, hence lies automatically in $\mathcal{N}^{\mathrm{even}}_{\Lambda_k}$. The correct codomain for factorization and kernel identification is therefore the intrinsic physical covariance-response space $\mathcal{T}^{\mathrm{phys}}_{\Lambda_k}$, not the even observable quotient.

Proposition 12.10 (Operational triviality is the kernel of the active response)**.** *Assume assumptions 12.1 and 12.2. Then*

$$\mathcal{T}^{\mathrm{op}}_{\Lambda_k} = \ker \rho_{\Lambda_k}.$$

Proof. Because $C^{\mathrm{act}}_{\Lambda_k} \neq 0$, the factorization identity (73) implies

$$R^{\mathrm{phys}}_{\Lambda_k}(F) = 0 \qquad \Longleftrightarrow \qquad \rho_{\Lambda_k}(F) = 0$$

for every $F \in E^-_{\Lambda_k}$. Thus $\ker R^{\mathrm{phys}}_{\Lambda_k} = \ker \rho_{\Lambda_k}$. Apply assumption 12.2. □

Corollary 12.11 (Rank-one active odd quotient)**.** *Assume assumptions 12.1 and 12.2. Then*

$$E^-_{\Lambda_k}/\mathcal{T}^{\mathrm{op}}_{\Lambda_k} \cong \mathbb{R},$$

and the quotient is generated by the class of the active odd generator $[J^{\mathrm{act}}_{\Lambda_k}]$.

Proof. By proposition 12.10, the quotient is canonically isomorphic to the image of the nonzero functional ρ_{Λ_k}. Since ρ_{Λ_k} is real-valued and $\rho_{\Lambda_k}(J^{\mathrm{act}}_{\Lambda_k}) = 1$, its image is $\mathbb{R}$ and the quotient is generated by $[J^{\mathrm{act}}_{\Lambda_k}]$. □

Proposition 12.12 (BRST/gauge-exact compatibility on the characterization layer)**.** *For each accepted tick k, the odd parts of the BRST/gauge-exact directions*

$$Q_t X + Q_u Y$$

lie in $\mathcal{T}^{\mathrm{op}}_{\Lambda_k}$.

Proof. The exported characterization observables a_{Λ_k} and S_{Λ_k} are physical finite-cutoff observables on the characterization layer. The BRST/gauge-exact odd directions $Q_t X + Q_u Y$ represent redundancies of that physical layer, so their Peierls action vanishes on $\mathcal{O}^{\mathrm{char}}_{\Lambda_k}$. By (71), their odd parts therefore lie in $\mathcal{T}^{\mathrm{op}}_{\Lambda_k}$. □

Proposition 12.13 (Boundary-admissible divergence compatibility)**.** *Let K^μ be a sufficiently local current whose renormalized boundary flux vanishes on the core characterization algebra $\mathcal{O}^{\mathrm{char}}_{\Lambda_k}$. Then the odd part of*

$$\partial_\mu K^\mu$$

lies in $\mathcal{T}^{\mathrm{op}}_{\Lambda_k}$.

Proof. For the admissible currents in the statement, the renormalized Stokes reduction of $\partial_\mu K^\mu$ produces no boundary term on the characterization layer. Hence its Peierls action on every $O \in \mathcal{O}^{\mathrm{char}}_{\Lambda_k}$ vanishes, and the odd part belongs to $\mathcal{T}^{\mathrm{op}}_{\Lambda_k}$ by (71). □

Definition 12.14 (Admissible odd counterterms on the characterization layer)**.** For each accepted tick k, an odd sufficiently local counterterm ambiguity

$$\Delta^-_{\mathrm{ct}} \in E^-_{\Lambda_k}$$

is called *admissible on the characterization layer* if its intrinsic covariance response has no physical projection:

$$\Pi_{\mathrm{phys},\Lambda_k}\, \mathcal{R}_{\Lambda_k}(\Delta^-_{\mathrm{ct}}) = 0.$$

Equivalently, admissible odd counterterms are precisely those local scheme shifts whose response is confined to the RP, Weyl, conformal, or other even-null monitor sectors and does not enter the compressed active odd line.

Proposition 12.15 (Counterterm rigidity on the characterization layer)**.** *Assume that the characterization observable a_{Λ_k} is extracted from the physical covariance response so that*

$$\Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}(F) = 0 \quad \Longrightarrow \quad \{F, a_{\Lambda_k}\}_{\mathrm{P},\Lambda_k} = 0 \qquad \forall F \in E^-_{\Lambda_k}.$$

Then every admissible odd counterterm ambiguity in the sense of definition 12.14 acts trivially on the core characterization algebra:

$$\Delta^-_{\mathrm{ct}} \in \mathcal{T}^{\mathrm{op}}_{\Lambda_k}.$$

Proof. If $F = \Delta^-_{\mathrm{ct}}$ is admissible, then this physical projection vanishes by definition. The displayed hypothesis therefore gives

$$\{\Delta^-_{\mathrm{ct}}, a_{\Lambda_k}\}_{\mathrm{P},\Lambda_k} = 0,$$

and since $S_{\Lambda_k} = a^2_{\Lambda_k}$, the derivation property yields

$$\{\Delta^-_{\mathrm{ct}}, S_{\Lambda_k}\}_{\mathrm{P},\Lambda_k} = 0.$$

By proposition 12.7, the first identity already implies $\Delta^-_{\mathrm{ct}} \in \mathcal{T}^{\mathrm{op}}_{\Lambda_k}$. □

Remark 12.16 (Active stiffness versus full calibration). The split/leak analysis determines the active stiffness $\widetilde{\alpha}(\Lambda_k)$ on the distinguished line generated by $J^{\mathrm{act}}_{\Lambda_k}$, with

$$\widetilde{\alpha}(\Lambda_k) \asymp \alpha(\Lambda_k) \sim \Lambda_k^{d+2}$$

by assumption 9.7. What UOM does *not* determine canonically is a positive bilinear form on the whole odd generator space $E^-_{\Lambda_k}$. That extension is auxiliary interface data, not a UOM-dynamical output.

Assumption 12.3 (Auxiliary coercive core on the operationally trivial sector)**.** For each accepted tick k, choose a continuous symmetric coercive bilinear form

$$\langle\cdot,\cdot\rangle^{\mathrm{aux}}_{\Lambda_k} : \mathcal{T}^{\mathrm{op}}_{\Lambda_k} \times \mathcal{T}^{\mathrm{op}}_{\Lambda_k} \to \mathbb{R}$$

whose induced norm Hilbertizes $\mathcal{T}^{\mathrm{op}}_{\Lambda_k}$.

Proposition 12.17 (Strong-package calibration extension from active stiffness)**.** *Assume assumptions 12.1 to 12.3. Set*

$$\chi_{\Lambda_k} := \rho_{\Lambda_k}, \qquad \mathcal{T}_{\Lambda_k} := \mathcal{T}^{\mathrm{op}}_{\Lambda_k}.$$

Then the formula

$$\mathsf{B}_{\Lambda_k}(F,G) := \widetilde{\alpha}(\Lambda_k)\,\chi_{\Lambda_k}(F)\chi_{\Lambda_k}(G) + \big\langle F - \chi_{\Lambda_k}(F)J^{\mathrm{act}}_{\Lambda_k},\, G - \chi_{\Lambda_k}(G)J^{\mathrm{act}}_{\Lambda_k}\big\rangle^{\mathrm{aux}}_{\Lambda_k} \tag{74}$$

defines a continuous symmetric positive definite bilinear form on $E^-_{\Lambda_k}$ satisfying

$$\mathsf{B}_{\Lambda_k}(J^{\mathrm{act}}_{\Lambda_k}, J^{\mathrm{act}}_{\Lambda_k}) = \widetilde{\alpha}(\Lambda_k), \qquad \mathsf{B}_{\Lambda_k}(J^{\mathrm{act}}_{\Lambda_k}, T) = 0 \quad \forall T \in \mathcal{T}_{\Lambda_k}.$$

Proof. By proposition 12.10, every $F \in E^-_{\Lambda_k}$ has a unique decomposition

$$F = \chi_{\Lambda_k}(F)\, J^{\mathrm{act}}_{\Lambda_k} + T_F, \qquad T_F \in \mathcal{T}_{\Lambda_k}.$$

Equation (74) is therefore well-defined, continuous, and symmetric. Positivity is immediate from $\widetilde{\alpha}(\Lambda_k) > 0$ together with coercivity of $\langle\cdot,\cdot\rangle^{\mathrm{aux}}_{\Lambda_k}$ on $\mathcal{T}_{\Lambda_k}$. If $\mathsf{B}_{\Lambda_k}(F,F) = 0$, then both $\chi_{\Lambda_k}(F) = 0$ and $\langle T_F, T_F\rangle^{\mathrm{aux}}_{\Lambda_k} = 0$, hence $T_F = 0$ and $F = 0$. The displayed normalization and orthogonality properties follow by substitution into (74). □

Corollary 12.18 (Descendant ambiguity compatibility on the characterization layer)**.** *For each accepted tick k, the odd-projected ambiguity directions coming from*

$$Q_t X + Q_u Y, \qquad \partial_\mu K^\mu, \qquad \Delta^-_{\mathrm{ct}}$$

belong to $\mathcal{T}^{\mathrm{op}}_{\Lambda_k}$ whenever K^μ is taken from the boundary-admissible class of proposition 12.13 and Δ^-_{ct} is admissible in the sense of definition 12.14.

Proof. Combine propositions 12.12, 12.13 and 12.15. □

Theorem 12.19 (Strong rank-one realization of the characterization layer)**.** *Assume assumptions 9.4, 9.6, 9.7 and 12.1 to 12.3. Then for each accepted tick k, the split/leak construction exports the UOM-side data*

$$\big(a_{\Lambda_k},\, S_{\Lambda_k},\, J^{\mathrm{act}}_{\Lambda_k},\, C^{\mathrm{act}}_{\Lambda_k},\, \widetilde{\alpha}(\Lambda_k),\, \rho_{\Lambda_k}\big), \tag{75}$$

where:

1. *$\mathcal{O}^{\mathrm{char}}_{\Lambda_k} = \mathrm{alg}\{a_{\Lambda_k}, S_{\Lambda_k}\}$ is the minimal completion-sector characterization algebra;*
2. *the operationally trivial subspace satisfies*

$$\mathcal{T}_{\Lambda_k} := \mathcal{T}^{\mathrm{op}}_{\Lambda_k} = \ker R^{\mathrm{phys}}_{\Lambda_k} = \ker \rho_{\Lambda_k};$$

3. *the intrinsic physical response is one-dimensional on the strong sector:*

$$R^{\mathrm{phys}}_{\Lambda_k}(F) = \rho_{\Lambda_k}(F)\, C^{\mathrm{act}}_{\Lambda_k}, \qquad R^{\mathrm{phys}}_{\Lambda_k}(J^{\mathrm{act}}_{\Lambda_k}) = C^{\mathrm{act}}_{\Lambda_k} \neq 0;$$

4. *the active odd quotient is rank one and is generated by $[J^{\mathrm{act}}_{\Lambda_k}]$;*
5. *the abstract orientation functional of the transported interface is realized by*

$$\chi_{\Lambda_k} := \rho_{\Lambda_k};$$

6. *any admissible calibration form* B_{Λ_k} *in this strong package may be obtained by extending the active stiffness* $\widetilde{\alpha}(\Lambda_k)$ *with proposition 12.17.*

Proof. Definition 12.1 packages the active completion-sector channel extracted from the split/leak construction: a_{Λ_k} is the distinguished odd split coordinate singled out by sections 9.1 and 9.3, while $S_{\Lambda_k} = a_{\Lambda_k}^2$ is the even stock entering (47) and (50). The active odd generator $J_{\Lambda_k}^{\mathrm{act}}$, active physical covariance tangent $C_{\Lambda_k}^{\mathrm{act}}$, and scalar response coefficient ρ_{Λ_k} are exported by assumption 12.1; proposition 12.10 and corollary 12.11 then identify the abstract interface-trivial subspace and the corresponding rank-one quotient. The split/leak law fixes the active stiffness $\widetilde{\alpha}(\Lambda_k)$, and proposition 12.17 upgrades that one-line datum to an admissible full calibration form once the auxiliary coercive core is chosen. Finally, corollary 12.18 places the odd-projected descendant ambiguities in $\mathcal{T}_{\Lambda_k}$, yielding the stated ambiguity compatibility on the characterization layer. □

Theorem 12.20 (Intrinsic weak transported descendant-class interface)**.** *Assume the hypotheses of theorem 13.56 on a connected accepted-branch segment* I*. Then for every* $\Lambda \in I$*, the sufficiently local odd generator layer carries the weak transported interface data*

$$\big(E_\Lambda^-,\, \mathcal{A}_\Lambda^-,\, [\Xi_\Lambda^{(1)}],\, \chi_\Lambda,\, \varepsilon_\Lambda\big)$$

with

$$\chi_\Lambda := \nu_\Lambda^{-1}\rho_\Lambda^{\mathrm{dom}},$$

and the following conclusions hold.

1. *there exists a representative*

$$J_\Lambda^{\mathrm{desc}} \in [\Xi_\Lambda^{(1)}]$$

such that

$$\rho_\Lambda^{\mathrm{dom}}\big(J_\Lambda^{\mathrm{desc}}\big) \neq 0;$$

equivalently, the transported descendant sector has nonzero overlap with the canonically selected compressed dominant odd line;

2. *the orientation functional is realized intrinsically on the sufficiently local generator layer:*

$$\mathcal{A}_\Lambda^- \subset \ker \chi_\Lambda, \qquad \chi_\Lambda(J) = \varepsilon_\Lambda \quad \forall J \in [\Xi_\Lambda^{(1)}];$$

3. *along* I*, the sign* ε_Λ *is locally constant.*

In particular, the weak transported interface is realized intrinsically, without appeal to completion coordinates, under the stated hypotheses.

Proof. The functional $\rho_\Lambda^{\mathrm{dom}}$ is defined intrinsically from the sufficiently local generator layer by definition 12.3 together with the dominant-line prescription of definition 13.3. Item (1) is exactly the nonzero-overlap conclusion of item (3) in theorem 13.56, item (2) is exactly item (4) there, and item (3) is item (5). □

Theorem 12.21 (Quadratic-probe realization of the weak activity supplement)**.** *Assume the hypotheses of theorem 13.56 on a connected accepted-branch segment* I*. For each* $\Lambda \in I$*, define the even quadratic probe sector by*

$$\mathcal{P}_\Lambda^+ := \operatorname{span}\big\{\mathcal{Q}_{f,g}^\Lambda : f, g \text{ admissible band-limited receiver probes}\big\} \subset E_\Lambda^+.$$

Then, for every $F \in E_\Lambda^-$,

$$\{F, A\}_{\mathrm{P},\Lambda} = 0 \qquad \forall A \in \mathcal{P}_\Lambda^+$$

implies

$$\chi_\Lambda(F) = 0.$$

Equivalently, if $\chi_\Lambda(F) \neq 0$, then there exists a sufficiently local even quadratic probe $A \in \mathcal{P}_\Lambda^+$ such that

$$\{F, A\}_{\mathrm{P},\Lambda} \neq 0.$$

In particular, the weak interface data of theorem 12.20 carry a canonical activity witness sector built from the intrinsic quadratic receiver probes.

Proof. Assume

$$\{F, A\}_{\mathrm{P},\Lambda} = 0 \qquad \forall A \in \mathcal{P}_\Lambda^+.$$

In particular,

$$\{F, \mathcal{Q}_{f,g}^\Lambda\}_{\mathrm{P},\Lambda} = 0$$

for every admissible pair f, g. By definition 12.3, this means

$$\langle f, \mathcal{R}_\Lambda(F) g\rangle = \left\langle \{F, \mathcal{Q}_{f,g}^\Lambda\}_{\mathrm{P},\Lambda} \right\rangle_{\rho_{*,\Lambda}} = 0$$

for all admissible f, g, hence $\mathcal{R}_\Lambda(F) = 0$. Therefore

$$\Pi_{\mathrm{phys},\Lambda} \mathcal{R}_\Lambda(F) = 0,$$

and the definition of the raw dominant-response coefficient in definition 13.3 gives

$$\widetilde{\rho}_\Lambda^{\mathrm{dom}}(F) = 0.$$

By item (1) of theorem 13.56, $\delta C_\Lambda^{\mathrm{LoG}} \neq 0$, so dominant-line realization holds and $\rho_\Lambda^{\mathrm{dom}}$ is well defined. Hence

$$\rho_\Lambda^{\mathrm{dom}}(F) = 0.$$

Now apply theorem 12.20, which identifies $\chi_\Lambda = \nu_\Lambda^{-1} \rho_\Lambda^{\mathrm{dom}}$, to obtain

$$\chi_\Lambda(F) = 0.$$

The contrapositive gives the equivalent detection statement. □

Remark 12.22 (Correct scalar form of representative comparisons)*.* The weak export theorem should be read ρ^{dom}-first on the ambiguity quotient. Because

$$\chi_\Lambda := \nu_\Lambda^{-1} \rho_\Lambda^{\mathrm{dom}},$$

any comparison statement that depends only on dominant pairing is naturally a statement about the descended scalar

$$\overline{\rho}_\Lambda^{\mathrm{dom}} : E_\Lambda^- / \mathcal{A}_\Lambda^- \to \mathbb{R}$$

or, equivalently, about its value on the transported descendant affine class. Such a statement may be rewritten in χ_Λ form only when the normalization is made explicit; for example, if $\nu_\Lambda = 1$, then the same scalar value is represented by a descended $\bar{\chi}_\Lambda$. Without that normalization the stronger unscaled χ-compatibility form is false in general.

Corollary 12.23 (UOM-stiffness specialization of the weak transported calibration)**.** *Assume assumption 9.7 and fix an accepted tick k for which the conclusions of theorem 12.20 hold. Suppose* $\ker \chi_{\Lambda_k}$ *admits a continuous symmetric coercive bilinear form whose induced norm Hilbertizes it, and choose any*

$$J^\sharp_{\Lambda_k} \in E^-_{\Lambda_k} \qquad \textit{with} \qquad \chi_{\Lambda_k}(J^\sharp_{\Lambda_k}) = 1.$$

Then proposition 13.58 may be applied with

$$\beta_{\Lambda_k} := \widetilde{\alpha}(\Lambda_k)$$

to produce a continuous symmetric positive definite bilinear form

$$\mathsf{B}^{\widetilde{\alpha}}_{\Lambda_k} : E^-_{\Lambda_k} \times E^-_{\Lambda_k} \to \mathbb{R}$$

such that

$$\mathsf{B}^{\widetilde{\alpha}}_{\Lambda_k}(J^\sharp_{\Lambda_k}, J^\sharp_{\Lambda_k}) = \widetilde{\alpha}(\Lambda_k), \qquad \mathsf{B}^{\widetilde{\alpha}}_{\Lambda_k}(J^\sharp_{\Lambda_k}, K) = 0 \quad \forall K \in \ker \chi_{\Lambda_k}.$$

In particular,

$$\mathcal{A}^-_{\Lambda_k} \subset \ker \chi_{\Lambda_k}$$

implies

$$\mathsf{B}^{\widetilde{\alpha}}_{\Lambda_k}(J^\sharp_{\Lambda_k}, A) = 0 \qquad \forall A \in \mathcal{A}^-_{\Lambda_k}.$$

Proof. By remark 12.16, the split/leak package fixes the positive stiffness scale $\widetilde{\alpha}(\Lambda_k)$, and assumption 9.7 records its positivity and asymptotic size. Apply proposition 13.58 with $\beta_{\Lambda_k} = \widetilde{\alpha}(\Lambda_k)$. The final orthogonality statement follows from item (2) of theorem 12.20. □

Remark 12.24 (Inherited ledger compatibility)*.* The inherited commutative ledger algebra is intentionally not part of the core characterization algebra $\mathcal{O}^{\mathrm{char}}_{\Lambda_k}$. Its role is separate: the split model requires the active odd representative to be completion-sector nontrivial while acting trivially, to leading order, on the inherited classical ledger. This is the UOM-side compatibility statement behind the slogan "oddness in the completion sector, classicality in the accessible ledger."

13 Finite-Cutoff Active Response Transport Theorem

This section assembles a finite-cutoff transport theorem for the odd descendant / Hamiltonization lane of the 2T Main assumption package Definition 4.2. Accordingly, the phrase "accepted tick" remains literal UOM cutoff language throughout this note: the transport theorem is pointwise on the accepted branch at fixed Λ_k.

Remark 13.1 (Physical odd sector and branchwise ambient-space conventions)*.* Throughout this note, the superscript $^-$ refers to physical Γ-odd parity on the characterization layer, not to cohomological oddness. Every ambiguity direction appearing below is therefore understood after passage to the physically odd characterization sector $E^-_{\Lambda_k}$.

All branchwise continuity statements are made after the standard accepted-branch identifications of the relevant finite-cutoff source spaces, covariance-tangent spaces, and physical subspaces with fixed ambient finite-cutoff spaces. Operator-norm and Fisher-norm continuity are always understood with respect to those identifications.

Definition 13.2 (Odd ambiguity subspace on the characterization layer)**.** For each accepted tick k, let

$$\mathcal{A}^-_{\Lambda_k} \subset E^-_{\Lambda_k}$$

denote the closed linear span of the odd characterization-layer projections of the descendant ambiguity directions

$$\big(Q_t X + Q_u Y\big)^-, \qquad \big(\partial_\mu K^\mu\big)^-, \qquad \Delta^-_{\mathrm{ct}},$$

where K^μ is boundary-admissible on the characterization layer and Δ^-_{ct} is admissible in the sense of definition 12.14. Equivalently, $\mathcal{A}^-_{\Lambda_k}$ is the odd ambiguity space already placed inside $\mathcal{T}^{\mathrm{op}}_{\Lambda_k}$ by corollary 12.18.

Definition 13.3 (Dominant compressed response line and scalar coefficient)**.** Fix an accepted tick k. Let

$$u^{\mathrm{LoG}}_{\Lambda_k}$$

and

$$P^{\mathrm{LoG}}_{-,\Lambda_k}$$

be the canonically selected compressed odd line and projector furnished by corollary 10.10. Define the physical covariance tangent induced by that line by

$$\delta C^{\mathrm{LoG}}_{\Lambda_k} := \Pi_{\mathrm{phys},\Lambda_k}\, \mathsf{T}_{\Lambda_k}[u^{\mathrm{LoG}}_{\Lambda_k}],$$

and, whenever this tangent is nonzero, write

$$L^{\mathrm{dom}}_{\Lambda_k} := \mathbb{R}\, \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}$$

for its Fisher-normalized line. The raw dominant-response coefficient is

$$\widetilde{\rho}^{\mathrm{dom}}_{\Lambda_k}(F) := \big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}(F),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\big\rangle\!\big\rangle_{C_{*,\Lambda_k}}, \qquad F \in E^-_{\Lambda_k}. \tag{76}$$

Whenever $\delta C^{\mathrm{LoG}}_{\Lambda_k} \neq 0$, equivalently whenever dominant-line realization holds at Λ_k, define the normalized dominant-response coefficient by

$$\rho^{\mathrm{dom}}_{\Lambda_k}(F) := \kappa^{-1/2}_{\Lambda_k}\, \widetilde{\rho}^{\mathrm{dom}}_{\Lambda_k}(F), \qquad F \in E^-_{\Lambda_k}. \tag{77}$$

Proposition 13.4 (Well-posedness and continuity of the dominant-response coefficients)**.** *Fix an accepted tick k. Then the raw dominant-response coefficient*

$$\widetilde{\rho}^{\mathrm{dom}}_{\Lambda_k} : E^-_{\Lambda_k} \to \mathbb{R}$$

is continuous and linear. If dominant-line realization holds at Λ_k, then the normalized dominant-response coefficient

$$\rho^{\mathrm{dom}}_{\Lambda_k} : E^-_{\Lambda_k} \to \mathbb{R}$$

is also continuous and linear.

Proof. By proposition 12.4, the intrinsic response map $\mathcal{R}_{\Lambda_k} : E^-_{\Lambda_k} \to \mathcal{T}^{\mathrm{cov}}_{\Lambda_k}$ is continuous and linear. Pairing its physical projection against the fixed covariance tangent $\delta C^{\mathrm{LoG}}_{\Lambda_k}$ in the Fisher metric therefore gives a continuous linear functional, which is exactly $\widetilde{\rho}^{\mathrm{dom}}_{\Lambda_k}$. If dominant-line realization holds, then $\kappa_{\Lambda_k} > 0$ by proposition 13.7, so $\rho^{\mathrm{dom}}_{\Lambda_k}$ is obtained from $\widetilde{\rho}^{\mathrm{dom}}_{\Lambda_k}$ by multiplication by the fixed scalar $\kappa^{-1/2}_{\Lambda_k}$. □

Dominant-line realization

The compressed spectral-selection note already supplies the one-dimensional source line $\mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}$. The present transport note needs only the nonvanishing and branchwise continuity of its *physical* covariance push-forward.

Definition 13.5 (Compressed-line transversality)**.** For an accepted tick k, say that the welded response is *transversely physical on the compressed line* if

$$\Pi_{\mathrm{phys},\Lambda_k}\,\mathsf{T}_{\Lambda_k}\restriction_{\mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}} \neq 0. \tag{78}$$

Equivalently, the unique generator $u^{\mathrm{LoG}}_{\Lambda_k}$ of the compressed odd line has nonzero physical covariance response.

Definition 13.6 (Dominant-response norm)**.** Whenever $\delta C^{\mathrm{LoG}}_{\Lambda_k}$ is defined, set

$$\kappa_{\Lambda_k} := \langle\!\langle \delta C^{\mathrm{LoG}}_{\Lambda_k},\, \delta C^{\mathrm{LoG}}_{\Lambda_k} \rangle\!\rangle_{C_{*,\Lambda_k}}. \tag{79}$$

This is the Fisher squared norm of the physical covariance tangent induced by the compressed welded LoG source line.

Proposition 13.7 (Equivalent formulations of dominant-line realization)**.** *For a fixed accepted tick k, the following are equivalent.*

1. *the compressed welded LoG line is transversely physical in the sense of definition 13.5;*
2. *the distinguished physical covariance tangent is nonzero,*
$$\delta C^{\mathrm{LoG}}_{\Lambda_k} \neq 0;$$
3. *the dominant-response norm is strictly positive,*
$$\kappa_{\Lambda_k} > 0;$$
4. *there exists a physical covariance tangent $B_{\Lambda_k} \in \mathcal{T}^{\mathrm{phys}}_{\Lambda_k}$ such that*
$$\langle\!\langle \delta C^{\mathrm{LoG}}_{\Lambda_k},\, B_{\Lambda_k} \rangle\!\rangle_{C_{*,\Lambda_k}} \neq 0;$$
5. *in any Gaussian completion realization of corollary 10.11, the completion-sector covariance tangent representing $\delta C^{\mathrm{LoG}}_{\Lambda_k}$ is nonzero.*

Proof. The equivalence of (1) and (2) is immediate from (78) and the definition

$$\delta C^{\mathrm{LoG}}_{\Lambda_k} = \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}[u^{\mathrm{LoG}}_{\Lambda_k}],$$

because $\mathcal{H}^{\mathrm{LoG}}_{\Lambda_k}$ is the one-dimensional line generated by $u^{\mathrm{LoG}}_{\Lambda_k}$.

The equivalence of (2) and (3) follows from positive definiteness of the Fisher metric on the covariance tangent space: a vector has positive Fisher norm if and only if it is nonzero.

Statement (3) implies (4) by taking $B_{\Lambda_k} := \delta C^{\mathrm{LoG}}_{\Lambda_k}$, which belongs to the physical tangent space by construction. Conversely, if (4) holds then $\delta C^{\mathrm{LoG}}_{\Lambda_k}$ cannot vanish, so (2) holds.

Finally, proposition 12.4 and corollary 10.11 identify the abstract covariance tangent with any Gaussian completion representative of the same tangent object. Therefore the abstract tangent is nonzero exactly when its completion realization is nonzero, giving the equivalence of (2) and (5). □

Theorem 13.8 (Branchwise realization and persistence of the dominant line)**.** *Let I be a connected accepted-branch segment in cutoff space. Assume that:*

1. *the compressed projector $P^{\mathrm{LoG}}_{-,\Lambda}$ varies continuously on I;*
2. *the physical welded response operator*
$$\Lambda \longmapsto \Pi_{\mathrm{phys},\Lambda}\mathsf{T}_\Lambda$$
is continuous on I in the operator topology induced by the Fisher norm on covariance tangents;
3. *there exists a constant $c_I > 0$ such that*
$$\kappa_\Lambda \geq c_I \qquad \forall \Lambda \in I.$$

Then:

1. *$\delta C^{\mathrm{LoG}}_\Lambda \neq 0$ for every $\Lambda \in I$;*
2. *the normalized dominant vector*
$$\widehat{\delta C}^{\mathrm{LoG}}_\Lambda := \kappa_\Lambda^{-1/2}\,\delta C^{\mathrm{LoG}}_\Lambda$$
is continuous on I;
3. *the dominant physical line*
$$L^{\mathrm{dom}}_\Lambda = \mathbb{R}\,\widehat{\delta C}^{\mathrm{LoG}}_\Lambda$$
is canonically transported along I;
4. *the Fisher-orthogonal rank-one projector*
$$\Pi^{\mathrm{dom}}_\Lambda := \widehat{\delta C}^{\mathrm{LoG}}_\Lambda \otimes \widehat{\delta C}^{\mathrm{LoG}}_\Lambda$$
depends continuously on $\Lambda \in I$.

Proof. By corollary 10.10, the compressed projector $P^{\mathrm{LoG}}_{-,\Lambda}$ and hence the normalized source vector u^{LoG}_Λ vary continuously on the branch. By the assumed continuity of $\Pi_{\mathrm{phys},\Lambda}\mathsf{T}_\Lambda$, the map
$$\Lambda \longmapsto \delta C^{\mathrm{LoG}}_\Lambda = \Pi_{\mathrm{phys},\Lambda}\mathsf{T}_\Lambda[u^{\mathrm{LoG}}_\Lambda]$$
is continuous in the Fisher norm.

The lower bound $\kappa_\Lambda \geq c_I > 0$ implies $\delta C^{\mathrm{LoG}}_\Lambda \neq 0$ for every $\Lambda \in I$, proving item (1). Since the normalization map $v \mapsto v/\|v\|_{C_*}$ is continuous on the complement of the zero section, continuity of $\delta C^{\mathrm{LoG}}_\Lambda$ and the positive lower bound on its norm imply continuity of $\widehat{\delta C}^{\mathrm{LoG}}_\Lambda$, proving item (2). Items (3) and (4) are immediate consequences of item (2). □

Corollary 13.9 (Local persistence from one-tick transversality)**.** *Assume the continuity hypotheses of theorem 13.8. If the compressed line is transversely physical at one accepted tick Λ_{k_0}, then there exists a neighborhood I_{k_0} of Λ_{k_0} on the same accepted branch such that*
$$\delta C^{\mathrm{LoG}}_\Lambda \neq 0 \qquad \forall \Lambda \in I_{k_0}.$$
Hence the dominant physical line is locally well defined and continuously transported near Λ_{k_0}.

Proof. By proposition 13.7, transversality at Λ_{k_0} is equivalent to $\kappa_{\Lambda_{k_0}} > 0$. Continuity of κ_Λ then gives a neighborhood I_{k_0} and a constant $c_{k_0} > 0$ such that $\kappa_\Lambda \geq c_{k_0}$ for all $\Lambda \in I_{k_0}$. Apply theorem 13.8. □

Remark 13.10 (Practical witness criteria). For applications, any one of the following suffices to discharge dominant-line realization at a fixed accepted tick:

1. a direct lower bound $\kappa_{\Lambda_k} > 0$;
2. a physical witness tangent $B_{\Lambda_k} \in \mathcal{T}^{\mathrm{phys}}_{\Lambda_k}$ with nonzero Fisher pairing against $\delta C^{\mathrm{LoG}}_{\Lambda_k}$;
3. a Gaussian completion calculation showing that the completion covariance of the realized active odd mode changes nontrivially under the compressed welded LoG source.

Each witness proves the same theorem-level fact by proposition 13.7.

Descent to the ambiguity quotient

Once the dominant line has been realized, the next task is purely structural: show that the dominant-response coefficient is insensitive to ambiguity moves. This is the exact point at which the transported descendant class becomes an affine class in the sense required by Main rather than a collection of representatives.

Definition 13.11 (Dominant-response quotient and descended descendant point)**.** For each accepted tick k, define the dominant-response quotient by

$$\mathfrak{D}^-_{\Lambda_k} := E^-_{\Lambda_k}/\mathcal{A}^-_{\Lambda_k},$$

with quotient map

$$q_{\Lambda_k} : E^-_{\Lambda_k} \to \mathfrak{D}^-_{\Lambda_k}.$$

If

$$[\Xi^{(1)}_{\Lambda_k}] = J_{0,\Lambda_k} + \mathcal{A}^-_{\Lambda_k},$$

then its image in the quotient is a single point,

$$[\![\Xi^{(1)}_{\Lambda_k}]\!] := q_{\Lambda_k}(J_{0,\Lambda_k}) \in \mathfrak{D}^-_{\Lambda_k},$$

independent of the chosen representative J_{0,Λ_k}.

Proposition 13.12 (Universal descent criterion for the dominant-response coefficient)**.** *Assume dominant-line realization at an accepted tick k, so that $\rho^{\mathrm{dom}}_{\Lambda_k}$ is defined by (77). Then the following are equivalent.*

1. *there exists a unique continuous linear functional*

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k} : \mathfrak{D}^-_{\Lambda_k} \to \mathbb{R}$$

   such that

$$\rho^{\mathrm{dom}}_{\Lambda_k} = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k} \circ q_{\Lambda_k};$$

2. *the ambiguity subspace is contained in the kernel:*

$$\mathcal{A}^-_{\Lambda_k} \subset \ker \rho^{\mathrm{dom}}_{\Lambda_k};$$

3. the dominant-response coefficient is constant on each ambiguity coset:

$$\rho^{\mathrm{dom}}_{\Lambda_k}(F+A) = \rho^{\mathrm{dom}}_{\Lambda_k}(F) \qquad \forall F \in E^-_{\Lambda_k},\ \forall A \in \mathcal{A}^-_{\Lambda_k}.$$

Proof. The equivalence of (2) and (3) is immediate from linearity:

$$\rho^{\mathrm{dom}}_{\Lambda_k}(F+A) - \rho^{\mathrm{dom}}_{\Lambda_k}(F) = \rho^{\mathrm{dom}}_{\Lambda_k}(A).$$

If (2) holds, define

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big(q_{\Lambda_k}(F)\big) := \rho^{\mathrm{dom}}_{\Lambda_k}(F).$$

This is well defined precisely because changing representatives by an element of $\mathcal{A}^-_{\Lambda_k}$ does not change the value of $\rho^{\mathrm{dom}}_{\Lambda_k}$. Linearity and continuity are inherited from $\rho^{\mathrm{dom}}_{\Lambda_k}$, and uniqueness follows because q_{Λ_k} is surjective.

Conversely, if (1) holds and $A \in \mathcal{A}^-_{\Lambda_k}$, then $q_{\Lambda_k}(A) = 0$, hence

$$\rho^{\mathrm{dom}}_{\Lambda_k}(A) = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big(q_{\Lambda_k}(A)\big) = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}(0) = 0.$$

Thus (1) implies (2). □

Corollary 13.13 (Constancy on the transported descendant affine class)**.** *Assume proposition 13.12. Then*

$$\rho^{\mathrm{dom}}_{\Lambda_k}(J) = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big) \qquad \forall J \in [\Xi^{(1)}_{\Lambda_k}].$$

In particular, once the descended quotient functional exists, the descendant affine class carries a single unambiguous dominant-response value.

Proof. Every $J \in [\Xi^{(1)}_{\Lambda_k}]$ has the form $J = J_{0,\Lambda_k} + A$ with $A \in \mathcal{A}^-_{\Lambda_k}$, so

$$q_{\Lambda_k}(J) = q_{\Lambda_k}(J_{0,\Lambda_k}) = [\![\Xi^{(1)}_{\Lambda_k}]\!].$$

Apply the factorization identity of proposition 13.12. □

Remark 13.14 (False stronger form and normalized special case)*.* The quotient statement above is intrinsically a statement about the descended dominant scalar

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big),$$

not about a bare χ-compatibility condition. At the weak transported-interface layer one has

$$\chi_{\Lambda_k} = \nu^{-1}_{\Lambda_k}\rho^{\mathrm{dom}}_{\Lambda_k},$$

so a representative-search condition of the form

$$\rho^{\mathrm{dom}}_{\Lambda_k}(J) = s$$

is equivalent to

$$\chi_{\Lambda_k}(J) = \nu^{-1}_{\Lambda_k} s,$$

not in general to $\chi_{\Lambda_k}(J) = s$. Only after an explicit normalization, for example $\nu_{\Lambda_k} = 1$, may the same scalar comparison be rewritten in descended $\bar{\chi}_{\Lambda_k}$ form without changing the value. The finite-cutoff Wolfram audit contains exact countermodels to the stronger unscaled χ-level claim.

Proposition 13.15 (Generator-by-generator reduction)**.** *Assume dominant-line realization at an accepted tick k. Suppose the dominant-response coefficient vanishes on the three generating ambiguity families:*

1. *for every odd-projected BRST/gauge-exact direction $\big(Q_tX + Q_uY\big)^-$,*

$$\rho^{\mathrm{dom}}_{\Lambda_k}\big(\big(Q_tX + Q_uY\big)^-\big) = 0;$$

2. *for every odd-projected boundary-admissible divergence $\big(\partial_\mu K^\mu\big)^-$,*

$$\rho^{\mathrm{dom}}_{\Lambda_k}\big(\big(\partial_\mu K^\mu\big)^-\big) = 0;$$

3. *for every admissible odd counterterm Δ^-_{ct},*

$$\rho^{\mathrm{dom}}_{\Lambda_k}(\Delta^-_{\mathrm{ct}}) = 0.$$

Then

$$\mathcal{A}^-_{\Lambda_k} \subset \ker \rho^{\mathrm{dom}}_{\Lambda_k},$$

hence $\rho^{\mathrm{dom}}_{\Lambda_k}$ descends uniquely to the quotient $\mathfrak{D}^-_{\Lambda_k}$.

Proof. By definition 13.2, $\mathcal{A}^-_{\Lambda_k}$ is the closed linear span of the three families listed in the statement. Since $\rho^{\mathrm{dom}}_{\Lambda_k}$ is linear, it vanishes on the algebraic span of those generators. Since it is also continuous, it vanishes on the closure of that span. Therefore every element of $\mathcal{A}^-_{\Lambda_k}$ lies in the kernel. Now apply proposition 13.12. □

Theorem 13.16 (Descent to the ambiguity quotient)**.** *Fix an accepted tick k and assume dominant-line realization. If the three vanishing statements of proposition 13.15 hold, then there exists a unique descended dominant functional*

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k} : \mathfrak{D}^-_{\Lambda_k} \to \mathbb{R}$$

such that

$$\rho^{\mathrm{dom}}_{\Lambda_k} = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k} \circ q_{\Lambda_k},$$

and the transported descendant affine class has a single well-defined quotient value

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big).$$

Proof. Apply proposition 13.15 and corollary 13.13. □

Nonzero overlap on the descendant class

After dominant-line realization and quotient descent, the descendant class defines a single point in the ambiguity quotient, and the dominant-response functional descends to that quotient. This subsection studies whether that descended descendant point is *active*, i.e. not annihilated by the descended dominant functional.

Definition 13.17 (Dominant covariance projector)**.** Assume dominant-line realization at an accepted tick k. Define the Fisher-orthogonal rank-one projector onto the dominant physical line by

$$\Pi^{\mathrm{dom}}_{\Lambda_k}(X) := \big\langle\!\big\langle X,\, \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k} \big\rangle\!\big\rangle_{C_{*,\Lambda_k}} \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}, \qquad X \in \mathcal{T}^{\mathrm{cov}}_{\Lambda_k}. \tag{80}$$

Proposition 13.18 (Representative-level witness criteria)**.** *Assume dominant-line realization at an accepted tick k. For any odd generator $F \in E^-_{\Lambda_k}$, the following are equivalent.*

1. *the dominant-response coefficient is nonzero:*

$$\rho^{\mathrm{dom}}_{\Lambda_k}(F) \neq 0;$$

2. *the dominant projection of the physical covariance response is nonzero:*

$$\Pi^{\mathrm{dom}}_{\Lambda_k}\Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}(F) \neq 0;$$

3. *the physical covariance response has nonzero Fisher pairing with the dominant tangent:*

$$\langle\!\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}(F),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\rangle\!\rangle_{C_{*,\Lambda_k}} \neq 0;$$

4. *in any Gaussian completion realization of corollary 10.11, the completion representative of $\Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}(F)$ has nonzero component on the realized active odd line.*

Proof. By definition 13.17,

$$\Pi^{\mathrm{dom}}_{\Lambda_k}\Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}(F) = \rho^{\mathrm{dom}}_{\Lambda_k}(F)\,\widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}.$$

Since $\widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k} \neq 0$, this proves the equivalence of (1) and (2).

Statement (3) is equivalent to (1) because

$$\langle\!\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}(F),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\rangle\!\rangle_{C_{*,\Lambda_k}} = \|\delta C^{\mathrm{LoG}}_{\Lambda_k}\|_{C_{*,\Lambda_k}}\,\rho^{\mathrm{dom}}_{\Lambda_k}(F),$$

and the prefactor is strictly positive by dominant-line realization.

Finally, by proposition 12.4 and corollary 10.11, the abstract covariance tangent and its Gaussian completion representative describe the same physical response object in different coordinates. Therefore the abstract dominant projection is nonzero if and only if the completion representative has nonzero component on the realized active line, proving the equivalence of (2) and (4). □

Definition 13.19 (Descendant overlap margin)**.** Assume dominant-line realization and quotient descent at an accepted tick k. Define the descendant overlap margin by

$$\nu_{\Lambda_k} := \left|\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big)\right|. \tag{81}$$

We say that the transported descendant class is *dominant-active* if $\nu_{\Lambda_k} > 0$.

Proposition 13.20 (Equivalent formulations of nonzero overlap on the descendant class)**.** *Assume dominant-line realization and quotient descent at an accepted tick k. Then the following are equivalent.*

1. *the transported descendant sector has nonzero overlap with the canonically selected compressed dominant odd line;*

2. *there exists one representative $J^{\mathrm{desc}}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$ such that*

$$\Pi^{\mathrm{dom}}_{\Lambda_k}\Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}(J^{\mathrm{desc}}_{\Lambda_k}) \neq 0;$$

3. *every representative* $J \in [\Xi^{(1)}_{\Lambda_k}]$ *has the same nonzero dominant-response value:*

$$\rho^{\mathrm{dom}}_{\Lambda_k}(J) = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big) \neq 0;$$

4. *the descended descendant point is not annihilated by the descended dominant functional:*

$$[\![\Xi^{(1)}_{\Lambda_k}]\!] \notin \ker \overline{\rho}^{\mathrm{dom}}_{\Lambda_k};$$

5. *the descendant overlap margin is strictly positive:*

$$\nu_{\Lambda_k} > 0.$$

Proof. The equivalence of (1) and (2) follows immediately from proposition 13.18.

If (2) holds, then $\rho^{\mathrm{dom}}_{\Lambda_k}(J^{\mathrm{desc}}_{\Lambda_k}) \neq 0$ by proposition 13.18. By corollary 13.13, every other representative of $[\Xi^{(1)}_{\Lambda_k}]$ has the same value, proving (3). Conversely, (3) obviously implies (1).

The equivalence of (3) and (4) is immediate from the identity

$$\rho^{\mathrm{dom}}_{\Lambda_k}(J) = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big) \qquad \forall J \in [\Xi^{(1)}_{\Lambda_k}]$$

of corollary 13.13.

Finally, (4) and (5) are equivalent by the definition of ν_{Λ_k} in definition 13.19. □

Theorem 13.21 (Nonzero overlap on the descendant class)**.** *Fix an accepted tick k, and assume dominant-line realization together with quotient descent. If any one of the equivalent conditions in proposition 13.20 holds, then the following conclusions obtain. In particular, it is enough to construct a single representative*

$$J^{\mathrm{desc}}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$$

whose dominant projected covariance response is nonzero.

When this holds, the quotient descendant point is dominant-active, the scalar value

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big)$$

is nonzero, and the sign

$$\varepsilon_{\Lambda_k} := \mathrm{sgn}\Big(\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big)\Big) \in \{\pm 1\} \tag{82}$$

is well defined. Thus the transported descendant sector has nonzero overlap with the canonically selected compressed dominant odd line.

Proof. The equivalence package is exactly proposition 13.20. Once one representative has nonzero dominant projected response, the same proposition gives the nonzero scalar value on the descended descendant point. The sign in (82) is therefore well defined. □

Conditional mod-null reentry on the selected scalar line

Definition 13.22 (Parity-stable mod-null subspace transverse to the selected scalar line)**.** Fix an accepted tick k, assume dominant-line realization and quotient descent, and let

$$\overline{L}^{\mathrm{Main}}_{\Lambda_k} = \mathbb{R}\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_k} \subset \mathfrak{D}^{-}_{\Lambda_k}$$

be a one-dimensional selected scalar line in the ambiguity quotient. A *parity-stable mod-null datum transverse to* $\overline{L}^{\mathrm{Main}}_{\Lambda_k}$ consists of a linear subspace

$$N^{\mathrm{mn}}_{\Lambda_k} \subset \mathfrak{D}^{-}_{\Lambda_k}$$

such that:

(a) $N^{\mathrm{mn}}_{\Lambda_k}$ is stable under the descended physical parity on $\mathfrak{D}^-_{\Lambda_k}$;

(b) the selected scalar line is transverse to $N^{\mathrm{mn}}_{\Lambda_k}$:

$$\overline{L}^{\mathrm{Main}}_{\Lambda_k} \cap N^{\mathrm{mn}}_{\Lambda_k} = \{0\}.$$

Define

$$\mathcal{V}^{1,\mathrm{mn}}_{\Lambda_k} := \mathfrak{D}^-_{\Lambda_k}/N^{\mathrm{mn}}_{\Lambda_k}$$

with quotient map

$$\pi^{\mathrm{mn}}_{\Lambda_k} : \mathfrak{D}^-_{\Lambda_k} \to \mathcal{V}^{1,\mathrm{mn}}_{\Lambda_k}.$$

Because $\mathfrak{D}^-_{\Lambda_k}$ is itself a quotient of the physical Γ-odd sector, the descended parity acts on $\mathfrak{D}^-_{\Lambda_k}$ by $-\mathrm{id}$. Thus the parity-stability clause yields a descended involution

$$\Gamma^{\mathrm{mn}}_{\Lambda_k}\big(\pi^{\mathrm{mn}}_{\Lambda_k}(x)\big) := \pi^{\mathrm{mn}}_{\Lambda_k}(-x), \qquad x \in \mathfrak{D}^-_{\Lambda_k}.$$

Definition 13.23 (Dominant-null parity-stable mod-null datum)**.** Fix an accepted tick k, assume dominant-line realization and quotient descent, and let

$$\overline{L}^{\mathrm{Main}}_{\Lambda_k} = \mathbb{R}\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}$$

be a one-dimensional selected scalar line in the ambiguity quotient. A parity-stable mod-null datum

$$N^{\mathrm{mn}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}$$

is called *dominant-null* if it is contained in the kernel of the descended dominant functional:

$$N^{\mathrm{mn}}_{\Lambda_k} \subset \ker \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}.$$

It is understood that $N^{\mathrm{mn}}_{\Lambda_k}$ is stable under the descended physical parity on $\mathfrak{D}^-_{\Lambda_k}$, and that the associated quotient $\mathfrak{D}^-_{\Lambda_k}/N^{\mathrm{mn}}_{\Lambda_k}$ carries the descended involution from definition 13.22. The canonical dominant-kernel choice is isolated separately in definition 13.24. The transversality lemma below uses only this kernel-subordination property.

Definition 13.24 (Canonical dominant-null datum)**.** Fix an accepted tick k and assume dominant-line realization together with quotient descent. Define the *canonical dominant-null datum* by

$$N^{\mathrm{dom-null}}_{\Lambda_k} := \ker \overline{\rho}^{\mathrm{dom}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}.$$

This datum depends only on the descended dominant functional on the ambiguity quotient. No additional reduced-welded or completion-side choice is built into the definition.

Lemma 13.25 (Dominant-kernel criterion for selected-line transversality)**.** *Fix an accepted tick k, assume dominant-line realization together with quotient descent, and let*

$$\overline{L}^{\mathrm{Main}}_{\Lambda_k} = \mathbb{R}\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}$$

be a one-dimensional selected scalar line with

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}\big) \neq 0.$$

If

$$N^{\mathrm{mn}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}$$

is a dominant-null parity-stable mod-null datum in the sense of definition 13.23, then

$$\overline{L}^{\mathrm{Main}}_{\Lambda_k} \cap N^{\mathrm{mn}}_{\Lambda_k} = \{0\}.$$

So every dominant-null parity-stable mod-null datum is automatically transverse to the selected scalar line.

Proof. Let $x \in \overline{L}^{\mathrm{Main}}_{\Lambda_k} \cap N^{\mathrm{mn}}_{\Lambda_k}$. Because $\overline{L}^{\mathrm{Main}}_{\Lambda_k} = \mathbb{R}\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}$, there exists $a \in \mathbb{R}$ with $x = a\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}$. Since $N^{\mathrm{mn}}_{\Lambda_k} \subset \ker \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}$, we have

$$0 = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}(x) = a\,\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}\big).$$

The coefficient $\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k})$ is nonzero by hypothesis, so $a = 0$. Therefore $x = 0$, proving $\overline{L}^{\mathrm{Main}}_{\Lambda_k} \cap N^{\mathrm{mn}}_{\Lambda_k} = \{0\}$. □

Theorem 13.26 (The dominant kernel is the canonical dominant-null datum)**.** *Fix an accepted tick k, assume dominant-line realization together with quotient descent, and let*

$$\overline{L}^{\mathrm{Main}}_{\Lambda_k} = \mathbb{R}\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}$$

be a one-dimensional selected scalar line with

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}\big) \neq 0.$$

Define

$$N^{\mathrm{dom-null}}_{\Lambda_k} := \ker \overline{\rho}^{\mathrm{dom}}_{\Lambda_k} \subset \mathfrak{D}^-_{\Lambda_k}.$$

Then:

1. *$N^{\mathrm{dom-null}}_{\Lambda_k}$ is stable under the descended physical parity on $\mathfrak{D}^-_{\Lambda_k}$;*
2. *$N^{\mathrm{dom-null}}_{\Lambda_k}$ is a dominant-null parity-stable mod-null datum in the sense of definition 13.23;*
3. *the selected scalar line is automatically transverse to the canonical dominant-null datum:*

$$\overline{L}^{\mathrm{Main}}_{\Lambda_k} \cap N^{\mathrm{dom-null}}_{\Lambda_k} = \{0\}.$$

Hence the canonical dominant kernel is admissible as the mod-null datum for the conditional reentry theorem.

Proof. Let $x \in N^{\mathrm{dom-null}}_{\Lambda_k}$. Because the descended physical parity acts on $\mathfrak{D}^-_{\Lambda_k}$ by $-\mathrm{id}$, we have

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}(-x) = -\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}(x) = 0.$$

So $-x \in N^{\mathrm{dom-null}}_{\Lambda_k}$, proving parity stability. Item (2) is then exactly definition 13.23 with $N^{\mathrm{mn}}_{\Lambda_k} = N^{\mathrm{dom-null}}_{\Lambda_k}$. Item (3) is lemma 13.25 applied to this canonical choice. □

Lemma 13.27 (Transversality reduces mod-null reentry to quotient construction)**.** *Fix an accepted tick k, assume dominant-line realization together with quotient descent, and let*

$$\overline{L}^{\mathrm{Main}}_{\Lambda_k} = \mathbb{R}\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_k} \subset \mathfrak{D}^{-}_{\Lambda_k}$$

be a one-dimensional selected scalar line with

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}\big) \neq 0.$$

Assume moreover a parity-stable mod-null datum transverse to $\overline{L}^{\mathrm{Main}}_{\Lambda_k}$ in the sense of definition 13.22. Define

$$\ell^{\mathrm{sel,mn}}_{\Lambda_k} := \pi^{\mathrm{mn}}_{\Lambda_k}\big(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}\big)$$

and

$$L^{\mathrm{sel,mn}}_{\Lambda_k} := \mathbb{R}\,\ell^{\mathrm{sel,mn}}_{\Lambda_k} \subset \mathcal{V}^{1,\mathrm{mn}}_{\Lambda_k}.$$

Then:

1. *the restriction of $\pi^{\mathrm{mn}}_{\Lambda_k}$ to $\overline{L}^{\mathrm{Main}}_{\Lambda_k}$ is injective;*
2. *$L^{\mathrm{sel,mn}}_{\Lambda_k}$ is a well-defined one-dimensional line in the mod-null quotient, generated by the nonzero class $\ell^{\mathrm{sel,mn}}_{\Lambda_k}$;*
3. *the mod-null selected realization map*

$$\mathfrak{R}^{\mathrm{mn}}_{\Lambda_k} : L^{\mathrm{sel,mn}}_{\Lambda_k} \longrightarrow \mathbb{R}\,u^{\mathrm{LoG}}_{\Lambda_k}, \qquad \mathfrak{R}^{\mathrm{mn}}_{\Lambda_k}\big(a\,\ell^{\mathrm{sel,mn}}_{\Lambda_k}\big) := a\,\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}\big)\,u^{\mathrm{LoG}}_{\Lambda_k}$$

   is well defined and faithful;
4. *the descended involution acts by sign:*

$$\Gamma^{\mathrm{mn}}_{\Lambda_k}\big(\ell^{\mathrm{sel,mn}}_{\Lambda_k}\big) = -\ell^{\mathrm{sel,mn}}_{\Lambda_k}.$$

In particular, once the transversality condition $\overline{L}^{\mathrm{Main}}_{\Lambda_k} \cap N^{\mathrm{mn}}_{\Lambda_k} = \{0\}$ is known, the remaining mod-null reentry step reduces to quotient construction plus parity descent.

Proof. If $x \in \overline{L}^{\mathrm{Main}}_{\Lambda_k}$ and $\pi^{\mathrm{mn}}_{\Lambda_k}(x) = 0$, then $x \in N^{\mathrm{mn}}_{\Lambda_k}$ by definition of the quotient kernel. Hence $x \in \overline{L}^{\mathrm{Main}}_{\Lambda_k} \cap N^{\mathrm{mn}}_{\Lambda_k} = \{0\}$, which proves injectivity. Since $\overline{\ell}^{\mathrm{Main}}_{\Lambda_k} \neq 0$, its image $\ell^{\mathrm{sel,mn}}_{\Lambda_k}$ is therefore nonzero, and the image of a one-dimensional line under an injective linear map is again one-dimensional. This proves items (1) and (2).

Injectivity on $\overline{L}^{\mathrm{Main}}_{\Lambda_k}$ implies that every element of $L^{\mathrm{sel,mn}}_{\Lambda_k}$ has a unique expression $a\,\ell^{\mathrm{sel,mn}}_{\Lambda_k}$, so the displayed formula defines $\mathfrak{R}^{\mathrm{mn}}_{\Lambda_k}$ unambiguously. Its coefficient $\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k})$ is nonzero by hypothesis, and $u^{\mathrm{LoG}}_{\Lambda_k} \neq 0$ by corollary 10.10. Therefore $\mathfrak{R}^{\mathrm{mn}}_{\Lambda_k}(a\,\ell^{\mathrm{sel,mn}}_{\Lambda_k}) = 0$ implies $a = 0$, so the map is faithful. This proves item (3).

By construction of the descended involution,

$$\Gamma^{\mathrm{mn}}_{\Lambda_k}\big(\ell^{\mathrm{sel,mn}}_{\Lambda_k}\big) = \Gamma^{\mathrm{mn}}_{\Lambda_k}\Big(\pi^{\mathrm{mn}}_{\Lambda_k}\big(\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}\big)\Big) = \pi^{\mathrm{mn}}_{\Lambda_k}\big(-\overline{\ell}^{\mathrm{Main}}_{\Lambda_k}\big) = -\ell^{\mathrm{sel,mn}}_{\Lambda_k},$$

which proves item (4). □

Theorem 13.28 (Conditional mod-null reentry on the selected scalar line)**.** *Fix an accepted tick k and the next accepted receiver step $k+1$. Assume the hypotheses and notation of* **??**. *In particular, let*

$$\overline{L}^{\mathrm{Main}}_{\Lambda_{k+1}} = \mathbb{R}\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}} \subset \mathfrak{D}^{-}_{\Lambda_{k+1}}$$

be the nonzero selected scalar descendant line supplied there. Assume moreover a parity-stable mod-null datum transverse to $\overline{L}^{\mathrm{Main}}_{\Lambda_{k+1}}$ in the sense of definition 13.22. Define

$$\ell^{\mathrm{sel,mn}}_{\Lambda_{k+1}} := \pi^{\mathrm{mn}}_{\Lambda_{k+1}}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}})$$

and

$$L^{\mathrm{sel,mn}}_{\Lambda_{k+1}} := \mathbb{R}\,\ell^{\mathrm{sel,mn}}_{\Lambda_{k+1}} \subset \mathcal{V}^{1,\mathrm{mn}}_{\Lambda_{k+1}}.$$

Then:

1. *$L^{\mathrm{sel,mn}}_{\Lambda_{k+1}}$ is a well-defined one-dimensional selected descendant line in the mod-null quotient;*
2. *the induced mod-null selected realization map*

$$\mathfrak{R}^{\mathrm{mn}}_{\Lambda_{k+1}} : L^{\mathrm{sel,mn}}_{\Lambda_{k+1}} \longrightarrow \mathbb{R}\,u^{\mathrm{LoG}}_{\Lambda_{k+1}}, \qquad \mathfrak{R}^{\mathrm{mn}}_{\Lambda_{k+1}}(a\,\ell^{\mathrm{sel,mn}}_{\Lambda_{k+1}}) := a\,\overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}})\,u^{\mathrm{LoG}}_{\Lambda_{k+1}}$$

   is well defined and faithful;
3. *the descended involution acts by sign:*

$$\Gamma^{\mathrm{mn}}_{\Lambda_{k+1}}(\ell^{\mathrm{sel,mn}}_{\Lambda_{k+1}}) = -\ell^{\mathrm{sel,mn}}_{\Lambda_{k+1}}.$$

Proof. By **??**,

$$\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}} \neq 0 \qquad \text{and} \qquad \overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}}) \neq 0.$$

So every hypothesis of lemma 13.27 is in force at Λ_{k+1}. Applying that lemma gives items (1)–(3) immediately. □

Corollary 13.29 (Canonical dominant-kernel discharge of the mod-null reentry step)**.** *Fix an accepted tick k and the next accepted receiver step $k+1$. Assume the hypotheses and notation of* **??**. *Define the canonical dominant-null datum*

$$N^{\mathrm{dom-null}}_{\Lambda_{k+1}} := \ker \overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}} \subset \mathfrak{D}^{-}_{\Lambda_{k+1}}.$$

Then $N^{\mathrm{dom-null}}_{\Lambda_{k+1}}$ is admissible as the mod-null datum of theorem 13.28, the transversality hypothesis of that theorem is automatic, and hence the conclusions of that theorem hold with this canonical datum.

Proof. By **??**,

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}}) \neq 0.$$

So theorem 13.26 applies at Λ_{k+1}. It shows that $N^{\mathrm{dom-null}}_{\Lambda_{k+1}}$ is a dominant-null parity-stable mod-null datum and that

$$\overline{L}^{\mathrm{Main}}_{\Lambda_{k+1}} \cap N^{\mathrm{dom-null}}_{\Lambda_{k+1}} = \{0\}.$$

This is exactly the admissibility and transversality clause required by theorem 13.28, so that theorem now applies directly with the canonical datum. □

Remark 13.30 (Conditional character of the reentry step)*.* theorem 13.28 is a quotient-level reentry statement. After lemma 13.25, theorem 13.26, and corollary 13.29, the canonical dominant-null datum is fixed as $\ker \overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}}$ on $\mathfrak{D}^{-}_{\Lambda_{k+1}}$. What remains is passage to the mod-null quotient together with parity descent on the odd line. Accordingly, the theorem should not be read as a statement on the raw descendant ambiguity quotient or on any stronger physical quotient.

Branchwise orientation transport

This subsection studies transport of the descended descendant sign along a connected accepted branch. The sign can be transported once the descended descendant value varies continuously with the cutoff. The compressed spectral-selection theorem already gives continuity of the dominant projector, and theorem 13.8 gives continuity of the dominant physical line. We combine those inputs with continuity of the projected descendant response.

Definition 13.31 (Branchwise descendant overlap function)**.** Let I be a connected accepted-branch segment. Assume dominant-line realization and quotient descent for every $\Lambda \in I$. Suppose there is a branchwise choice of representatives

$$J_\Lambda^{\mathrm{desc}} \in [\Xi_\Lambda^{(1)}] \qquad (\Lambda \in I)$$

such that the projected covariance response

$$\Lambda \longmapsto \Pi_{\mathrm{phys},\Lambda}\mathcal{R}_\Lambda(J_\Lambda^{\mathrm{desc}})$$

is continuous in the Fisher topology. Define the associated branchwise descendant overlap function by

$$m_I(\Lambda) := \rho_\Lambda^{\mathrm{dom}}(J_\Lambda^{\mathrm{desc}}) = \overline{\rho}_\Lambda^{\mathrm{dom}}\big([\![\Xi_\Lambda^{(1)}]\!]\big). \tag{83}$$

Proposition 13.32 (Continuity of the descended overlap scalar)**.** *Let I be a connected accepted-branch segment. Assume:*

1. *the hypotheses of theorem 13.8 on I;*
2. *quotient descent holds for every $\Lambda \in I$;*
3. *the branchwise choice of representatives in definition 13.31 exists.*

Then the scalar function $m_I : I \to \mathbb{R}$ is well defined, independent of the chosen branchwise representatives, and continuous. Moreover,

$$|m_I(\Lambda)| = \nu_\Lambda \qquad \forall \Lambda \in I. \tag{84}$$

Proof. Independence of the chosen branchwise representatives is exactly the coset constancy proved in corollary 13.13: if $J_\Lambda'^{\mathrm{desc}} - J_\Lambda^{\mathrm{desc}} \in \mathcal{A}_\Lambda^-$, then

$$\rho_\Lambda^{\mathrm{dom}}(J_\Lambda'^{\mathrm{desc}}) = \rho_\Lambda^{\mathrm{dom}}(J_\Lambda^{\mathrm{desc}}) = \overline{\rho}_\Lambda^{\mathrm{dom}}\big([\![\Xi_\Lambda^{(1)}]\!]\big).$$

Thus m_I is well defined.

By theorem 13.8, the normalized dominant vector $\widehat{\delta C}_\Lambda^{\mathrm{LoG}}$ and the dominant projector $\Pi_\Lambda^{\mathrm{dom}}$ depend continuously on $\Lambda \in I$. By assumption, the response vector

$$X_\Lambda := \Pi_{\mathrm{phys},\Lambda}\mathcal{R}_\Lambda(J_\Lambda^{\mathrm{desc}})$$

is also continuous on I. Using definition 13.17, we have

$$\Pi_\Lambda^{\mathrm{dom}} X_\Lambda = m_I(\Lambda)\, \widehat{\delta C}_\Lambda^{\mathrm{LoG}}.$$

The left-hand side is continuous because both factors are continuous, while $\widehat{\delta C}_\Lambda^{\mathrm{LoG}} \neq 0$ for all $\Lambda \in I$ by theorem 13.8. Therefore the scalar coefficient $m_I(\Lambda)$ is continuous.

Finally, (84) is just the definition of ν_Λ from definition 13.19 together with (83). □

Theorem 13.33 (Branchwise orientation transport on the descendant class)**.** *Let I be a connected accepted-branch segment and assume the hypotheses of proposition 13.32. Then:*

1. *on every connected subsegment $I' \subset I$ such that*

$$\nu_\Lambda > 0 \qquad \forall \Lambda \in I',$$

the sign

$$\varepsilon_\Lambda := \operatorname{sgn} m_I(\Lambda) = \operatorname{sgn}\Big(\overline{\rho}_\Lambda^{\mathrm{dom}}\big([\![\Xi_\Lambda^{(1)}]\!]\big)\Big) \in \{\pm 1\} \tag{85}$$

is constant on I';

2. *if $\nu_{\Lambda_0} > 0$ at one point $\Lambda_0 \in I$, then there exists a neighborhood $I_0 \subset I$ of Λ_0 on which ε_Λ is constant and equal to ε_{Λ_0};*

3. *under the present hypotheses, a change of sign can occur only at a cutoff where the descendant overlap closes $\nu_\Lambda = 0$.*

In particular, once dominant-line realization, quotient descent, and nonzero overlap are established on an accepted branch segment and the projected descendant response varies continuously there, the branchwise sign statement of theorem 13.56 follows.

Proof. By proposition 13.32, $m_I(\Lambda)$ is a continuous real-valued function on I. On any connected subsegment I' where $\nu_\Lambda = |m_I(\Lambda)| > 0$, the function m_I never vanishes. A continuous nonvanishing real-valued function on a connected set has constant sign, proving item (1).

Item (2) is the local version of the same argument: if $m_I(\Lambda_0) \neq 0$, continuity gives a neighborhood I_0 on which m_I keeps the sign it has at Λ_0.

For item (3), any sign change would force m_I to pass through zero by the intermediate value theorem, hence $\nu_\Lambda = |m_I(\Lambda)| = 0$ at some intermediate cutoff. □

Corollary 13.34 (Local orientation persistence from one-tick activity)**.** *Let I be a connected accepted-branch segment and assume the hypotheses of proposition 13.32. If the transported descendant class is dominant-active at one cutoff $\Lambda_0 \in I$, i.e.*

$$\nu_{\Lambda_0} > 0,$$

then there exists a neighborhood $I_0 \subset I$ of Λ_0 such that

$$\varepsilon_\Lambda = \varepsilon_{\Lambda_0} \qquad \forall \Lambda \in I_0.$$

Proof. This is exactly item (2) of theorem 13.33. □

Witness-indexed seam realization and scalar bridge

The bridge used below factors through a common LS witness datum rather than a direct identification of a UOM seam source with a descendant representative. On the UOM side one realizes a filler tag together with a detector witness by an admissible odd welded source, measures the resulting dominant physical pairing, and then compares that scalar with the odd representative attached to the same witness.

Definition 13.35 (Accepted record payload of a welded odd source)**.** Fix an accepted tick k. For an admissible welded odd trace-channel source τ, let

$$\rho_{\Lambda_k}[\tau]$$

denote the receiver state at cutoff Λ_k produced by that source in the welded UOM channel. Define its accepted record payload by

$$\mathcal{P}^{\mathrm{acc}}_{\Lambda_k}[\tau] := \Pi_{\Delta(\Lambda_k)}\big(\rho_{\Lambda_k}[\tau]\big). \tag{86}$$

This is the diagonal/record part retained by the finite-cutoff acceptance map.

Definition 13.36 (Witness-indexed seam realization at an accepted tick)**.** Fix an accepted tick k. Let y be a witness on the LS difference object, and write $\mathrm{flip}(y)$ for the induced LS involution. A *witness-indexed seam realization* at Λ_k is a choice of admissible welded odd trace-channel sources

$$\tau^{f,y}_{\Lambda_k}, \qquad f \in \{\mathrm{diag}, \mathrm{bridge}\},$$

such that:

1. **filler toggle changes the odd seam sign:**

$$\tau^{\mathrm{bridge},y}_{\Lambda_k} = -\,\tau^{\mathrm{diag},y}_{\Lambda_k};$$

2. **witness flip changes the odd seam sign:**

$$\tau^{f,\mathrm{flip}(y)}_{\Lambda_k} = -\,\tau^{f,y}_{\Lambda_k} \qquad \forall f \in \{\mathrm{diag}, \mathrm{bridge}\};$$

3. **accepted record payload is unchanged under filler toggle:**

$$\mathcal{P}^{\mathrm{acc}}_{\Lambda_k}\big[\tau^{\mathrm{bridge},y}_{\Lambda_k}\big] = \mathcal{P}^{\mathrm{acc}}_{\Lambda_k}\big[\tau^{\mathrm{diag},y}_{\Lambda_k}\big].$$

Remark 13.37 (Accepted-record invariance versus seam-scalar use). The accepted-record invariance clause in definition 13.36 encodes the physical indistinguishability of filler toggle at the accepted-record layer. The seam-scalar theorems below use only the induced odd sign relations in the source channel.

Definition 13.38 (Raw and normalized seam pairings)**.** Fix an accepted tick k and a witness-indexed seam realization as in definition 13.36. For $f \in \{\mathrm{diag}, \mathrm{bridge}\}$ and witness y, define the *raw seam pairing* by

$$\widetilde{s}_{\Lambda_k}(f,y) := \big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}\big[\tau^{f,y}_{\Lambda_k}\big],\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\big\rangle\!\big\rangle_{C_{*,\Lambda_k}}. \tag{87}$$

Whenever dominant-line realization holds so that $\widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}$ is defined, also set

$$s_{\Lambda_k}(f,y) := \big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}\big[\tau^{f,y}_{\Lambda_k}\big],\, \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}\big\rangle\!\big\rangle_{C_{*,\Lambda_k}}. \tag{88}$$

Proposition 13.39 (Filler and witness parity of the seam scalar)**.** *Fix an accepted tick k and assume a witness-indexed seam realization at Λ_k. Then the raw seam pairing satisfies*

$$\widetilde{s}_{\Lambda_k}(\mathrm{bridge}, y) = -\,\widetilde{s}_{\Lambda_k}(\mathrm{diag}, y), \qquad \widetilde{s}_{\Lambda_k}(f, \mathrm{flip}(y)) = -\,\widetilde{s}_{\Lambda_k}(f, y). \tag{89}$$

If dominant-line realization also holds, then the same identities hold for the normalized seam pairing:

$$s_{\Lambda_k}(\mathrm{bridge}, y) = -\,s_{\Lambda_k}(\mathrm{diag}, y), \qquad s_{\Lambda_k}(f, \mathrm{flip}(y)) = -\,s_{\Lambda_k}(f, y). \tag{90}$$

Proof. The map $\tau \mapsto \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}[\tau]$ is linear because T_{Λ_k} and $\Pi_{\mathrm{phys},\Lambda_k}$ are linear. The Fisher pairing is bilinear. Applying the sign relations of definition 13.36 therefore gives

$$\Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}\big[\tau^{\mathrm{bridge},y}_{\Lambda_k}\big] = -\,\Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}\big[\tau^{\mathrm{diag},y}_{\Lambda_k}\big]$$

and

$$\Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}\big[\tau^{f,\mathrm{flip}(y)}_{\Lambda_k}\big] = -\,\Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}\big[\tau^{f,y}_{\Lambda_k}\big].$$

Pairing with $\delta C^{\mathrm{LoG}}_{\Lambda_k}$ proves (89). If dominant-line realization holds, then $\widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}$ is a fixed nonzero normalization of the same vector, so the same argument proves (90). □

Definition 13.40 (Detector-factorized seam coupling)**.** Fix an accepted tick k. A witness-indexed seam realization is called *detector-factorized* if there exists a single admissible welded odd trace-channel source

$$\tau^{\mathrm{unit}}_{\Lambda_k}$$

and the LS detector factor

$$\overline{\mathrm{obs}} : \text{LS difference object} \to \mathbb{R}$$

such that, for every filler tag f and witness y,

$$\tau^{f,y}_{\Lambda_k} = \sigma_{\mathrm{fill}}(f)\,\overline{\mathrm{obs}}(y)\,\tau^{\mathrm{unit}}_{\Lambda_k}, \qquad \sigma_{\mathrm{fill}}(\mathrm{diag}) = +1, \quad \sigma_{\mathrm{fill}}(\mathrm{bridge}) = -1. \tag{91}$$

Theorem 13.41 (Seam faithfulness scalar)**.** *Fix an accepted tick k and assume a detector-factorized witness-indexed seam realization at Λ_k. Define the fixed-cutoff seam coefficient by*

$$\widetilde{c}_{\Lambda_k} := \big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}\big[\tau^{\mathrm{unit}}_{\Lambda_k}\big],\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\big\rangle\!\big\rangle_{C_{*,\Lambda_k}}. \tag{92}$$

Then, for every filler tag f and witness y,

$$\widetilde{s}_{\Lambda_k}(f,y) = \sigma_{\mathrm{fill}}(f)\,\widetilde{c}_{\Lambda_k}\,\overline{\mathrm{obs}}(y). \tag{93}$$

In particular, if

$$\widetilde{c}_{\Lambda_k} \neq 0 \qquad \textit{and} \qquad \overline{\mathrm{obs}}(y) \neq 0,$$

then the raw seam pairing is nonzero:

$$\widetilde{s}_{\Lambda_k}(f,y) \neq 0.$$

Consequently the compressed welded LoG line is transversely physical at Λ_k, hence dominant-line realization holds at that cutoff by proposition 13.7. When dominant-line realization holds, the normalized seam coefficient

$$c_{\Lambda_k} := \kappa^{-1/2}_{\Lambda_k}\,\widetilde{c}_{\Lambda_k} \tag{94}$$

is defined and

$$s_{\Lambda_k}(f,y) = \sigma_{\mathrm{fill}}(f)\,c_{\Lambda_k}\,\overline{\mathrm{obs}}(y). \tag{95}$$

Proof. Substitute the detector-factorized form (91) into the definition (87). Linearity of T_{Λ_k} and bilinearity of the Fisher pairing give

$$\widetilde{s}_{\Lambda_k}(f,y) = \sigma_{\mathrm{fill}}(f)\,\overline{\mathrm{obs}}(y)\,\big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}\big[\tau^{\mathrm{unit}}_{\Lambda_k}\big],\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\big\rangle\!\big\rangle_{C_{*,\Lambda_k}},$$

which is exactly (93).

If $\widetilde{c}_{\Lambda_k} \neq 0$ and $\overline{\mathrm{obs}}(y) \neq 0$, then $\widetilde{s}_{\Lambda_k}(f,y) \neq 0$. Set

$$B_{\Lambda_k} := \Pi_{\mathrm{phys},\Lambda_k} \mathsf{T}_{\Lambda_k}[\tau^{f,y}_{\Lambda_k}].$$

Then

$$\langle\!\langle \delta C^{\mathrm{LoG}}_{\Lambda_k},\, B_{\Lambda_k} \rangle\!\rangle_{C_{*,\Lambda_k}} = \widetilde{s}_{\Lambda_k}(f,y) \neq 0.$$

By item (4) of proposition 13.7, this proves transversality of the compressed line and hence dominant-line realization at Λ_k.

Once dominant-line realization holds, divide (93) by $\|\delta C^{\mathrm{LoG}}_{\Lambda_k}\|_{C_{*,\Lambda_k}} = \kappa^{1/2}_{\Lambda_k}$ to obtain (95). □

Remark 13.42 (Downstream role of the seam-coefficient hypothesis). The fixed-cutoff nonvanishing hypothesis $\widetilde{c}_{\Lambda_k} \neq 0$ is exactly the input that, once dominant-line realization defines c_{Λ_k}, is imported downstream as the nondegeneracy clause $c_{\Lambda_k} \neq 0$ inside the delayed selected-line package. This note does not derive that clause from split-threshold analysis alone.

Definition 13.43 (Odd source representability on the characterization layer)**.** Fix an accepted tick k. Let τ be an admissible welded odd trace-channel source. We say that τ is *representable on the characterization layer* if there exists an odd sufficiently local generator

$$F_{\tau,\Lambda_k} \in E^-_{\Lambda_k}$$

such that its intrinsic physical covariance response agrees with the UOM welded response:

$$\Pi_{\mathrm{phys},\Lambda_k} \mathcal{R}_{\Lambda_k}(F_{\tau,\Lambda_k}) = \Pi_{\mathrm{phys},\Lambda_k} \mathsf{T}_{\Lambda_k}[\tau]. \tag{96}$$

This is the strong source-insertion realization statement.

Theorem 13.44 (UOM source-insertion representability theorem)**.** *Fix an accepted tick k. Let τ be an admissible welded odd trace-channel source at cutoff Λ_k, in the finite-cutoff UOM sense: its receiver-head profile is band-limited at Λ_k and compactly supported inside the accepted slab. Then there exists a canonically associated odd sufficiently local generator*

$$F_{\tau,\Lambda_k} \in E^-_{\Lambda_k}$$

such that

$$\Pi_{\mathrm{phys},\Lambda_k} \mathcal{R}_{\Lambda_k}(F_{\tau,\Lambda_k}) = \Pi_{\mathrm{phys},\Lambda_k} \mathsf{T}_{\Lambda_k}[\tau]. \tag{97}$$

In particular, every witness-indexed seam source used in the present note is representable on the characterization layer.

Proof. By the finite-cutoff UOM Peierls package, admissible receiver-head sources at cutoff Λ_k are band-limited boundary test data and the sufficiently local class $\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}$ contains all compact smearings of local boundary densities. Let

$$F_{\tau,\Lambda_k}$$

be the linear source functional obtained by pairing the odd trace-channel source profile τ with the corresponding odd trace-channel boundary field on the receiver head. Because this is the linear case of the sufficiently local form and τ is physically odd,

$$F_{\tau,\Lambda_k} \in E^-_{\Lambda_k}.$$

Now apply the UOM source insertion map from the finite-cutoff retarded-kernel package. The source τ is sent to the bulk forcing term in the linearized equation; the unique retarded solution is

obtained by the retarded Green operator; and boundary restriction gives the retarded covariance variation. By the UOM identification of the induced retarded kernel with the SK mixed second derivative, this covariance variation is exactly the welded source-to-covariance response

$$\mathsf{T}_{\Lambda_k}[\tau].$$

On the other hand, by the definition of the intrinsic covariance-response map, the matrix elements of $\mathcal{R}_{\Lambda_k}(F_{\tau,\Lambda_k})$ are computed by the Peierls action of F_{τ,Λ_k} on the sufficiently local quadratic probes representing boundary covariance entries. That Peierls variation is precisely the same retarded second-moment variation generated by the source insertion τ. Therefore

$$\mathcal{R}_{\Lambda_k}(F_{\tau,\Lambda_k}) = \mathsf{T}_{\Lambda_k}[\tau]$$

as covariance tangents. Applying the physical projector $\Pi_{\mathrm{phys},\Lambda_k}$ gives (97).

The witness-indexed seam sources of definition 13.36 are, by construction, live admissible welded odd trace-channel sources at the accepted tick, so the same argument applies to each of them. □

Proposition 13.45 (Dominant pairing for a representable odd source)**.** *Fix an accepted tick k and let τ be an admissible welded odd trace-channel source that is representable in the sense of theorem 13.44. Then*

$$\langle\!\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}(F_{\tau,\Lambda_k}),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\rangle\!\rangle_{C_{*,\Lambda_k}} = \langle\!\langle \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}[\tau],\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\rangle\!\rangle_{C_{*,\Lambda_k}}. \tag{98}$$

If dominant-line realization also holds, then

$$\rho^{\mathrm{dom}}_{\Lambda_k}(F_{\tau,\Lambda_k}) = \langle\!\langle \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}[\tau],\, \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}\rangle\!\rangle_{C_{*,\Lambda_k}}. \tag{99}$$

Proof. Equation (96) identifies the two physical covariance tangents. Pairing both sides with $\delta C^{\mathrm{LoG}}_{\Lambda_k}$ gives (98). If dominant-line realization holds, pair with the normalized vector $\widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}$ instead and use the definition (77) of $\rho^{\mathrm{dom}}_{\Lambda_k}$ to obtain (99). □

Corollary 13.46 (Unit welded odd source representability and scalar identification)**.** *Fix an accepted tick k. Assume a detector-factorized witness-indexed seam realization at Λ_k in the sense of definition 13.40, and assume dominant-line realization at Λ_k. Then the unit welded odd source $\tau^{\mathrm{unit}}_{\Lambda_k}$ is representable on the characterization layer, with canonical representable generator*

$$F_{\tau^{\mathrm{unit}}_{\Lambda_k},\Lambda_k} \in E^-_{\Lambda_k},$$

and

$$\rho^{\mathrm{dom}}_{\Lambda_k}(F_{\tau^{\mathrm{unit}}_{\Lambda_k},\Lambda_k}) = \langle\!\langle \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}[\tau^{\mathrm{unit}}_{\Lambda_k}],\, \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}\rangle\!\rangle_{C_{*,\Lambda_k}} = c_{\Lambda_k}. \tag{100}$$

Proof. Representability of $\tau^{\mathrm{unit}}_{\Lambda_k}$ is the specialization of theorem 13.44 to the admissible welded odd trace source singled out by definition 13.40. Because dominant-line realization holds, proposition 13.45 applies to that representable source and gives the first equality in (100). The second equality is exactly the definition of the normalized seam coefficient c_{Λ_k} from theorem 13.41. □

Definition 13.47 (Scalar comparison shadow of the witness-indexed bridge)**.** Fix an accepted tick k. Let y be an LS witness and let $\tau^{f,y}_{\Lambda_k}$ be a witness-indexed seam realization source for a filler tag f. A *scalar comparison shadow* for that data is a choice of representative

$$J^{f,y}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$$

such that the dominant raw pairing of the 2T-side representative agrees with the dominant raw pairing of the corresponding representable UOM source:

$$\langle\!\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}(J^{f,y}_{\Lambda_k}),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\rangle\!\rangle_{C_{*,\Lambda_k}} = \langle\!\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\rangle\!\rangle_{C_{*,\Lambda_k}}. \tag{101}$$

When dominant-line realization holds, this is equivalently the equality of normalized dominant pairings

$$\rho^{\mathrm{dom}}_{\Lambda_k}(J^{f,y}_{\Lambda_k}) = \rho^{\mathrm{dom}}_{\Lambda_k}(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}). \tag{102}$$

Definition 13.48 (Witness-indexed source-compatible descendant realization)**.** Fix an accepted tick k, a filler tag f, a witness y, and a witness-indexed seam source $\tau^{f,y}_{\Lambda_k}$. A representative

$$J^{f,y}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$$

is called a *witness-indexed source-compatible descendant realization* if it realizes, on the same LS witness datum, the Limited Scope output and odd Hamiltonization clauses from item **(M1)** and item **(M4)** of Definition 4.2 and differs from the source-insertion representative only by an ambiguity direction:

$$J^{f,y}_{\Lambda_k} - F_{\tau^{f,y}_{\Lambda_k},\Lambda_k} \in \mathcal{A}^{-}_{\Lambda_k}, \tag{103}$$

where $F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}$ is the representable source generator furnished by theorem 13.44. Equivalently, the witness-indexed transported descendant class and the witness-indexed UOM source-insertion realization define the same point of the ambiguity quotient $E^{-}_{\Lambda_k}/\mathcal{A}^{-}_{\Lambda_k}$.

Theorem 13.49 (Same-witness descended-scalar comparison from a source-compatible descendant realization)**.** *Fix an accepted tick k, assume dominant-line realization at Λ_k, and let $\tau^{f,y}_{\Lambda_k}$ be a witness-indexed seam realization source. Assume:*

1. *the source $\tau^{f,y}_{\Lambda_k}$ is represented by the canonical generator $F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}$ from theorem 13.44;*
2. *there exists a witness-indexed source-compatible descendant realization $J^{f,y}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$ in the sense of definition 13.48;*
3. *the three vanishing statements of proposition 13.15 hold at Λ_k.*

Then the raw and normalized comparison identities of definition 13.47 hold:

$$\langle\!\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}(J^{f,y}_{\Lambda_k}),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\rangle\!\rangle_{C_{*,\Lambda_k}} = \langle\!\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\rangle\!\rangle_{C_{*,\Lambda_k}}, \tag{104}$$

$$\rho^{\mathrm{dom}}_{\Lambda_k}(J^{f,y}_{\Lambda_k}) = \rho^{\mathrm{dom}}_{\Lambda_k}(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}). \tag{105}$$

Moreover,

$$\rho^{\mathrm{dom}}_{\Lambda_k}(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}) = \langle\!\langle \Pi_{\mathrm{phys},\Lambda_k}\mathsf{T}_{\Lambda_k}[\tau^{f,y}_{\Lambda_k}],\, \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}\rangle\!\rangle_{C_{*,\Lambda_k}}. \tag{106}$$

If quotient descent also holds at Λ_k, then the common value equals the descended dominant scalar on the transported descendant affine class:

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_k}([\![\Xi^{(1)}_{\Lambda_k}]\!]) = \rho^{\mathrm{dom}}_{\Lambda_k}(J^{f,y}_{\Lambda_k}) = \rho^{\mathrm{dom}}_{\Lambda_k}(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}). \tag{107}$$

Proof. Set

$$A^{f,y}_{\Lambda_k} := J^{f,y}_{\Lambda_k} - F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}.$$

By source compatibility, $A^{f,y}_{\Lambda_k} \in \mathcal{A}^-_{\Lambda_k}$.

Because dominant-line realization holds and the three vanishing statements of proposition 13.15 are assumed, that proposition gives

$$\mathcal{A}^-_{\Lambda_k} \subset \ker \rho^{\mathrm{dom}}_{\Lambda_k}.$$

Hence

$$\rho^{\mathrm{dom}}_{\Lambda_k}\big(J^{f,y}_{\Lambda_k}\big) = \rho^{\mathrm{dom}}_{\Lambda_k}\big(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k} + A^{f,y}_{\Lambda_k}\big) = \rho^{\mathrm{dom}}_{\Lambda_k}\big(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}\big),$$

proving (105).

Since dominant-line realization implies $\delta C^{\mathrm{LoG}}_{\Lambda_k} \neq 0$ and $\rho^{\mathrm{dom}}_{\Lambda_k} = \kappa^{-1/2}_{\Lambda_k}\widetilde{\rho}^{\mathrm{dom}}_{\Lambda_k}$, the raw pairing equality (104) is just the same identity multiplied by the positive factor $\kappa^{1/2}_{\Lambda_k}$.

Equation (106) is exactly the normalized pairing formula of proposition 13.45 applied to the representable source $\tau^{f,y}_{\Lambda_k}$.

If quotient descent also holds, then corollary 13.13 gives

$$\rho^{\mathrm{dom}}_{\Lambda_k}\big(J^{f,y}_{\Lambda_k}\big) = \overline{\rho}^{\mathrm{dom}}_{\Lambda_k}\big([\![\Xi^{(1)}_{\Lambda_k}]\!]\big)$$

for every representative $J^{f,y}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$, which yields (107) after combining with (105). □

Theorem 13.50 (Fixed-cutoff seam witness criterion for descendant overlap)**.** *Fix an accepted tick k. Assume:*

1. *a detector-factorized witness-indexed seam realization at Λ_k;*
2. *a witness y with*
$$\overline{\mathrm{obs}}(y) \neq 0;$$
3. *the seam coefficient is nonzero,*
$$\widetilde{c}_{\Lambda_k} \neq 0;$$
4. *for the chosen filler tag f, the source $\tau^{f,y}_{\Lambda_k}$ is representable on the characterization layer in the sense of definition 13.43;*
5. *there exists a witness-indexed source-compatible descendant realization in the sense of definition 13.48.*

Then:

1. *the compressed welded LoG line is transversely physical at Λ_k, hence dominant-line realization holds at that cutoff;*
2. *the chosen 2T representative satisfies*
$$\rho^{\mathrm{dom}}_{\Lambda_k}\big(J^{f,y}_{\Lambda_k}\big) \neq 0;$$
3. *therefore the transported descendant class has nonzero overlap with the canonically selected compressed dominant odd line at Λ_k;*

4. *if quotient descent also holds at Λ_k, then theorem 13.21 applies and the transported descendant class is dominant-active with well-defined sign ε_{Λ_k}.*

Proof. By theorem 13.41,

$$\widetilde{s}_{\Lambda_k}(f,y) = \sigma_{\mathrm{fill}}(f)\,\widetilde{c}_{\Lambda_k}\,\overline{\mathrm{obs}}(y) \neq 0.$$

The same theorem therefore gives item (1): the compressed line is transversely physical, hence dominant-line realization holds at Λ_k.

By proposition 13.45,

$$\big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}\big(F_{\tau^{f,y}_{\Lambda_k},\Lambda_k}\big),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\big\rangle\!\big\rangle_{C_{*,\Lambda_k}} = \widetilde{s}_{\Lambda_k}(f,y) \neq 0.$$

Applying the comparison identity (101) now gives

$$\big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda_k}\mathcal{R}_{\Lambda_k}\big(J^{f,y}_{\Lambda_k}\big),\, \delta C^{\mathrm{LoG}}_{\Lambda_k}\big\rangle\!\big\rangle_{C_{*,\Lambda_k}} \neq 0.$$

Because dominant-line realization has already been established, $\delta C^{\mathrm{LoG}}_{\Lambda_k}$ is nonzero and $\widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k}$ is defined. By proposition 13.18, this is equivalent to $\rho^{\mathrm{dom}}_{\Lambda_k}(J^{f,y}_{\Lambda_k}) \neq 0$, proving item (2).

Item (3) is exactly the nonzero-overlap condition from proposition 13.20. If quotient descent is also available, apply theorem 13.21 to obtain item (4). □

Smooth-taper branch continuity and witness transport

The previous fixed-cutoff theorem isolates the nonzero witness needed for activity. To transport that witness along an accepted branch one needs a continuity statement for the welded response. At theorem level, the clean analytic choice is the smooth receiver taper. The hard projector may still be used, but only on branch segments where its rank is constant.

Lemma 13.51 (Smooth-taper continuity of the receiver band projector)**.** *Let I be a connected accepted-branch segment. Suppose the receiver band is implemented by a smooth taper*

$$P^{(\mathrm{rec})}_{\mathrm{band}}(\Lambda) = \sum_{\ell,m} w_\ell(\Lambda)\,|Y_{\ell m}\rangle\langle Y_{\ell m}|, \qquad 0 \le w_\ell(\Lambda) \le 1,$$

on the receiver one-particle space, where

$$\sup_{\ell\ge 0} |w_\ell(\Lambda') - w_\ell(\Lambda)| \longrightarrow 0 \qquad \text{as } \Lambda' \to \Lambda$$

for every $\Lambda \in I$. Then

$$\Lambda \longmapsto P^{(\mathrm{rec})}_{\mathrm{band}}(\Lambda)$$

is continuous on I in operator norm. If instead the hard projector $\sum_{\ell\le\ell_\star(\Lambda)} |Y_{\ell m}\rangle\langle Y_{\ell m}|$ is used, the same conclusion holds on every connected subsegment of I on which $\ell_\star(\Lambda)$ is constant.

Proof. In the spherical-harmonic basis $\{Y_{\ell m}\}$, the difference

$$P^{(\mathrm{rec})}_{\mathrm{band}}(\Lambda') - P^{(\mathrm{rec})}_{\mathrm{band}}(\Lambda)$$

is diagonal with eigenvalues $w_\ell(\Lambda') - w_\ell(\Lambda)$. Therefore

$$\big\|P^{(\mathrm{rec})}_{\mathrm{band}}(\Lambda') - P^{(\mathrm{rec})}_{\mathrm{band}}(\Lambda)\big\|_{\mathrm{op}} = \sup_{\ell\ge 0} |w_\ell(\Lambda') - w_\ell(\Lambda)|.$$

The assumed uniform continuity of the taper coefficients is exactly the statement that this operator norm tends to zero as $\Lambda' \to \Lambda$.

For the hard projector, the coefficients take only the values 0 and 1, so discontinuities can occur only when $\ell_\star(\Lambda)$ jumps. On a subsegment where $\ell_\star(\Lambda)$ is constant, the projector is constant and therefore continuous. □

Theorem 13.52 (Smooth-taper branch continuity of the welded physical response)**.** *Let I be a connected accepted-branch segment. Write the welded source-to-covariance map in the UOM factorized form*

$$\mathsf{T}_\Lambda = \Pi_{L(\Lambda)} M_{\varepsilon,\Lambda} \circ \mathsf{K}_{\mathrm{comp},\Lambda} \circ \mathsf{G}_{\mathrm{ret},\Lambda} \circ \mathsf{W}_{\mathrm{src},\Lambda}. \tag{108}$$

Assume:

1. *the receiver band is implemented by a smooth taper satisfying lemma 13.51;*
2. *the operator families*

$$\Lambda \longmapsto \Pi_{L(\Lambda)} M_{\varepsilon,\Lambda}, \qquad \Lambda \longmapsto \mathsf{K}_{\mathrm{comp},\Lambda}, \qquad \Lambda \longmapsto \mathsf{G}_{\mathrm{ret},\Lambda}, \qquad \Lambda \longmapsto \mathsf{W}_{\mathrm{src},\Lambda}, \qquad \Lambda \longmapsto \Pi_{\mathrm{phys},\Lambda}$$

   are continuous on I in operator norm on the corresponding band-limited source and covariance tangent spaces;

3. *the source family*

$$\Lambda \longmapsto \tau_\Lambda$$

   is continuous on I in the welded source norm.

Then the physical covariance response

$$X_\Lambda[\tau_\Lambda] := \Pi_{\mathrm{phys},\Lambda} \mathsf{T}_\Lambda[\tau_\Lambda] \tag{109}$$

depends continuously on $\Lambda \in I$ in the Fisher norm. If the hard receiver projector is used instead, the same conclusion holds on each connected subsegment where $\ell_\star(\Lambda)$ is constant.

Proof. By lemma 13.51, the first factor $\Pi_{L(\Lambda)} M_{\varepsilon,\Lambda}$ is operator-norm continuous along the branch. By assumption, the remaining three UOM factors in (108) and the physical projector $\Pi_{\mathrm{phys},\Lambda}$ are also operator-norm continuous. Hence the composed operator family

$$\Lambda \longmapsto \Pi_{\mathrm{phys},\Lambda} \mathsf{T}_\Lambda$$

is continuous in operator norm on I. Applying this continuous family to the continuous source vector τ_Λ gives continuity of $X_\Lambda[\tau_\Lambda]$ in the Fisher norm.

For the hard projector, the same argument applies on every connected subsegment where $\ell_\star(\Lambda)$ is constant, because there the first factor is constant. □

Corollary 13.53 (Branchwise continuity of the seam coefficient)**.** *Let I be a connected accepted-branch segment and assume:*

1. *the hypotheses of theorem 13.8 on I;*
2. *a detector-factorized branchwise seam realization on I with continuous unit source*

$$\Lambda \longmapsto \tau_\Lambda^{\mathrm{unit}};$$

3. *the hypotheses of theorem 13.52 for that unit source.*

Define the branchwise normalized seam coefficient by

$$c_\Lambda := \langle\!\langle \Pi_{\mathrm{phys},\Lambda} \mathsf{T}_\Lambda[\tau_\Lambda^{\mathrm{unit}}],\, \widehat{\delta C}_\Lambda^{\mathrm{LoG}} \rangle\!\rangle_{C_{*,\Lambda}}. \tag{110}$$

Then c_Λ is continuous on I. Consequently, for every fixed filler tag f and witness y, the normalized seam scalar

$$s_\Lambda(f,y) = \sigma_{\mathrm{fill}}(f)\, c_\Lambda\, \overline{\mathrm{obs}}(y)$$

is continuous on I.

Proof. By theorem 13.52, the response vector

$$\Lambda \longmapsto \Pi_{\mathrm{phys},\Lambda} \mathsf{T}_\Lambda[\tau_\Lambda^{\mathrm{unit}}]$$

is continuous in the Fisher norm. By theorem 13.8, the normalized dominant vector $\widehat{\delta C}_\Lambda^{\mathrm{LoG}}$ is also continuous on I. Pairing these two continuous vectors in the Fisher metric gives continuity of c_Λ. The final statement is then immediate from (95). □

Theorem 13.54 (Branchwise seam witness transport)**.** *Let I be a connected accepted-branch segment. Assume:*

1. *the hypotheses of corollary 13.53;*
2. *quotient descent holds for every $\Lambda \in I$;*
3. *there exists a fixed LS witness y with*

$$\overline{\mathrm{obs}}(y) \neq 0;$$

4. *for some fixed filler tag f, there is a branchwise family of representatives*

$$J_\Lambda^{f,y} \in [\Xi_\Lambda^{(1)}] \qquad (\Lambda \in I)$$

such that, for every $\Lambda \in I$, the corresponding seam source $\tau_\Lambda^{f,y}$ is representable on the characterization layer and the 2T-side witness-to-descendant comparison identity (102) holds.

Then:

1. *the branchwise dominant-response scalar of the chosen descendant representatives is*

$$\rho_\Lambda^{\mathrm{dom}}\big(J_\Lambda^{f,y}\big) = \sigma_{\mathrm{fill}}(f)\, c_\Lambda\, \overline{\mathrm{obs}}(y); \tag{111}$$

2. *the scalar function $\Lambda \mapsto \rho_\Lambda^{\mathrm{dom}}(J_\Lambda^{f,y})$ is continuous on I;*
3. *on every connected subsegment $I' \subset I$ on which $c_\Lambda \neq 0$, the sign*

$$\mathrm{sgn}\big(\rho_\Lambda^{\mathrm{dom}}(J_\Lambda^{f,y})\big)$$

is constant;

4. *because quotient descent holds, the same constant sign is the transported orientation sign of the descendant affine class on I'.*

Proof. For each $\Lambda \in I$, representability of $\tau_\Lambda^{f,y}$ and proposition 13.45 give

$$\rho_\Lambda^{\mathrm{dom}}\big(F_{\tau_\Lambda^{f,y},\Lambda}\big) = \big\langle\!\big\langle \Pi_{\mathrm{phys},\Lambda}\mathsf{T}_\Lambda\big[\tau_\Lambda^{f,y}\big],\, \widehat{\delta C}_\Lambda^{\mathrm{LoG}} \big\rangle\!\big\rangle_{C_{*,\Lambda}} = s_\Lambda(f,y).$$

Applying the branchwise comparison identity (102) then yields

$$\rho_\Lambda^{\mathrm{dom}}\big(J_\Lambda^{f,y}\big) = s_\Lambda(f,y).$$

Now use corollary 13.53 to substitute $s_\Lambda(f,y) = \sigma_{\mathrm{fill}}(f)c_\Lambda\overline{\mathrm{obs}}(y)$, which proves item (1). Item (2) follows because c_Λ is continuous and $\overline{\mathrm{obs}}(y)$ is a fixed nonzero scalar. On any connected subsegment where $c_\Lambda \neq 0$, the right-hand side of (111) is a continuous nonvanishing real-valued function, so its sign is constant. This proves item (3).

Finally, quotient descent implies that $\rho_\Lambda^{\mathrm{dom}}$ is constant on $[\Xi_\Lambda^{(1)}]$, so the sign of one representative is the sign of the whole descendant class. This gives item (4). □

Corollary 13.55 (Seam package discharge of the minimal scalar transport input)**.** *Let I be a connected accepted-branch segment and assume the hypotheses of theorem 13.54. Then the branchwise family*

$$J_\Lambda^{f,y} \in [\Xi_\Lambda^{(1)}] \qquad (\Lambda \in I)$$

satisfies

$$\Lambda \longmapsto \rho_\Lambda^{\mathrm{dom}}\big(J_\Lambda^{f,y}\big) = \sigma_{\mathrm{fill}}(f)\, c_\Lambda\, \overline{\mathrm{obs}}(y)$$

continuously on I. On every connected subsegment $I' \subset I$ such that

$$c_\Lambda \neq 0 \qquad \forall \Lambda \in I',$$

assumption (3) *of theorem 13.56 holds in its minimal scalar form with*

$$J_\Lambda^{\mathrm{desc}} := J_\Lambda^{f,y} \qquad (\Lambda \in I').$$

If, in addition, the three generator-wise vanishing statements of proposition 13.15 hold for every $\Lambda \in I'$, then theorem 13.56 applies on I'.

Proof. Continuity of the displayed scalar function is exactly item (2) of theorem 13.54, together with the explicit formula of item (1). On I', the hypotheses $c_\Lambda \neq 0$ and $\overline{\mathrm{obs}}(y) \neq 0$ imply

$$\rho_\Lambda^{\mathrm{dom}}\big(J_\Lambda^{f,y}\big) \neq 0 \qquad \forall \Lambda \in I'.$$

Together with the continuity statement already proved above, this is exactly assumption (3) of theorem 13.56 in its minimal scalar form. The hypotheses of theorem 13.54 already include those of corollary 13.53, and hence the dominant-line persistence input of theorem 13.56. If the three generator-wise vanishing statements also hold on I', then every hypothesis of theorem 13.56 is in force there. □

Theorem 13.56 (Finite-cutoff Active Response Transport Theorem)**.** *Let I be a connected accepted-branch segment. For each $\Lambda \in I$, let*

$$[\Xi_\Lambda^{(1)}] := J_{0,\Lambda} + \mathcal{A}_\Lambda^- \subset E_\Lambda^-$$

be the parity-transported odd descendant affine class modulo the ambiguity space of definition 13.2. Assume:

1. *the hypotheses of theorem 13.8 hold on I;*

2. *for every $\Lambda \in I$, the three vanishing statements of proposition 13.15 hold;*

3. *there exists a branchwise family of representatives*

$$J_\Lambda^{\mathrm{desc}} \in [\Xi_\Lambda^{(1)}] \qquad (\Lambda \in I)$$

such that

$$\rho_\Lambda^{\mathrm{dom}}(J_\Lambda^{\mathrm{desc}}) \neq 0 \qquad \forall \Lambda \in I,$$

and such that the dominant-response scalar

$$\Lambda \longmapsto \rho_\Lambda^{\mathrm{dom}}(J_\Lambda^{\mathrm{desc}})$$

is continuous on I.

Then, for every $\Lambda \in I$:

1. *the compressed dominant covariance tangent is nonzero,*

$$\delta C_\Lambda^{\mathrm{LoG}} \neq 0,$$

so the dominant line L_Λ^{dom} of definition 13.3 is canonically defined;

2. *the dominant-response coefficient $\rho_\Lambda^{\mathrm{dom}}$ annihilates the odd ambiguity space,*

$$\rho_\Lambda^{\mathrm{dom}}(A) = 0 \qquad \forall A \in \mathcal{A}_\Lambda^-;$$

3. *the transported descendant affine class carries a unique nonzero dominant-response value*

$$\rho_\Lambda^{\mathrm{dom}}(J) = \overline{\rho}_\Lambda^{\mathrm{dom}}\big([\![\Xi_\Lambda^{(1)}]\!]\big) \neq 0 \qquad \forall J \in [\Xi_\Lambda^{(1)}],$$

and hence a well-defined sign

$$\varepsilon_\Lambda := \mathrm{sgn}\Big(\overline{\rho}_\Lambda^{\mathrm{dom}}\big([\![\Xi_\Lambda^{(1)}]\!]\big)\Big) \in \{\pm 1\};$$

4. *the normalized orientation functional*

$$\chi_\Lambda := \nu_\Lambda^{-1} \rho_\Lambda^{\mathrm{dom}} \tag{112}$$

is well defined, annihilates $\mathcal{A}_\Lambda^-$, and satisfies

$$\chi_\Lambda(J) = \varepsilon_\Lambda \qquad \forall J \in [\Xi_\Lambda^{(1)}];$$

5. *along I, the sign ε_Λ is locally constant once an initial orientation is fixed.*

Proof. Item (1) is exactly the conclusion of theorem 13.8.

By assumption, the three generator-wise vanishing statements of proposition 13.15 hold for every $\Lambda \in I$. Therefore proposition 13.15 and theorem 13.16 give item (2) and the existence of the descended functional $\overline{\rho}_\Lambda^{\mathrm{dom}}$ at each cutoff.

Applying proposition 13.20 and theorem 13.21 pointwise to the nonvanishing scalar assumption $\rho^{\mathrm{dom}}_{\Lambda}(J^{\mathrm{desc}}_{\Lambda}) \neq 0$ now yields item (3): the entire descendant affine class has one common nonzero dominant-response value, and its sign ε_{Λ} is well defined.

Since $\nu_{\Lambda} > 0$ by item (3), the normalized functional χ_{Λ} of (112) is well defined. Item (2) implies that χ_{Λ} annihilates $\mathcal{A}^{-}_{\Lambda}$, and item (3) gives

$$\chi_{\Lambda}(J) = \frac{\rho^{\mathrm{dom}}_{\Lambda}(J)}{\left|\overline{\rho}^{\mathrm{dom}}_{\Lambda}\big([\![\Xi^{(1)}_{\Lambda}]\!]\big)\right|} = \varepsilon_{\Lambda} \qquad \forall J \in [\Xi^{(1)}_{\Lambda}],$$

proving item (4).

Finally, assumption (3) provides a continuous nonvanishing real-valued branchwise descendant scalar, while items (1)–(3) above identify that scalar with the transported orientation sign of the full descendant affine class. Since $\Lambda \mapsto \rho^{\mathrm{dom}}_{\Lambda}(J^{\mathrm{desc}}_{\Lambda})$ is continuous and nonzero on the connected set I, its sign is constant, hence in particular locally constant. Using item (3), this is exactly the transported orientation sign of the full descendant affine class, proving item (5). □

Proposition 13.57 (Ambiguity-triviality on the compressed welded image)**.** *Assume the hypotheses of theorem 13.56 at an accepted tick k. Define the reduced welded image of an odd generator $F \in E^{-}_{\Lambda_k}$ by*

$$\mathsf{W}^{\mathrm{red}}_{\Lambda_k}(F) := \rho^{\mathrm{dom}}_{\Lambda_k}(F)\, P^{\mathrm{LoG}}_{-,\Lambda_k},$$

where $P^{\mathrm{LoG}}_{-,\Lambda_k}$ is the canonical compressed odd projector of corollary 10.10. Then

$$\mathsf{W}^{\mathrm{red}}_{\Lambda_k}(F + A) = \mathsf{W}^{\mathrm{red}}_{\Lambda_k}(F) \qquad \forall F \in E^{-}_{\Lambda_k},\ \forall A \in \mathcal{A}^{-}_{\Lambda_k}.$$

Equivalently, $\mathsf{W}^{\mathrm{red}}_{\Lambda_k}$ factors through the ambiguity quotient $E^{-}_{\Lambda_k}/\mathcal{A}^{-}_{\Lambda_k}$. In particular, if $J_1, J_2 \in [\Xi^{(1)}_{\Lambda_k}]$, then

$$\mathsf{W}^{\mathrm{red}}_{\Lambda_k}(J_1) = \mathsf{W}^{\mathrm{red}}_{\Lambda_k}(J_2).$$

Proof. Item (2) of theorem 13.56 gives

$$\rho^{\mathrm{dom}}_{\Lambda_k}(A) = 0 \qquad \forall A \in \mathcal{A}^{-}_{\Lambda_k}.$$

Therefore, for every $F \in E^{-}_{\Lambda_k}$ and $A \in \mathcal{A}^{-}_{\Lambda_k}$,

$$\mathsf{W}^{\mathrm{red}}_{\Lambda_k}(F + A) = \rho^{\mathrm{dom}}_{\Lambda_k}(F + A)\, P^{\mathrm{LoG}}_{-,\Lambda_k} = \big(\rho^{\mathrm{dom}}_{\Lambda_k}(F) + \rho^{\mathrm{dom}}_{\Lambda_k}(A)\big) P^{\mathrm{LoG}}_{-,\Lambda_k} = \rho^{\mathrm{dom}}_{\Lambda_k}(F)\, P^{\mathrm{LoG}}_{-,\Lambda_k}.$$

Since corollary 10.10 fixes $P^{\mathrm{LoG}}_{-,\Lambda_k}$ canonically on the compressed, co-centered, band-locked welded channel, the right-hand side depends only on the ambiguity class of F. If $J_2 - J_1 \in \mathcal{A}^{-}_{\Lambda_k}$, apply the same identity with $F = J_1$ and $A = J_2 - J_1$. □

Proposition 13.58 (Construction of an admissible calibration for the minimal transported interface)**.** *Fix an accepted tick k. Suppose a weak transported-interface package is given:*

$$\mathcal{A}^{-}_{\Lambda_k} \subset E^{-}_{\Lambda_k}, \qquad [\Xi^{(1)}_{\Lambda_k}] = J_{0,\Lambda_k} + \mathcal{A}^{-}_{\Lambda_k},$$

together with a continuous nonzero linear functional

$$\chi_{\Lambda_k} : E^{-}_{\Lambda_k} \to \mathbb{R}$$

and a sign $\varepsilon_{\Lambda_k} \in \{\pm 1\}$ *such that*

$$\mathcal{A}^-_{\Lambda_k} \subset \ker \chi_{\Lambda_k}, \qquad \chi_{\Lambda_k}(J) = \varepsilon_{\Lambda_k} \quad \forall J \in [\Xi^{(1)}_{\Lambda_k}].$$

Assume in addition that $\ker \chi_{\Lambda_k}$ *admits a continuous symmetric coercive bilinear form whose induced norm Hilbertizes it. Choose any vector* $J^\sharp_{\Lambda_k} \in E^-_{\Lambda_k}$ *satisfying*

$$\chi_{\Lambda_k}(J^\sharp_{\Lambda_k}) = 1,$$

any scale $\beta_{\Lambda_k} > 0$*, and any continuous symmetric coercive bilinear form*

$$\langle \cdot, \cdot \rangle^{\mathrm{aux},\chi}_{\Lambda_k} : \ker \chi_{\Lambda_k} \times \ker \chi_{\Lambda_k} \to \mathbb{R}$$

whose induced norm Hilbertizes $\ker \chi_{\Lambda_k}$*. Define*

$$\mathsf{B}_{\Lambda_k}(F, G) := \beta_{\Lambda_k}\, \chi_{\Lambda_k}(F)\chi_{\Lambda_k}(G) + \Big\langle F - \chi_{\Lambda_k}(F) J^\sharp_{\Lambda_k},\, G - \chi_{\Lambda_k}(G) J^\sharp_{\Lambda_k} \Big\rangle^{\mathrm{aux},\chi}_{\Lambda_k}. \tag{113}$$

Then B_{Λ_k} *is a continuous symmetric positive definite bilinear form on* $E^-_{\Lambda_k}$ *whose induced norm Hilbertizes* $E^-_{\Lambda_k}$*, and*

$$\mathsf{B}_{\Lambda_k}(J^\sharp_{\Lambda_k}, K) = 0 \qquad \forall K \in \ker \chi_{\Lambda_k}.$$

Hence $\mathsf{B}_{\Lambda_k}(J^\sharp_{\Lambda_k}, K) = 0$ *in particular for every* $K \in \mathcal{A}^-_{\Lambda_k}$*, and*

$$\mathsf{B}_{\Lambda_k}(J^\sharp_{\Lambda_k}, J^\sharp_{\Lambda_k}) = \beta_{\Lambda_k}.$$

Moreover, for any representative $J^\Xi_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$*, the choice*

$$J^\sharp_{\Lambda_k} := \varepsilon_{\Lambda_k} J^\Xi_{\Lambda_k}$$

is admissible.

Proof. Because $\chi_{\Lambda_k}(J^\sharp_{\Lambda_k}) = 1$, the linear map

$$P^\chi_{\Lambda_k}(F) := F - \chi_{\Lambda_k}(F) J^\sharp_{\Lambda_k}$$

takes values in $\ker \chi_{\Lambda_k}$:

$$\chi_{\Lambda_k}\big(P^\chi_{\Lambda_k}(F)\big) = \chi_{\Lambda_k}(F) - \chi_{\Lambda_k}(F)\chi_{\Lambda_k}(J^\sharp_{\Lambda_k}) = 0.$$

Thus every $F \in E^-_{\Lambda_k}$ has the decomposition

$$F = \chi_{\Lambda_k}(F) J^\sharp_{\Lambda_k} + P^\chi_{\Lambda_k}(F), \qquad P^\chi_{\Lambda_k}(F) \in \ker \chi_{\Lambda_k}.$$

This decomposition is unique: if $F = aJ^\sharp_{\Lambda_k} + K$ with $K \in \ker \chi_{\Lambda_k}$, applying χ_{Λ_k} gives $a = \chi_{\Lambda_k}(F)$.

Equation (113) is therefore well defined, continuous, and symmetric. If $\mathsf{B}_{\Lambda_k}(F, F) = 0$, then

$$\beta_{\Lambda_k} \chi_{\Lambda_k}(F)^2 = 0$$

and

$$\Big\langle P^\chi_{\Lambda_k}(F), P^\chi_{\Lambda_k}(F) \Big\rangle^{\mathrm{aux},\chi}_{\Lambda_k} = 0.$$

Since $\beta_{\Lambda_k} > 0$, we get $\chi_{\Lambda_k}(F) = 0$, hence $P^{\chi}_{\Lambda_k}(F) = F$. Coercivity on $\ker \chi_{\Lambda_k}$ then implies $F = 0$, proving positive definiteness.

The map

$$F \longmapsto \big(\chi_{\Lambda_k}(F), P^{\chi}_{\Lambda_k}(F)\big)$$

is a continuous linear isomorphism from $E^-_{\Lambda_k}$ onto $\mathbb{R} \oplus \ker \chi_{\Lambda_k}$, with inverse

$$(a, K) \longmapsto aJ^{\sharp}_{\Lambda_k} + K.$$

Under this isomorphism, B_{Λ_k} becomes the product Hilbert form

$$(a, K), (b, L) \longmapsto \beta_{\Lambda_k} ab + \langle K, L\rangle^{\mathrm{aux},\chi}_{\Lambda_k},$$

so its induced norm Hilbertizes $E^-_{\Lambda_k}$.

Finally, if $K \in \ker \chi_{\Lambda_k}$, then $P^{\chi}_{\Lambda_k}(J^{\sharp}_{\Lambda_k}) = 0$ and $\chi_{\Lambda_k}(K) = 0$, hence

$$\mathsf{B}_{\Lambda_k}(J^{\sharp}_{\Lambda_k}, K) = 0.$$

Setting $K = J^{\sharp}_{\Lambda_k}$ gives $\mathsf{B}_{\Lambda_k}(J^{\sharp}_{\Lambda_k}, J^{\sharp}_{\Lambda_k}) = \beta_{\Lambda_k}$. If $J^{\Xi}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$, then $\chi_{\Lambda_k}(J^{\Xi}_{\Lambda_k}) = \varepsilon_{\Lambda_k}$, so $J^{\sharp}_{\Lambda_k} := \varepsilon_{\Lambda_k} J^{\Xi}_{\Lambda_k}$ satisfies $\chi_{\Lambda_k}(J^{\sharp}_{\Lambda_k}) = 1$. □

Remark 13.59 (Calibration existence versus stiffness matching). Proposition 13.58 is a construction theorem, not an unconditional existence theorem: it extends a previously chosen Hilbertizing form on $\ker \chi_{\Lambda_k}$ to an admissible bilinear form on the whole weak transported interface. It does *not* identify the scale β_{Λ_k} with the split/leak stiffness $\widetilde{\alpha}(\Lambda_k)$. That physical identification belongs to a later strong or physical layer.

Theorem 13.60 (Rank-one descendant-sector upgrade)**.** *Assume theorem 13.56, and fix an accepted tick k on the branch. Suppose in addition that on the transported descendant sector the projected response map*

$$F \longmapsto \Pi^{\mathrm{dom}}_{\Lambda_k} \Pi_{\mathrm{phys},\Lambda_k} \mathcal{R}_{\Lambda_k}(F), \qquad \Pi^{\mathrm{dom}}_{\Lambda_k} := \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k} \otimes \widehat{\delta C}^{\mathrm{LoG}}_{\Lambda_k},$$

has kernel exactly $\mathcal{A}^-_{\Lambda_k}$. Then the stronger conclusions follow:

1. *the descendant quotient is rank one,*

$$E^-_{\Lambda_k}/\mathcal{A}^-_{\Lambda_k} \cong \mathbb{R};$$

2. *the orientation kernel is exactly the ambiguity space,*

$$\ker \chi_{\Lambda_k} = \ker \rho^{\mathrm{dom}}_{\Lambda_k} = \mathcal{A}^-_{\Lambda_k},$$

   so the descendant sector satisfies the rank-one kernel-identification package with quotient generated by any vector $J^{\mathrm{act}}_{\Lambda_k}$ satisfying $\chi_{\Lambda_k}(J^{\mathrm{act}}_{\Lambda_k}) = 1$;

3. *after choosing any such $J^{\mathrm{act}}_{\Lambda_k}$, any positive scale β_{Λ_k}, and any auxiliary coercive Hilbertizing form on $\ker \chi_{\Lambda_k}$, proposition 13.58 yields an admissible calibration form. Recovering the stiffness-matched formula of proposition 12.17 requires additional physical identifications, namely $\chi_{\Lambda_k} = \rho_{\Lambda_k}$, $\ker \chi_{\Lambda_k} = \mathcal{T}^{\mathrm{op}}_{\Lambda_k}$, agreement of the chosen normalized generator with the split/leak active generator, and $\beta_{\Lambda_k} = \widetilde{\alpha}(\Lambda_k)$.*

Proof. By theorem 13.56, the dominant line is defined at Λ_k, the scalar ν_{Λ_k} is positive, and

$$\chi_{\Lambda_k} = \nu_{\Lambda_k}^{-1} \rho_{\Lambda_k}^{\mathrm{dom}}.$$

Hence $\ker \chi_{\Lambda_k} = \ker \rho_{\Lambda_k}^{\mathrm{dom}}$.

For every $F \in E_{\Lambda_k}^{-}$, the definition of $\rho_{\Lambda_k}^{\mathrm{dom}}$ gives

$$\Pi_{\Lambda_k}^{\mathrm{dom}} \Pi_{\mathrm{phys},\Lambda_k} \mathcal{R}_{\Lambda_k}(F) = \rho_{\Lambda_k}^{\mathrm{dom}}(F)\, \widehat{\delta C}_{\Lambda_k}^{\mathrm{LoG}}.$$

Therefore the kernel of the projected response map is exactly $\ker \rho_{\Lambda_k}^{\mathrm{dom}}$. By the additional hypothesis, that kernel is $\mathcal{A}_{\Lambda_k}^{-}$, proving item (2).

By theorem 13.56, there exists $J_{\Lambda_k}^{\mathrm{desc}} \in [\Xi_{\Lambda_k}^{(1)}]$ with $\rho_{\Lambda_k}^{\mathrm{dom}}(J_{\Lambda_k}^{\mathrm{desc}}) \neq 0$, hence the projected response map is nonzero. Its image is automatically contained in the one-dimensional line $L_{\Lambda_k}^{\mathrm{dom}}$, so the first isomorphism theorem gives

$$E_{\Lambda_k}^{-} / \mathcal{A}_{\Lambda_k}^{-} \cong \mathrm{im}\big(\Pi_{\Lambda_k}^{\mathrm{dom}} \Pi_{\mathrm{phys},\Lambda_k} \mathcal{R}_{\Lambda_k}\big) \cong \mathbb{R},$$

which is item (1). Any vector $J_{\Lambda_k}^{\mathrm{act}}$ with $\chi_{\Lambda_k}(J_{\Lambda_k}^{\mathrm{act}}) = 1$ lies outside $\ker \chi_{\Lambda_k} = \mathcal{A}_{\Lambda_k}^{-}$, so its quotient class generates this one-dimensional quotient, completing item (2).

Item (3) is now an application of proposition 13.58 with the weak-interface data supplied by theorem 13.56 and the chosen normalized vector $J_{\Lambda_k}^{\mathrm{act}}$. The final sentence identifies when this weak calibration construction coincides with the stiffness-matched formula of proposition 12.17. □

A Appendix: BY kernel discretization and scaling with ΔI

The discretized BY kernel constructs a welded time spectrum $\phi(u)$ and a welded spatial spectrum on S^3 harmonics; it then chooses an effective amplitude A_{eff} so that $A_{\mathrm{eff}}^2 \propto \Delta I / G(s_t, s_x; \Lambda)$ where G is a (no-amplitude) throughput. The integrated BY penalty is quadratic in the signal and thus linear in ΔI. This is the phenomenological origin of the manuscript factorization $\Theta_k = K(\bar{\Lambda}_k)\Delta I_k$.

B Appendix: Minimal bibliography

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